

Clash of Bubbles



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Declaration

We have read the project guidelines and we understand the meaning of academic dishonesty, in particular plagiarism and collusion. We hereby declare that the work we submitted for our final year project, entitled **Clash of Bubbles** is original work and has not been printed, published or submitted before as final year project, research work, publication or any other documentation.

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Statement of Submission

This is to certify that **Abdullah Noor Hussain** Roll No. **70109501**, **Zarar Azeem** Roll No. **70113410** and **Jawad Ahmad** Roll No. **70113698** have successfully submitted the final project named as: **Clash of Bubbles** at Computer Science & IT Department, The University of Lahore, Lahore Pakistan, to fulfill the partial requirement of the degree of **BS in Computer Science**.

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Signature:

Date:

Dedication

We dedicated this project to our Parents, Teachers, and Friends. Their unwavering support and patience were crucial in completing this project successfully. Special thanks also go to our dedicated group members who, through mutual support and collective effort, played a vital role in bringing this project to fruition.

Acknowledgement

We truly acknowledge the cooperation and help make by **Muhammad Zaman Aslam Lecturer** from **University of Lahore**, whose guidance and unwavering support have been instrumental in the successful completion of this final year project. Their expertise, insightful feedback, and encouragement have significantly enriched my academic experience, and I am truly thankful for their invaluable contributions.

Date:

Dec 19, 2023

Abstract

The "Clash of Bubble" project is a Unity-based game designed to provide an engaging and accessible gaming experience. Focused on the popular bubble-popping mechanic, the game introduces customizable AI difficulty levels and multiplayer functionality to cater to individual player preferences and foster social interaction. Leveraging Unity's capabilities, the project aims to deliver seamless cross-platform gameplay, visually appealing graphics, and an immersive user interface. In comparison to existing solutions like Bubble Witch Saga and Bubble Shooter Classic, "Clash of Bubble" stands out by combining traditional gameplay with innovative features, ensuring a personalized and dynamic gaming journey for players of all skill levels. The proposed solution emphasizes Unity's tools for 2D game development, responsive design, networking for multiplayer, and rigorous testing to optimize performance and gameplay satisfaction. The project commits to delivering a high-quality, continuously refined 2D game experience that captures player attention and sets itself apart in the competitive gaming industry.

Area of the Project

We are working on a project to make a **game** that can be played on different platforms like the **web** and **mobile**. We're adding features using **machine learning** to make the game more interesting. We're also making it **multiplayer** so that you can play with your friends.

Technologies used

- **Unity:** Unity is a powerful game development platform that allows creators to build interactive and engaging games for various devices.
- **C#:** C# is a programming language commonly used with Unity, enabling developers to write code and control the behavior of their games.
- **Lucid Chart:** Lucid Chart is a visual collaboration tool that helps in creating diagrams and charts for clear and organized project planning.
- **Word:** Word is a popular word processing software used for creating and editing documents with a wide range of formatting options.
- **Draw.io:** Draw.io is a user-friendly online tool for creating diagrams and flowcharts, making it easy to visualize and communicate ideas.
- **MongoDB:** MongoDB is a NoSQL database system that excels in handling large amounts of flexible, unstructured data with scalability and versatility.

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Chapter 1: Introduction to the Problem

1.1 Introduction

Clash of bubble is a Unity-based game designed to deliver an engaging and colorful gaming experience. In this game, players are transported to a lively world where they encounter unique and challenging levels. The game focuses on a simple yet addictive gameplay mechanic, requiring players to pop bubbles to progress through levels. Each level introduces new challenges and excitement, enhancing the overall experience. The intuitive controls add an extra layer of interest to the gameplay.

The beauty of "Clash of bubble" lies in its simplicity, making it accessible to players of all ages. This game not only provides entertainment but also offers an immersive and visually appealing gaming experience.

1.2 Purpose

As we know that gaming industry has experienced a remarkable surge in the popularity of web-based and mobile games, captivating players of all ages. However, a common challenge arises as many of these games lack the ability to provide a personalized and challenging experience tailored to individual player preferences. This often leads to players feeling either overwhelmed or bored, limiting their overall engagement and enjoyment.

Clash of bubble, is a dynamic gaming experience powered by the versatile Unity game engine. Clash of bubble aims to tackle this issue by offering a gaming experience that is both customizable and challenging. Unity, renowned for its capabilities in crafting immersive and interactive gaming environments, serves as the technological backbone of Clash of bubble, ensuring a seamless and visually stunning gameplay.

By harnessing Unity's extensive toolset and versatile features, Clash of bubble is set to revolutionize the way players interact with games. Players can anticipate highly customizable AI difficulty levels that adapt to their skills and preferences, providing a tailored experience. Additionally, Clash of bubble introduces a captivating multiplayer aspect, bringing together gamers for an engaging experience that transcends different platforms.

1.3 Objective

- Develop a cross-platform game using Unity 2D for seamless play on web browsers and mobile devices.
- Incorporate customizable AI difficulty levels (easy, hard, very hard) to allow players to tailor the challenge based on preference and skill level.
- Integrate multiplayer functionality for real-time matches, fostering social interaction and healthy competition among players.
- Design an immersive and visually appealing user interface to enhance the overall gaming experience.
- Optimize the game for smooth performance across different devices, minimizing loading times and lag, and conduct rigorous testing to ensure balanced and enjoyable gameplay based on player feedback.

1.4 Existing Solution

In the puzzle gaming domain, Clash of bubble stands out among competitors like Bubble Witch Saga and Bubble Shooter Classic by introducing unique features that elevate the player experience.

- **Bubble Witch Saga:**

Bubble Witch Saga, developed by King, boasts a substantial player base with its recognized bubble shooter gameplay, diverse levels, power-ups, and social features, Clash of bubble takes innovation a step further. Clash of bubble sets itself apart by integrating customizable AI difficulty levels and multiplayer functionality. This customization allows players to tailor the challenge level to their preferences and engage in real-time matches against friends or other players, enhancing the interactive and dynamic nature of the gameplay.

- **Bubble Shooter Classic:**

Bubble Shooter Classic offers a traditional bubble shooter experience with varied levels and challenges, but it lacks adaptive difficulty levels and multiplayer features. In response, Clash of bubble addresses these shortcomings by providing customizable AI difficulty levels and

multiplayer functionality. This strategic move enables players to immerse themselves in a challenging gameplay experience that adapts to their skill levels, fostering social connections and competitiveness as they compete against friends or other players.

The incorporation of customizable AI difficulty levels and multiplayer functionality is the hallmark of Clash of bubble, distinguishing it from its competitors. The game guarantees a personalized and engaging gaming journey, ensuring that players of all skill levels can immerse themselves fully in the gameplay. The development team remains dedicated to not only preserving the essence of the classic bubble shooter genre but also introducing innovative features that enhance the overall gaming experience, setting Clash of bubble as the superior choice for players seeking a blend of tradition and innovation.

1.5 Proposed Solution

- Leverage Unity for 2D game development, ensuring seamless cross-platform gameplay.
- Implement responsive design in Unity to adapt to diverse screen sizes and resolutions.
- Integrate customizable AI difficulty levels (easy, hard, and very hard) within Unity.
- Utilize Unity's networking for real-time multiplayer, fostering social interaction and competition.
- Design an intuitive 2D interface using Unity's UI tools for an immersive user experience.
- Capture player attention with visually appealing graphics, enhancing overall engagement.
- Optimize game performance on web browsers and mobile devices using Unity's features.
- Minimize loading times and eliminate lag for a seamless and enjoyable gaming experience.
- Conduct rigorous testing of 2D gameplay mechanics within Unity.
- Iterate based on player feedback to ensure balanced and satisfying gameplay.
- Commit to delivering a high-quality 2D game using Unity.
- Continuously refine gameplay and incorporate improvements based on player feedback and industry trends.

Chapter 2: Software Requirement Specification

2.1 Introduction

This chapter is like a helpful guide that tells everyone what the game needs to have and how it should work. In the game, you get to explore various levels filled with bubbles and exciting challenges. Your main aim is to pop the bubbles and progress through the levels. The chapter explains all the important things the game should do, like the actions you can take in the game (functional aspects), and also how the game should feel and look (non-functional aspects).

2.1.1 Purpose

The purpose of system requirement specification is that to ensure clear communication between stakeholders by documenting the requirements of this game development and provide a common understanding among developers, testers, projects managers, and other team members. it serves as a reference document for developers to understand the functional and non-functional requirements of the game clash of bubbles, it provides a basis for validation whether the delivered project meets the requirements.

2.1.2 Scope

The software product to be produced is a game titled "Clash of Bubbles". This game is designed to entertain users by providing an engaging and interactive experience through the Unity game development platform. This game allow players to aim and shoot bubbles to match and eliminate groups of the same color, AI or Multiple players playing as an competitor, Feature multiple levels of increasing difficulty and provide user interface for seamless gameplay. This game will not require internet for basic gameplay. The game will provide user with a fun and enjoyable gaming session and enhance user's hand-eye coordination and strategic thinking.

Benefit of playing this game is that it provides a visually appealing and enjoyable game for users of all ages. Objective of this game is to achieve user satisfaction through smooth gameplay, captivating graphics and increasing levels with different rewards. Goal for this game development is to gain user attention and interest by giving them different features like daily rewards on different tasks and competitive gaming with friends. Ultimately, aim of this game development is to provide user an interesting and relaxing game experience.

2.1.3 Definitions, acronyms, and abbreviations

Clash Of Bubbles: The software product to be developed, a game designed for entertainment purpose using the Unity game development platform.

Gameplay: The interactive and dynamic experience provided to players while navigating through different levels, shooting bubbles, and achieving objectives.

User Interface (UI): The graphical and interactive elements through which players interact with the game, including menus, buttons.

SRS: System Requirements Specification

ML: Machine Learning

AI: Artificial Intelligence

GUI: Graphical User Interface

UI: User Interface

SDK: Software Development Kit

FPS: Frames Per Second

2D: Two-dimensional

UDP: User Datagram Protocol

2.2. Overall description

This section provides a comprehensive overview of the "Clash of Bubbles" game, covering key aspects such as its technical framework, user interfaces, hardware and software compatibility, communication protocols, and memory requirements

2.2.1 Product perspective

- **System interfaces:**

The clash of bubbles game uses the Unity game engine for development and execution, which is a famous game development tool. The game will seamlessly function on different web browsers like chrome, safari etc. for web version and for mobile version the game is designed to work with both

IOS and Android operating System. The game ensures a consistent experience across various platforms, maintaining similar gameplay mechanics, features, and user interfaces.

- **User interfaces:**

The user interface of the "Clash of Bubble" game includes:

Responsive Design: The user interface adapts responsively to different screen sizes, ensuring a seamless experience on both web browsers and mobile devices.

Main Menu: Allows users to start or restart the game, access settings for like adjusting the sound, and view high scores, Consistent menu structure across web and mobile platforms.

Gameplay Screen: Displays the game grid, player current score on this level, and remaining lives.

End-of-Level Screen: Shows the player's performance at the end of each level, shows earned stars on the basis of the achieved score.

Touch Controls: Mobile users can interact with the game using touch gestures

- **Hardware interfaces:**

The game is designed to run on standard desktops, laptops and mobile devices including both IOS and Android platforms. Compatibility with a range of processors commonly found in modern desktops, laptops, smartphones and tablets. Game is able to operate on devices with at least 2 GB RAM. Keyboards and mouse required for desktop and laptops gameplay interaction and touchscreen gestures for smartphones and tablets. Requires a stable internet connection for loading game and using other features like multiplayer. Smartphones and tablets should have sufficient free storage space for app installation.

- **Software interfaces:**

The game relies on the Unity game engine for proper development and execution. Unity provides functionalities for implementing AI algorithms and decision-making processes.

- **Communications interfaces:**

The multiplayer feature incorporates networking protocol UDP (User Datagram Protocol) for real-time communication between players, including game state synchronization and player actions.

The AI competitor communicates with the game's core logic to analyze player moves and determine strategic responses. Both web and mobile versions require a stable internet connection for enabling multiplayer features communication.

- **Memory:**

Memory requirements vary based on the platform, with optimal performance on desktops and laptops must have minimum 4 GB RAM or more for the web version and at least 2 GB RAM for smartphones and tablet devices.

- **Operations:**

Normal operations are like Users initiate the game from the main menu, selecting options such as "Play" or "Start Game" to start playing, interacting with the game by aiming and shooting bubbles to match and eliminate groups of the same color to increase score, Users navigate through various menus, including settings, sound setting, and level selection, Users can customize settings such as sound volume, music, and visual effects for a personalized gaming experience. Special operations are like Users can join multiplayer sessions with other players and with players for competitive playing, special operations might include (In multiplayer mode) that players can communicate with each other using in-game chat or predefined messages.

Users engaging with the AI competitor during single-player session in which AI plays as an competitors and adjusting the difficulty level of AI, users will have the option to perform manual backups of their game progress like in case of data loss player can restore the game state from the backup, another special operation will be like user will receive notification for new levels and game updates.

- **Site adaptation requirements:**

Before playing this game users of desktops and laptops should have to check the availability of Internet for all type of features because on these platforms game is played on web on the other hand users of smartphones should require internet for installing of the game and using features like multiplayer gaming but no need of internet for single player gaming experience on smartphones.

2.2.2 Product functions

ID:	FR_01			
Name:	Create Account			
Description	Input	Output	Requirements	Basic Workflow
Enter details to create account	Name, Email, Password etc.	Account created	Internet Connectivity	Enter correct information and click submit button. System saves the record in database

Table 01 Functional Requirement create account

ID:	FR_02			
Name:	Login			
Description	Input	Output	Requirements	Basic Workflow
Enter Username & Password to Login	Username Or Email, Password	Profile displayed	Internet Connectivity	Enter correct information. click submit button. System Check the record in database and if exists display the Profile.

Table 02 Functional Requirement login

ID:	FR_03			
Name:	Manage Account			
Description	Input	Output	Requirements	Basic Workflow
Enter Username & Password to Login	Username Or Email, Password	Profile displayed	Internet Connectivity	Enter correct information. click submit button. System Display the Profile and options to modify your profile.

Table 03 Functional Requirement Manage Account

ID:	FR_04			
Name:	Starting the Game			
Description	Input	Output	Requirements	Basic Workflow
Users able to start new game session easily.	User initiating command "Start Game".	Display of Starting Visuals.	Starting Game option.	User selects "Start Game" from the main menu. System initializes the game grid, graphics, and other necessary components. Player enters the gameplay interface.

Table 04 Functional Requirement Starting the Game

ID:	FR_05			
Name:	Quitting the Game			
Description	Input	Output	Requirements	Basic Workflow
Users able to quit the game sessions.	User initiating Command "Quit Game".	Game session terminated; Main Menu screen displayed.	game quitting option	User selects "Quit" from the in-game menu. System prompts for confirmation. If confirmed, the game session terminates. the player returns to the main menu.

Table 05 Functional Requirement Quitting the Game

ID:	FR_06			
Name:	Gameplay			
Description	Input	Output	Requirements	Basic Workflow
Core game mechanics, such as launching bubbles, matching colors, and scoring points.	Player interactions such as mouse clicks for desktop and laptops, touching gestures for smartphones.	Updates game Visuals such as increasing score on matching.	Mousepad or Mouse for desktops or laptops and touchscreen for smartphones and tablets.	Player launches a bubble. System calculates matches and updates the game grid. Points are awarded based on matched bubbles.

Table 06 Functional Requirement Gameplay

ID:	FR_07			
Name:	Sound Adjustment			
Description	Input	Output	Requirements	Basic Workflow
Players having the ability to adjust sound settings to customize their auditory experience.	Player adjusts volume levels for music, effects, and overall game sound.	immediate application of changes.	no requirements	Player accesses the game settings menu. System provides options to adjust music, effects, and overall sound volume. Player modifies settings, and changes take effect.

Table 07 Functional Requirement Sound Adjustment

ID:	FR_08			
Name:	Rewards System			
Description	Input	Output	Requirements	Basic Workflow
tracking Users achievements, unlocks rewards, and supports a sense of progression within the game.	Completing tasks	Unlocked Rewards	Internet Connectivity, Registered User Account	Player completes a level or achieves a milestone. System updates player's progression status and awards rewards.

Table 08 Functional Requirement Reward System

ID:	FR_09			
Name:	Social Features			
Description	Input	Output	Requirements	Basic Workflow
Social Interaction like sending friend requests to friends, profile picture uploads.	Friend Requesting, chat messages.	New friends Lists, Updated Profile Pictures.	Internet Connectivity, Registered User Account	Player sends/receives friend requests accepts or delete it. updates new profile picture and selects done new profile picture uploaded

Table 09 Functional Requirement Social Features

ID:	FR_10			
Name:	Notification System			
Description	Input	Output	Requirements	Basic Workflow
Notification System to Alert Users for friend Requests, Match Invitations, in-game events, or game updates.	friends match request, in-game invitation or any game update.	Notification message pops on device.	Internet Connectivity	System generates notifications based on user activities. Notifications are displayed to the player.

Table 10 Functional Requirement Notification System

ID:	FR_11			
Name:	Security Measures			
Description	Input	Output	Requirements	Basic Workflow
Secure authentication and security features to ensure security of user's data.	User login details.	Secure authentication.	Internet Connectivity and valid login credentials	User provides login credentials. System authenticates and grant access.

Table 11 Functional Requirement Security Measures

ID:	FR_12			
Name:	Friend Request Handling			
Description	Input	Output	Requirements	Basic Workflow
users able to send and receive friend requests.	Player sends/receives friend requests.	Confirmation of friend request, updated friend list.	Internet connectivity, registered users.	user selects a user and sends a friend request. System notifies the recipient and updates friend lists

Table 12 Functional Requirement Friend Request Handling

ID:	FR_13			
Name:	Multiplayer Features			
Description	Input	Output	Requirements	Basic Workflow
real-time multiplayer interactions, including match making, leaderboards ranking, player disconnect handling.	Multiplayer matches	Match results, leaderboard ranking	Internet connectivity and both friends valid accounts	Player selects multiplayer mode, requests friend for match, Friend accepts request, Players interact in real-time game, and system updates leaderboard rankings.

Table 13 Functional Requirement Multiplayer Features

ID:	FR_14			
Name:	Matchmaking Request			
Description	Input	Output	Requirements	Basic Workflow
user able to request to be matched with opponents in multiplayer mode.	Player matchmaking request.	Confirmation of matchmaking request, opponent selected or accepted match request.	internet connectivity.	Player selects "Matchmaking" from the multiplayer menu, confirm or request for match, enters the game session with selected opponent.

Table 14 Functional Requirement Matchmaking Request

ID:	FR_15			
Name:	AI Competitor Module			
Description	Input	Output	Requirements	Basic Workflow
AI-driven competitor with adjustable difficulty levels, real-time decision-making, and continuous learning capability.	AI selection for as an opponent.	AI plays as opponent with decision making ability.	Internet connectivity.	user starts a game session with AI, AI competitor analyzes player moves and adapts strategies, System updates game based on AI and player actions.

Table 15 Functional Requirement AI Competitor Module

ID:	FR_16			
Name:	AI Difficulty Adjustment			
Description	Input	Output	Requirements	Basic Workflow
players able to adjust the difficulty level of AI competitor.	Player adjusts AI difficulty settings.	updated AI behavior.	Internet connectivity.	Player adjusts AI difficulty, System confirms the adjustment and AI adapts behavior accordingly.

Table 16 Functional Requirement AI Difficulty Adjustment

2.2.3 User characteristics

The game is suitable for users of all ages, Accessible to users with varying educational backgrounds, offering simple gameplay mechanics, the game appeals to users who enjoy puzzle solving and strategy in a gaming environment, game could be played by both casual players as well as experienced players, Suitable for users with different levels of technical expertise although, game's User-friendly interface minimizes the need for advanced technical knowledge, Adaptable to users with varying internet connectivity levels, this game is appealing to those users who prefer multiplayer interactions, game is designed for both mobile and desktop devices users, Appeals to users with different levels of experience in AI-driven gameplay.

2.2.4 Constraints

- **Hardware Limitations:**

The game must operate within the specified hardware requirements, considering limitations in processing power, memory, and graphics capabilities.

- **Database Limitation:**

The size of the database may be a limitation, especially if the game experiences rapid growth in terms of users and data volume.

- **Higher-Order Language Requirements:**

Higher-order language requirements can be a constraint if the chosen programming language for the "Clash Of bubbles" game lacks necessary libraries or community support, impacting development efficiency and flexibility.

- **Interfaces to Other Applications:**

Integration with external applications requires careful consideration of evolving API policies, because Integration with third-party applications or platforms must adhere to their respective APIs and guidelines.

- **Cross-Platform Compatibility:**

Ensuring a seamless experience across different platforms introduces constraints in terms of synchronization, gameplay parity, and feature implementation.

- **Communication Protocols:**

Selecting and implementing effective communication protocols for player interactions and real-time updates can be a constraint.

- **Internet Connectivity:**

The game may offer offline gameplay, certain features such as multiplayer modes or online updates may require a stable internet connection.

- **Storage Space:**

The game's size and storage requirements must be optimized, especially for mobile devices with limited storage capacity.

- **Budgetary Constraints:**

Financial limitations may affect the selection of technologies, tools, and third-party services, potentially impacting the overall quality and features of the game.

2.2.5 Assumptions and dependencies

- **Device Compatibility:**

Assumption: Users will access the game on devices meeting specified hardware and software requirements.

Dependency: The game's performance relies on user's device compatibility.

- **Real-time Decision Making:**

Assumption: The AI will make real-time decisions that enhance the user's gaming experience.

Dependency: The success of the AI feature relies on efficient decision-making algorithms.

- **Stable Internet Connection:**

Assumption: Users engaging in multiplayer feature have a stable internet connection.

Dependency: The effectiveness of multiplayer interactions depends on the quality and stability of users' internet connections.

- **Server Infrastructure:**

Assumption: The server infrastructure will support the expected user load and provide a reliable gaming experience.

Dependency: Server scalability and stability is important for accommodating user growth and maintaining gameplay performance.

2.2.6 Apportioning of requirements

This part talks about putting some features on hold for future updates. So, instead of trying to do everything at once, we focus on the most important things first. This way, we can make the game better over time, adding cool stuff based on what players want and new technology that becomes available. It's like building the game step by step to keep it fresh and exciting.

- **Advanced AI Enhancements:**

Advanced AI capabilities, such as more improved decision-making and learning algorithms, may be deferred to future versions for continuous improvement.

- **Expanded multiplayer features:**

Game could include organized tournaments, social hubs for player interaction, and customizable avatars to enhance the sense of community and personalization in future updates.

- **Integration with Emerging Technologies:**

Integration with emerging technologies, such as augmented reality or new gaming platforms, may be explored in future releases.

- **Enhanced Graphics and Animations:**

Advanced graphics and animation enhancements, such as more visual effects, may be deferred to future versions for continued refinement.

2.3 Specific requirements

2.3.1 Functional Requirement

User Account Management: Users can create accounts, log in securely, update profile information, and recover passwords.

Gameplay: Include the core game mechanics, such as launching bubbles, matching colors, and scoring points.

Rewards System: system that tracks player achievements, unlocks rewards, and supports a sense of progression within the game.

Multiplayer Features: user real-time multiplayer interactions, including matchmaking, leaderboards, and player disconnect handling.

AI Competitor Module: user can play with AI-driven competitor with adjustable difficulty levels, real-time decision-making, and continuous learning capability.

Social Features: Enabling users social interactions, including friend requests, profile picture upload, and a social hub for player engagement.

Notification System: notification system to alert users about friend requests, match invitations, and in-game events.

Cross-Platform Compatibility: Ensuring seamless accessibility and functionality across various platforms, including web browsers, IOS, and Android.

Game State Management: mechanisms for initializing and managing the game state, allowing users to start new games or resume from previous game.

2.3.2 Non-functional Requirements

- **Usability:**

The user interface (UI) should be intuitive, facilitating easy navigation and gameplay for players of varying skill levels.

- **Reliability:**

The game should be reliable means minimizing system crashes or unexpected downtime, ensuring a stable and consistent user experience.

- **Performance:**

The game should run smoothly with minimal lag, providing responsive gameplay even during multiplayer interactions and AI-driven components.

- **Portability:**

The game should be compatible with various platforms, including web browsers, IOS, and Android, maintaining consistent functionality across devices.

- **Maintainability:**

The game code should be easy to maintain and update. It should be well-documented and modular, so that changes can be made easily.

- **License Agreement:**

The game will have a user-friendly license agreement outlining terms of how to use the game, privacy policies, and any applicable legal restrictions.

Chapter 3: Use Case Analysis

General Use Case:

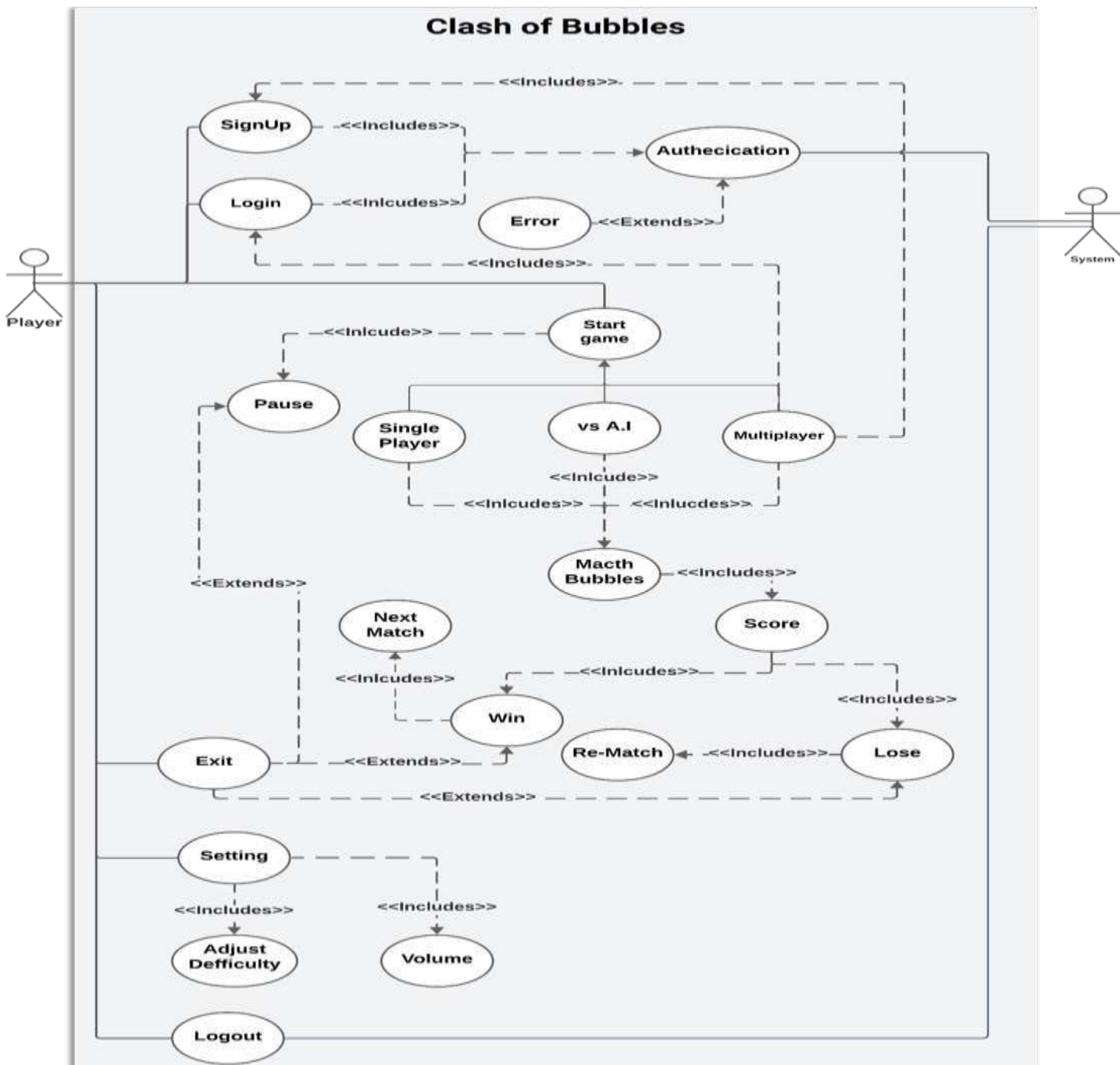


Figure 1 Use Case Diagram Game

1. Sign-Up (Create Account)

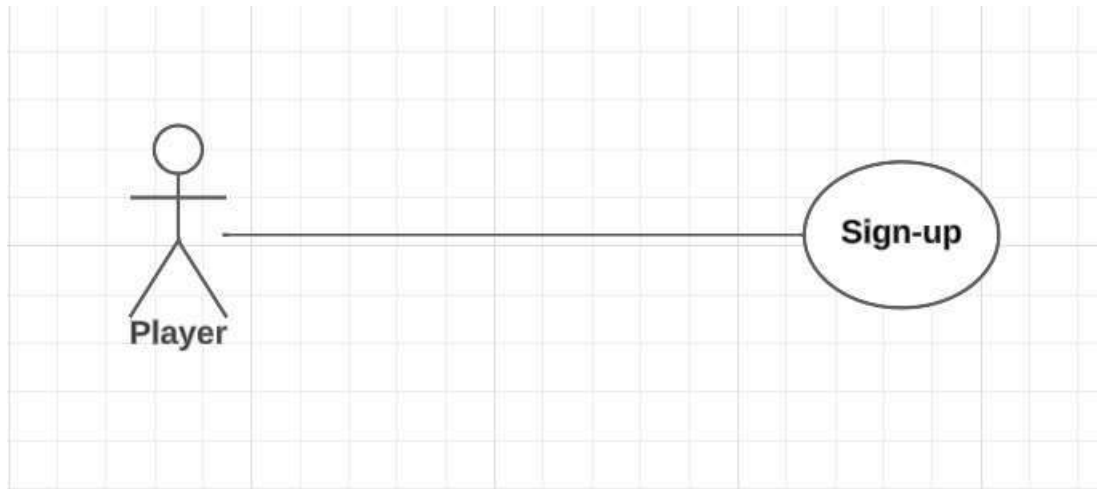


Figure 2 Use Case Diagram Sign-Up

Use Case ID	UC_01	
Use Case Name	Sing-Up	
Reference	See FR 1	
Description	Steps for creating a new Account	
Primary actor	Player	
Secondary Actor		
Pre-Conditions	1. Should have the application installed. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player has created an account within the game and can now enjoy multiplayer and other beneficial features. e.g. progress secured.	
Basic Flow	Actor Action	System Action
	Players enter his/her email ID and password and then the Player clicks on the Sign-Up button to create a New Account.	The system verifies the email ID and password and assigns a unique ID to the player signing up for the game.
Alternative Flow	Players can play the game without signing up for the game except for multiplayer.	

Table 17 Use Case Sign-Up

2. Sign-Up Authentication (Checks for valid credentials)

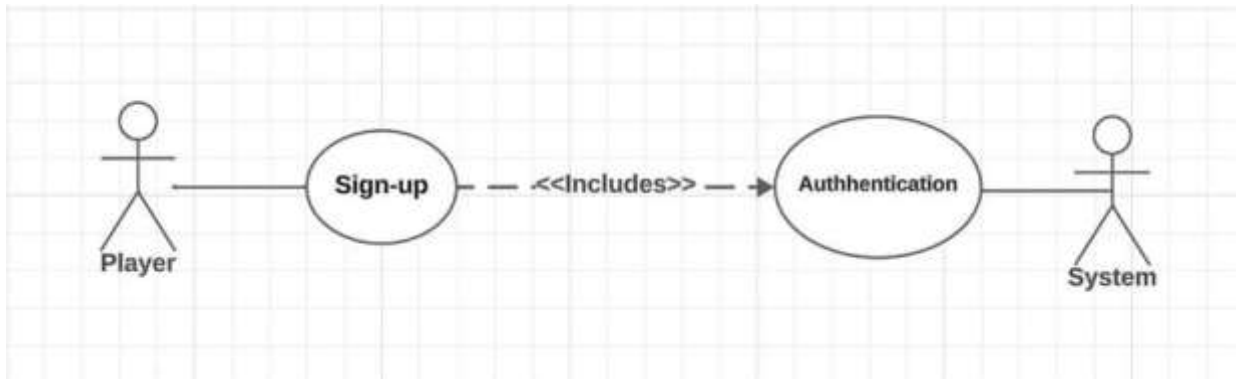


Figure 3 Use Case Diagram Sign-Up Authentication

Use Case ID	UC_02	
Use Case Name	Authentication	
Reference	See FR 2	
Description	The system checks whether the entered email/password is Authentic or not.	
Primary actor	Player	
Secondary Actor	System	
Pre-Conditions	1. The player should have entered a valid email and password. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player has entered his email ID/password and is now being verified for valid credentials.	
Basic Flow	Actor Action	System Action
	Players enter his/her email ID and password and click on the sign-up button.	The system verifies the email ID and password and if it checks out authentic will create an Account.
Alternative Flow	You can play the game without signing up for the game except for multiplayer.	

Table 18 Use Case Sign-Up Authentication (Sing-Up)

3. Sign-Up Error (Display an Error)

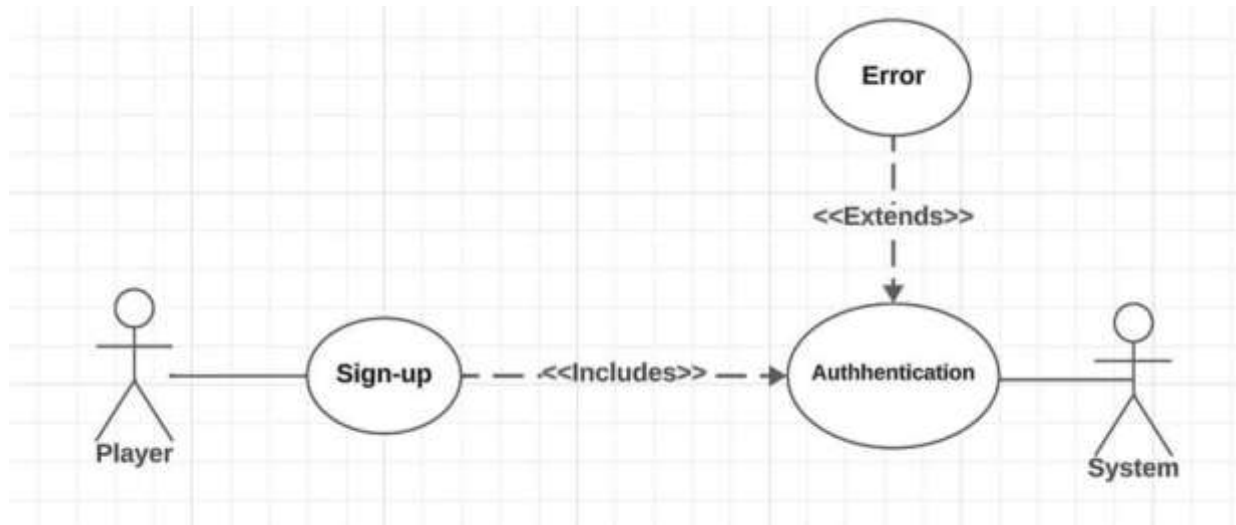


Figure 4 Use Case Diagram Sign-Up Error

Use Case ID	UC_03	
Use Case Name	Error	
Reference	See FR 3	
Description	If Authentication fails the system will display an error.	
Primary actor	Player	
Secondary Actor	System	
Pre-Conditions	1. The player should have entered an invalid email and password. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	An error message will be displayed about the player entering the incorrect email/password.	
Basic Flow	Actor Action	System Action
	Players enter an invalid email ID and password and click on the sign-up button.	The system verifies the email ID and password and displays an error message.
Alternative Flow	You can play the game without signing up for the game except for multiplayer.	

Table 19 Use Case Error (Sign-Up)

4. Login (Login to Account)



Figure 5 Use Case Diagram Login

Use Case ID	UC_04	
Use Case Name	Login	
Reference	See FR 4	
Description	Steps for logging into an existing account.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should have an Existing Account. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player has created an account within the game and can now enjoy multiplayer and other beneficial features. e.g. progress secured.	
Basic Flow	Actor Action	System Action
	Players enter his/her email ID and password and then the Player clicks on the login button to log into an existing Account.	The system verifies the email ID and password and assigns a unique ID to the player signing up for the game.
Alternative Flow	The player can play the game without logging in except for the multiplayer.	

Table 20 Use Case Login

5. Login Authentication (Verifies If Account Exist)

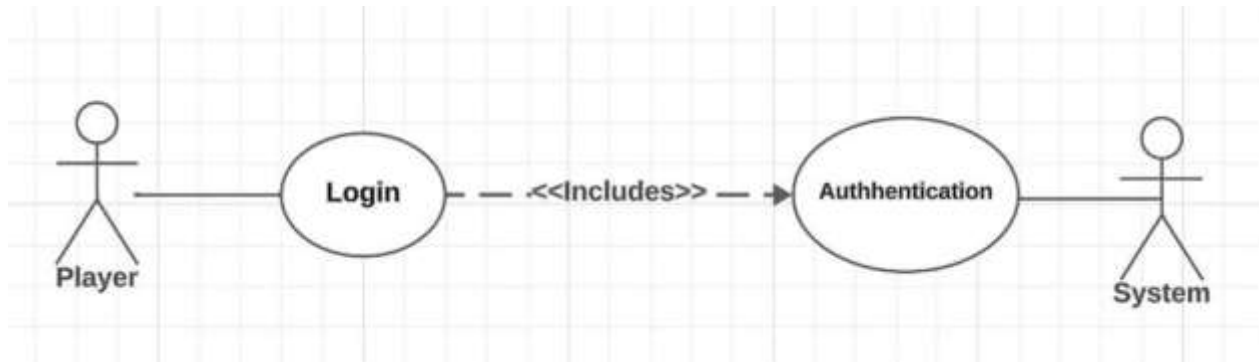


Figure 6 Use Case Diagram Login Authentication

Use Case ID	UC_05	
Use Case Name	Authentication	
Reference	See FR 5	
Description	Check if the entered login credentials match the with the existing account.	
Primary actor	Players	
Secondary Actor	System	
Pre-Conditions	1. Should have an Existing Account. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	If the credentials are correct then the player will be able to log into his account otherwise there will be an error displayed on his screen.	
Basic Flow	Actor Action	System Action
	Players enter his/her username/password into the login prompt and then click on the login button.	The system verifies the credentials and allows the player to access his/her account if correct.
Alternative Flow	The player can play the game without logging in except for the multiplayer.	

Table 21 Use Case Authentication (Login)

6. Login Error (Verifies If Account Exist)

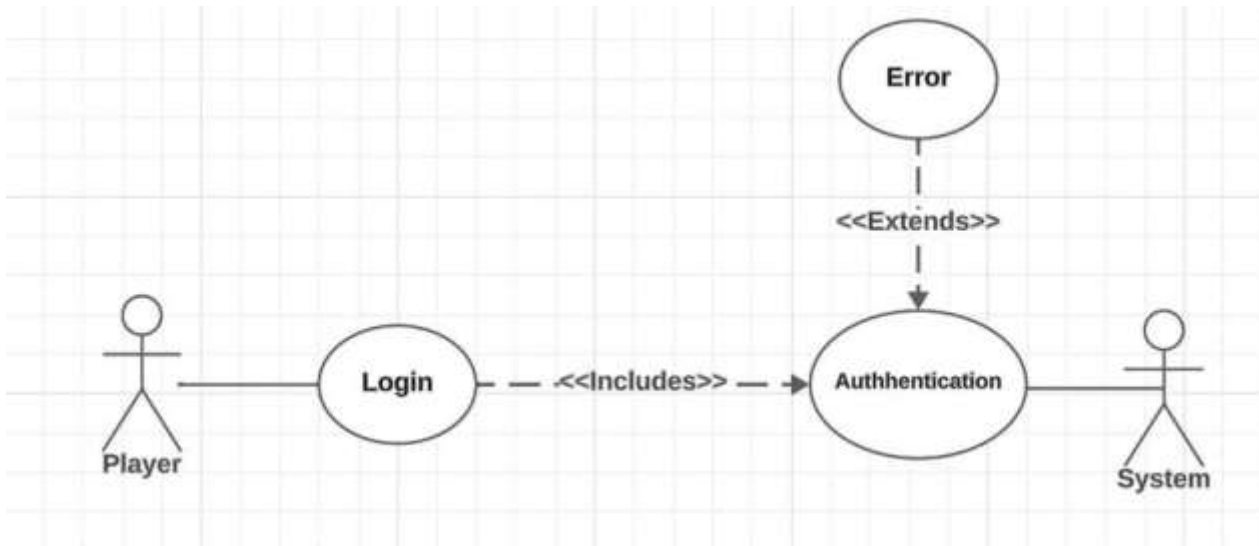


Figure 7 Use Case Diagram Login Error

Use Case ID	UC_06	
Use Case Name	Error	
Reference	See FR 6	
Description	Display an error if the credentials entered do not match the existing account credentials.	
Primary actor	Players	
Secondary Actor	System	
Pre-Conditions	1. Should have entered incorrect username/password. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	If the credentials are not correct then the player will see an error about the wrong entered credentials.	
Basic Flow	Actor Action	System Action
	Players enter his/her username/password into the login prompt and then click on the login button.	The system verifies the credentials and allows the player to access his/her account if correct.
Alternative Flow	The player can play the game without logging in except for the multiplayer.	

Table 22 Use Case Error (Login)

7. Start Game (Start Game)

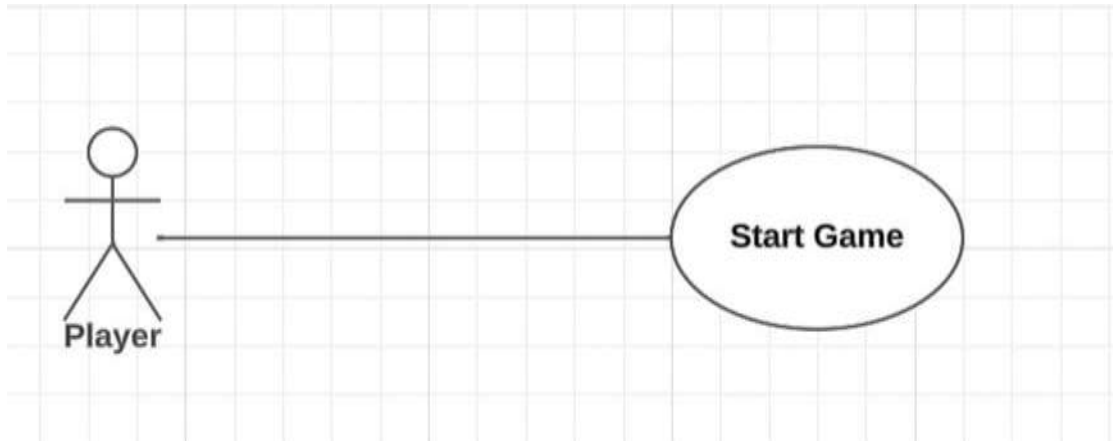


Figure 8 Use Case Diagram Start Game

Use Case ID	UC_07	
Use Case Name	Start Game	
Reference	See FR 7	
Description	This use case allows players to start a Game.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should be logged in. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	After clicking on the start button the player will be able to enjoy the different game modes available to him by the mods.	
Basic Flow	Actor Action	System Action
	Players click on the start game button with an already logged-in account.	The system will take the player towards the next screen where he will be able to choose different modes.
Alternative Flow	The player will be able to explore other choices.	

Table 23 Use Case Start Game

8. Multiplayer (Start Multiplayer)

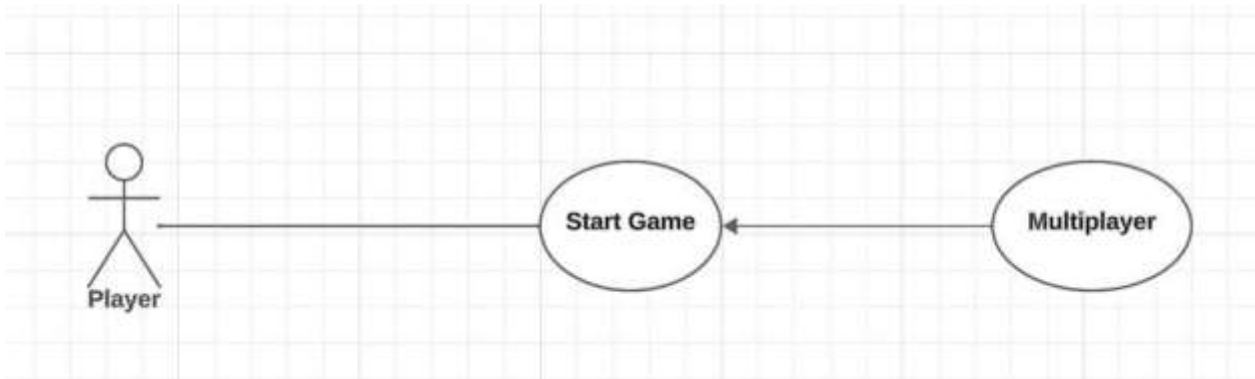


Figure 9 Use Case Diagram Multiplayer

Use Case ID	UC_08	
Use Case Name	Multiplayer	
Reference	See FR 8	
Description	This use case Is a generalized use case that will allow the player to choose the multiplayer option.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should be logged in. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After choosing the multiplayer option, the player can play against the other players online.	
Basic Flow	Actor Action	System Action
	Players click on the multiplayer option.	The system will match the player against another player online.
Alternative Flow	The player will be able to choose other two game modes.	

Table 24 Use Case Multiplayer

9. Multiplayer Sign-Up (check sign-Up for Multiplayer)

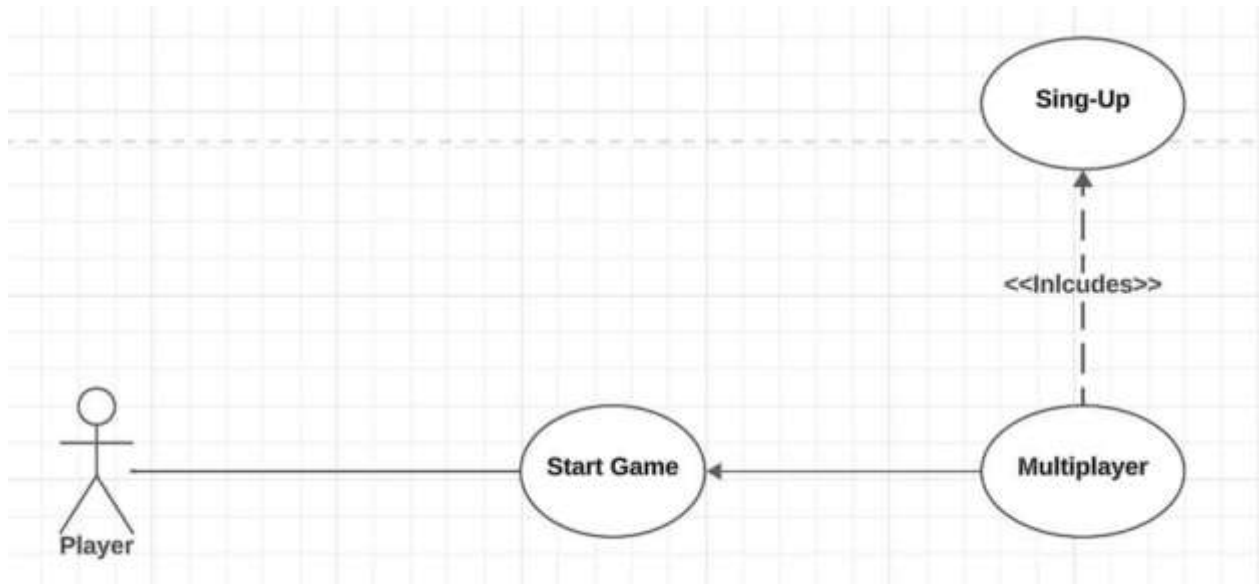


Figure 10 Use Case Diagram Multiplayer Sign-Up

Use Case ID	UC_09	
Use Case Name	Sign-Up	
Reference	See FR 9	
Description	Before allowing the player to enter the multiplayer mode this use case checks if the player is signed in or not.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should not be signed up. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	The player will be sent to the sign-up page to sign up before starting a multiplayer match.	
Basic Flow	Actor Action	System Action
	Players click on the multiplayer option when signing up.	The system will take the player to the sign-up page.
Alternative Flow	The player will have to choose other gaming modes.	

Table 25 Use Case Sign-up (Multiplayer)

10. Multiplayer Login (check login for Multiplayer)

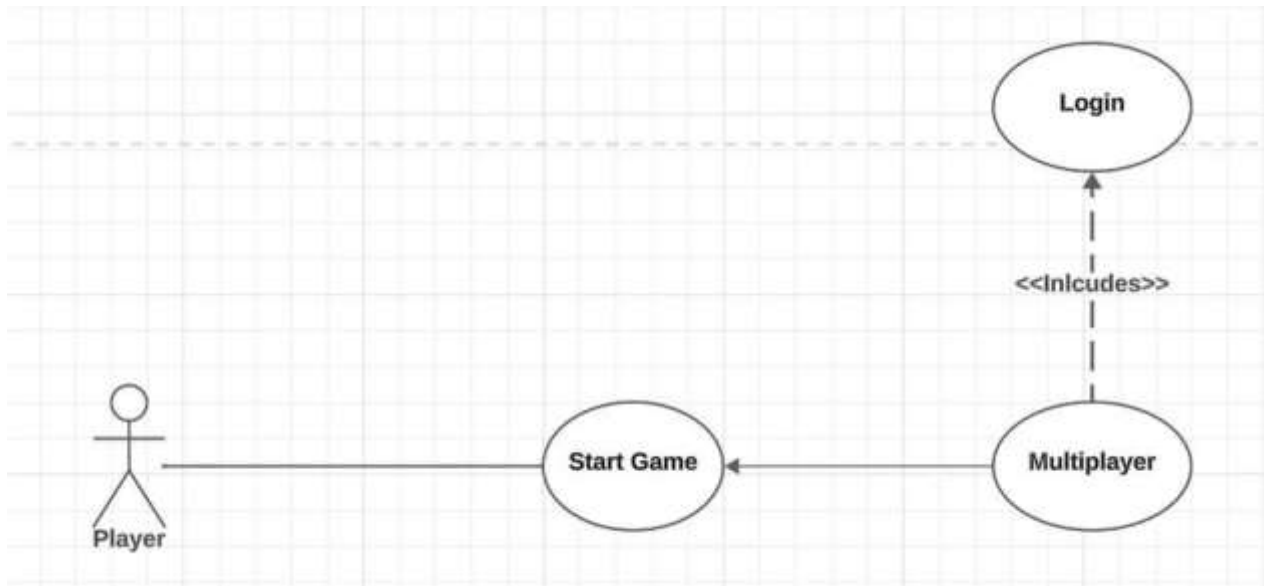


Figure 11 Use Case Diagram Multiplayer Login

Use Case ID	UC_10	
Use Case Name	Login	
Reference	See FR 10	
Description	Before allowing the player to enter the multiplayer mode this use case checks if the player is logged in or not.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should not be signed up. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	The player will be sent to the sign-up page to sign up before starting a multiplayer match.	
Basic Flow	Actor Action	System Action
	Players click on the multiplayer option when signing up.	The system will take the player to the sign-up page.
Alternative Flow	The player will have to choose other gaming modes.	

Table 26 Use Case Login (Multiplayer)

11. Multiplayer Match Bubbles (Match bubbles of the same color)

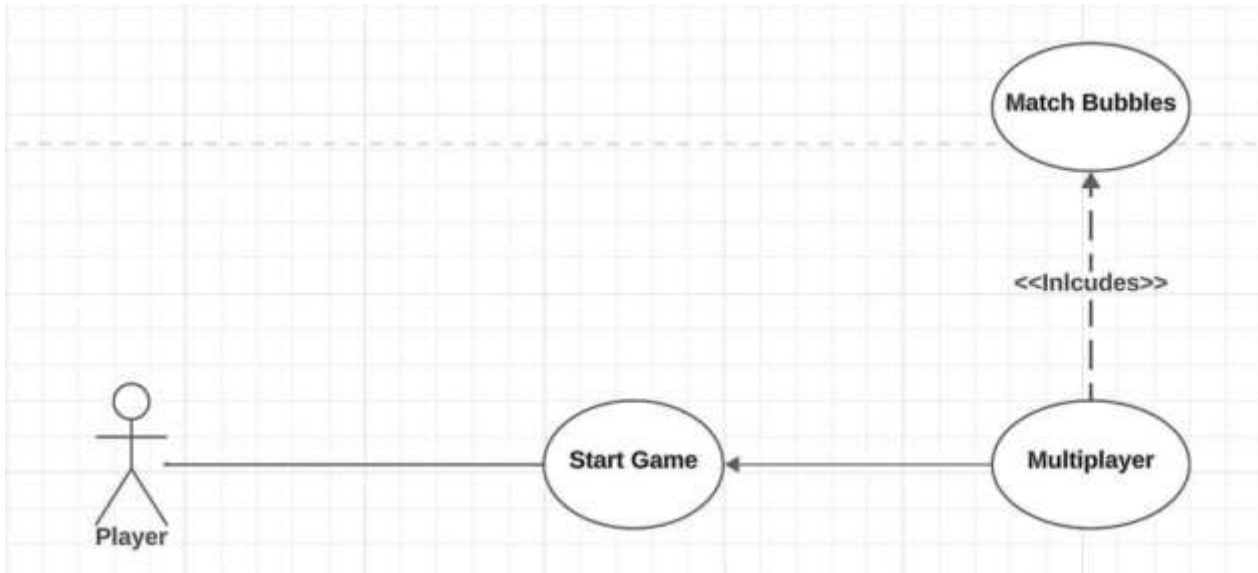


Figure 12 Use Case Diagram Multiplayer Match Bubbles

Use Case ID	UC_11	
Use Case Name	Match Bubbles	
Reference	See FR 11	
Description	After entering the multiplayer match, the player has to match the match with at least 3 bubbles of the same color.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should not be signed up/Logged In. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player will be able to enjoy the game.	
Basic Flow	Actor Action	System Action
	The player tries to match at least 3 bubbles of the same color.	The system will reward the player on how many matches he/she has achieved.
Alternative Flow	The player can exit the game if he wants.	

Table 27 Use Case Match Bubbles (Multiplayer)

12. Multiplayer Score (Score determines outcome)

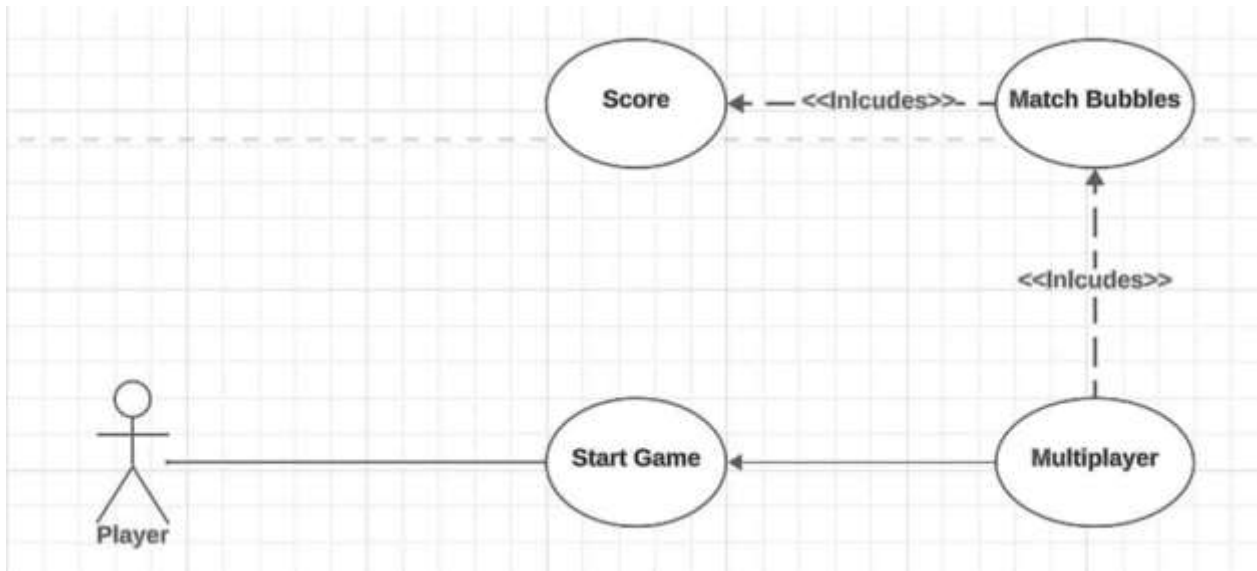


Figure 13 Use Case Diagram Multiplayer Score

Use Case ID	UC_12	
Use Case Name	Score	
Reference	See FR 12	
Description	After entering the multiplayer mode the player will be awarded with score on how well he performs.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should Match the same color bubbles. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player's score will increase.	
Basic Flow	Actor Action	System Action
	The player will try to score more by matching the same bubbles.	The system will reward the player with an increment in his/her score.
Alternative Flow	The player can exit the game if he wants.	

Table 28 Use Case Score (Multiplayer)

13. Multiplayer Wins (Score more to win)

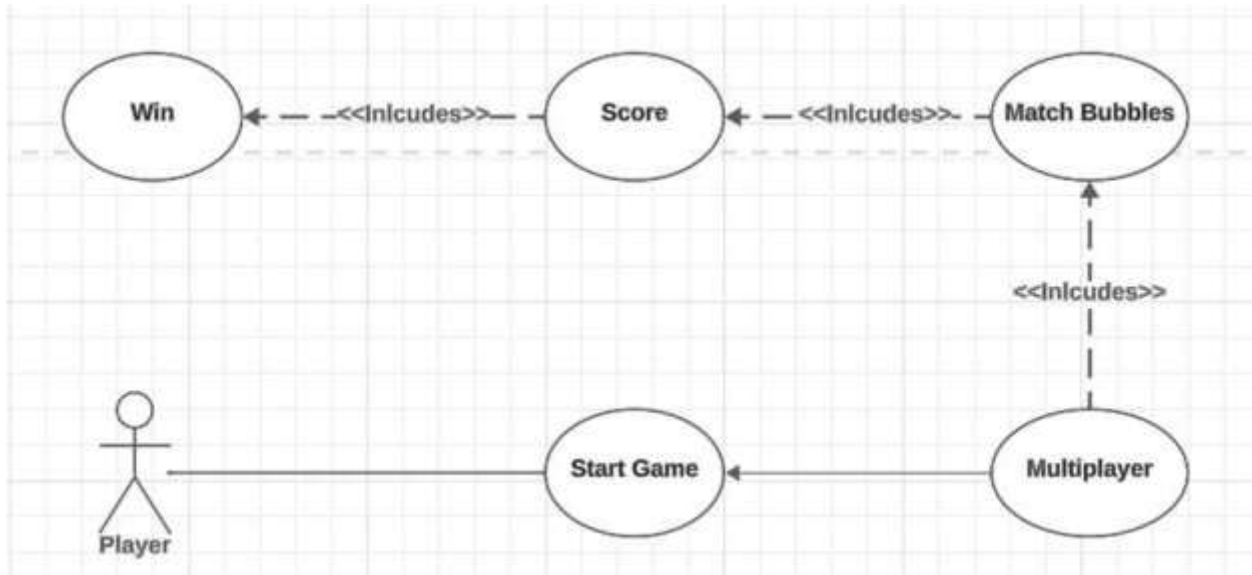


Figure 14 Use Case Diagram Multiplayer Wins

Use Case ID	UC_13	
Use Case Name	Win	
Reference	See FR 13	
Description	After scoring more against the opponent the player will win the game.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should score more than the opponent player. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring more the player will be sent to the winner screen and the reward point will be added to its profile.	
Basic Flow	Actor Action	System Action
	The player scores more than the opponent player.	The system will reward the player with experience points.
Alternative Flow	The player can choose to play the next round or exit.	

Table 29 Use Case Multiplayer Wins (Score more to win)

14. Multiplayer Next Match (goes to next match after winning)

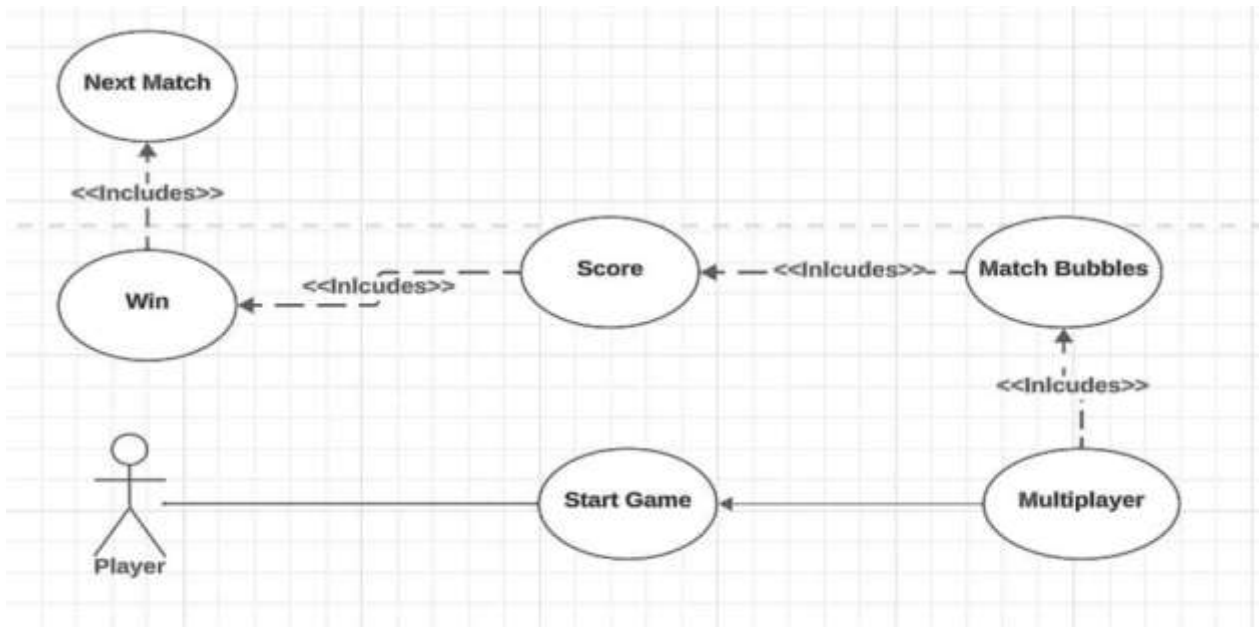


Figure 15 Use Case Diagram Multiplayer Next Match

Use Case ID	UC_14	
Use Case Name	Next Match	
Reference	See FR 14	
Description	After Winning the match the player will be transferred to the next more difficult match.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should win the Multiplayer Match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After winning the match the player will be sent to the next better match to challenge his/her new opponent.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the next match button.	The system will then send the player to another match with a better opponent.
Alternative Flow	The player can choose to exit.	

Table 30 Use Case Next Match (Multiplayer)

15. Multiplayer Exit (exit the game after winning)

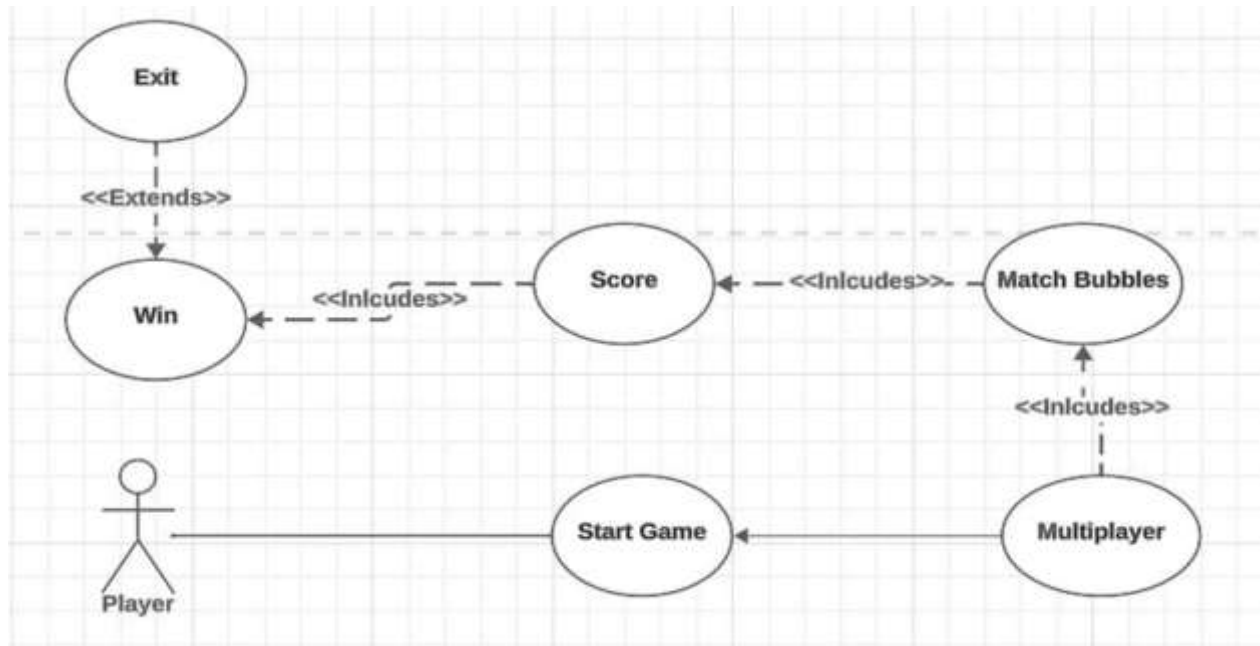


Figure 16 Use Case Diagram Multiplayer Exit (exit the game after winning)

Use Case ID	UC_15	
Use Case Name	Exit	
Reference	See FR 15	
Description	After Winning the match the player will be exited from the multiplayer mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should win the Multiplayer Match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	The player will be sent to the start game screen where he will be able to choose from other game modes.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the exit button.	The system will then send the player to another start game screen.
Alternative Flow	The player can choose to play the next match.	

Table 31 Use Case Exit (Multiplayer)

16. Multiplayer lose (Scored less)

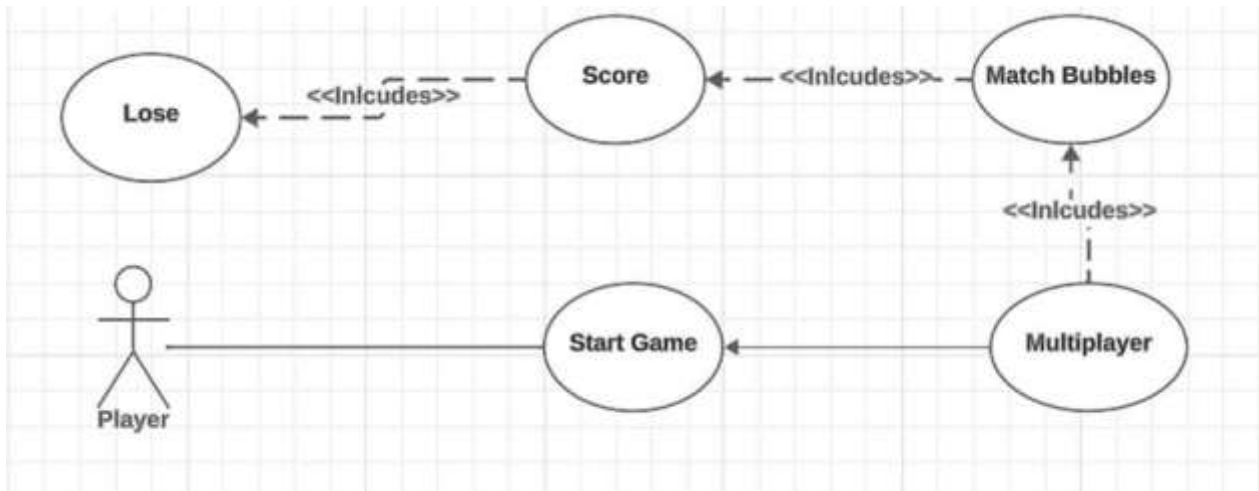


Figure 17 Use Case Diagram Multiplayer lose

Use Case ID	UC_16	
Use Case Name	Lose	
Reference	See FR 16	
Description	After entering the multiplayer the player lost the game against the other opponent.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Scored less than the opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring less than the opponent player the player will lose the game.	
Basic Flow	Actor Action	System Action
	The player scored less than the opponent he was competing against.	The system will send the player to the losing screen and will collect the point he lost.
Alternative Flow	The player can replay the game if he wants.	

Table 32 Use Case Lose (Multiplayer)

17. Multiplayer Rematch (Retry)

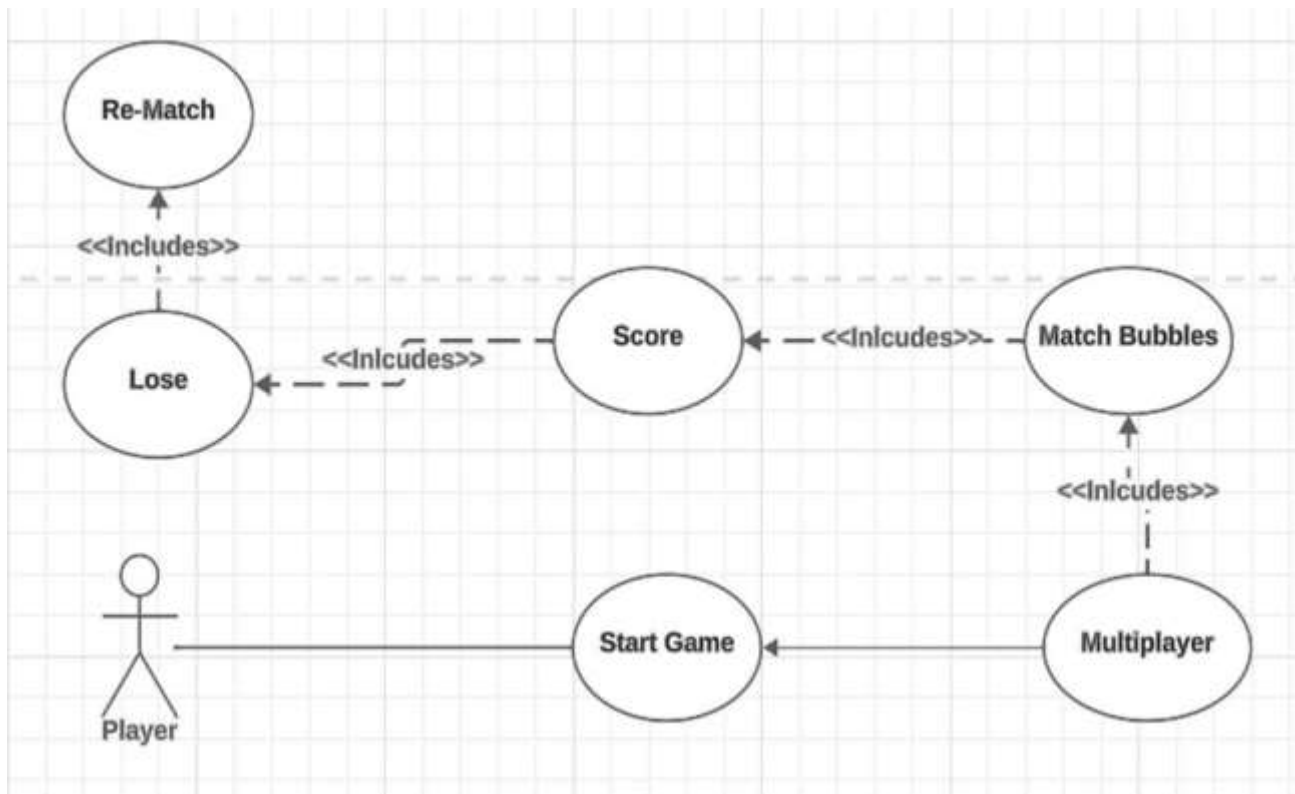


Figure 18 Use Case Diagram Multiplayer Rematch

Use Case ID	UC_17	
Use Case Name	Re-Match	
Reference	See FR 17	
Description	After losing the game in a multiplayer match the player is asking for a rematch.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should lose against the opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to play rematch.	
Basic Flow	Actor Action	System Action
	The player lost against the other player and clicked on the rematch button.	The system will arrange another match between the opponents.
Alternative Flow	The player can Exit the game mode.	

Table 33 Use Case Rematch (Multiplayer)

18. Multiplayer Exit (Exit the Multiplayer mode)

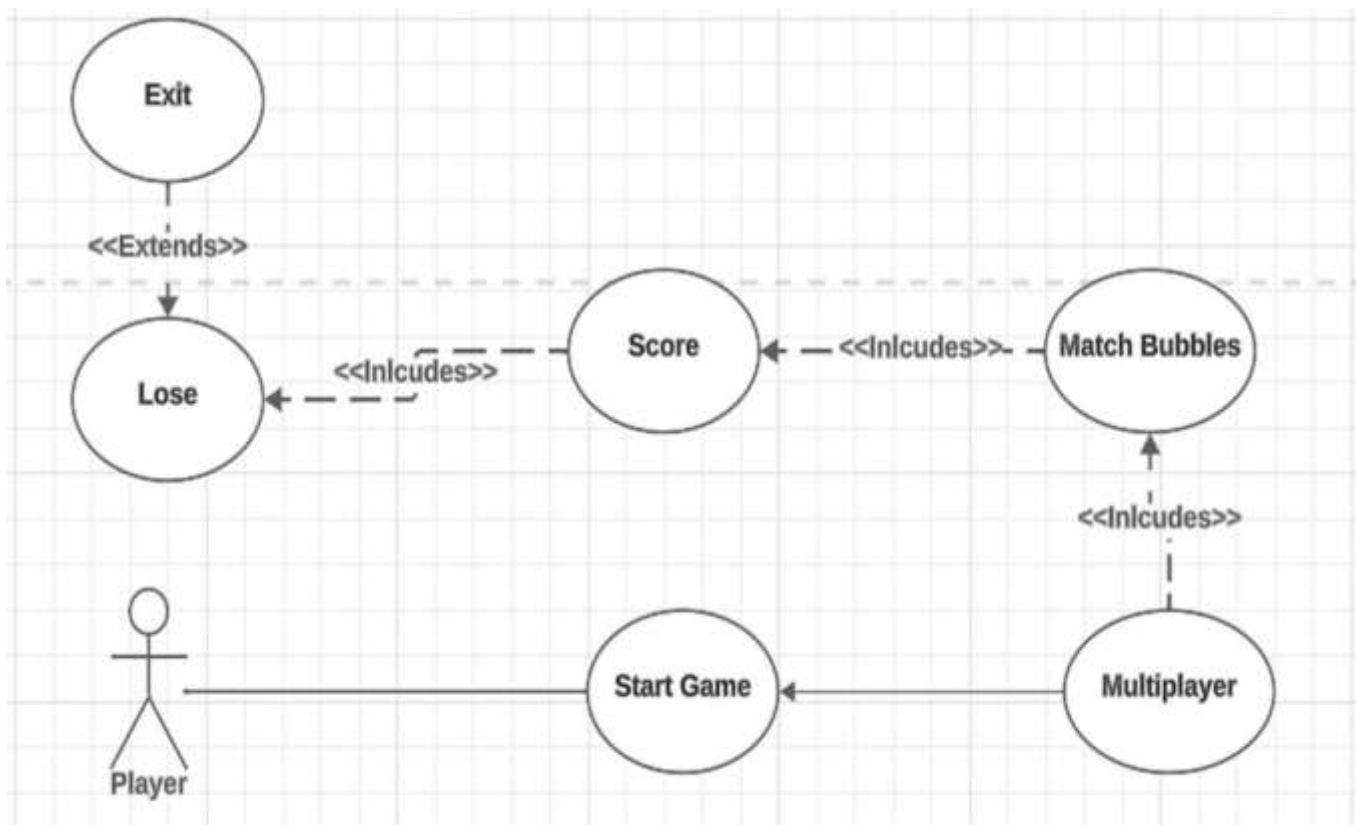


Figure 19 Use Case Diagram Multiplayer Exit

Use Case ID	UC_18	
Use Case Name	Exit	
Reference	See FR 18	
Description	After losing the game in a multiplayer match the player is exiting the multiplayer game mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should lose against the opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to exit the game mode and choose other game modes.	
Basic Flow	Actor Action	System Action
	The player lost against the other player and clicked on the Exit button.	The system will take the player to the game mode screen to choose from the other game modes.
Alternative Flow	The player can ask for a rematch.	

Table 34 Use Case Exit (Multiplayer)

19. vs A.I (Start vs AI mode)

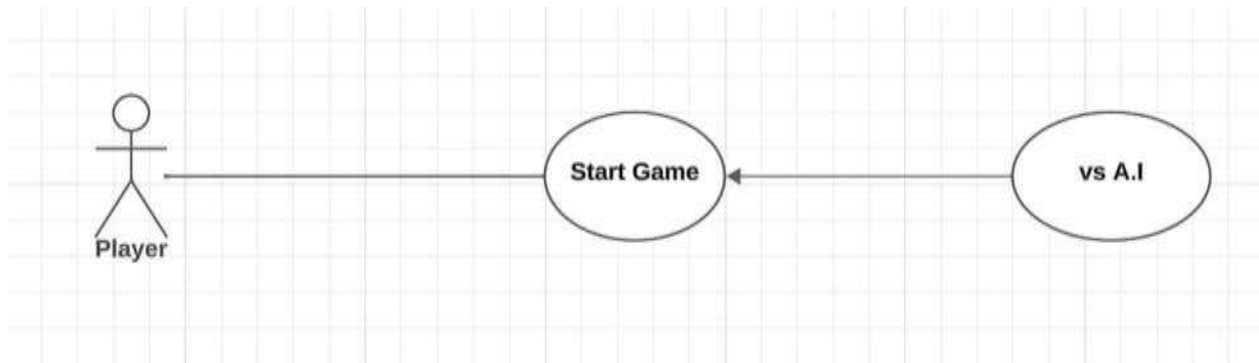


Figure 20 Use Case Diagram vs A.I

Use Case ID	UC_19	
Use Case Name	vs A.I	
Reference	See FR 19	
Description	This use case Is a generalized use case that will allow the player to choose the vs A. I option.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should be logged in. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After choosing the vs A. I option the player will be able to play against the Artificial intelligence Agent.	
Basic Flow	Actor Action	System Action
	Players click on the vs A. I option.	The system will match the player against an AI opponent.
Alternative Flow	The player will be able to choose other two game modes.	

Table 35 Use Case vs A.I

20. vs A.I Match (Match bubbles of the same color)

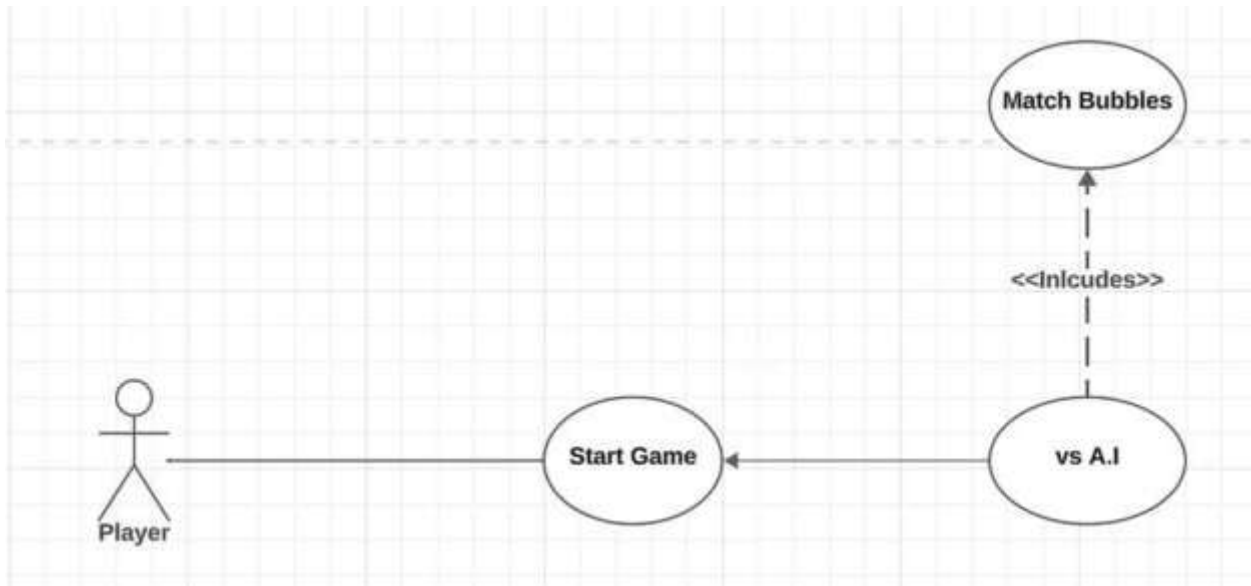


Figure 21 Use Case Diagram vs A.I Match

Use Case ID	UC_20	
Use Case Name	Match Bubbles	
Reference	See FR 20	
Description	After entering the vs A.I mode, the player has to match the match with at least 3 bubbles of the same color.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player will be able to challenge the game against an A.I opponent.	
Basic Flow	Actor Action	System Action
	The player tries to match at least 3 bubbles of the same color.	The system will reward the player on how many matches he/she has achieved.
Alternative Flow	The player can exit the game if he wants.	

Table 36 Use Case Match Bubbles (vs A.I)

21. vs A.I Score (Score determines outcome)

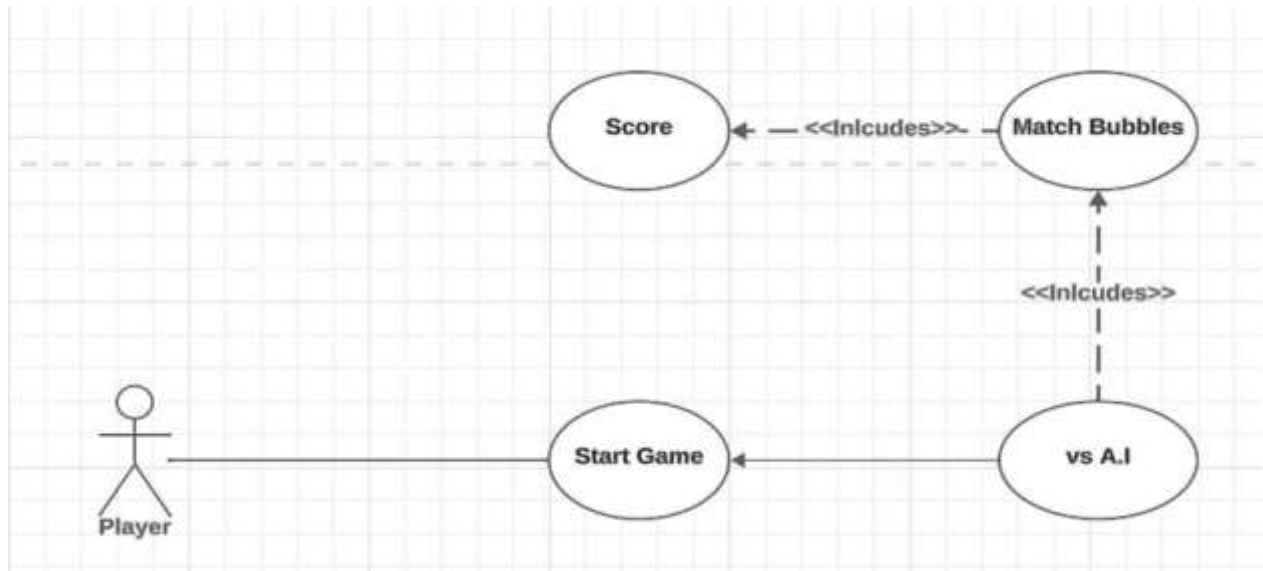


Figure 22 Use Case Diagram vs A.I Score

Use Case ID	UC_21	
Use Case Name	Score	
Reference	See FR 21	
Description	After entering the vs A.I mode, the player will be awarded a score on how well he performs.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Match the same color bubbles to increase the score. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player's score will increase.	
Basic Flow	Actor Action	System Action
	The player will try to score more by matching the same bubbles.	The system will reward the player with an increment in score giving an edge against A.I opponent.
Alternative Flow	The player can exit the game if he wants.	

Table 37 Use Case Score (vs A.I)

22. vs A.I Wins (Score more to win)

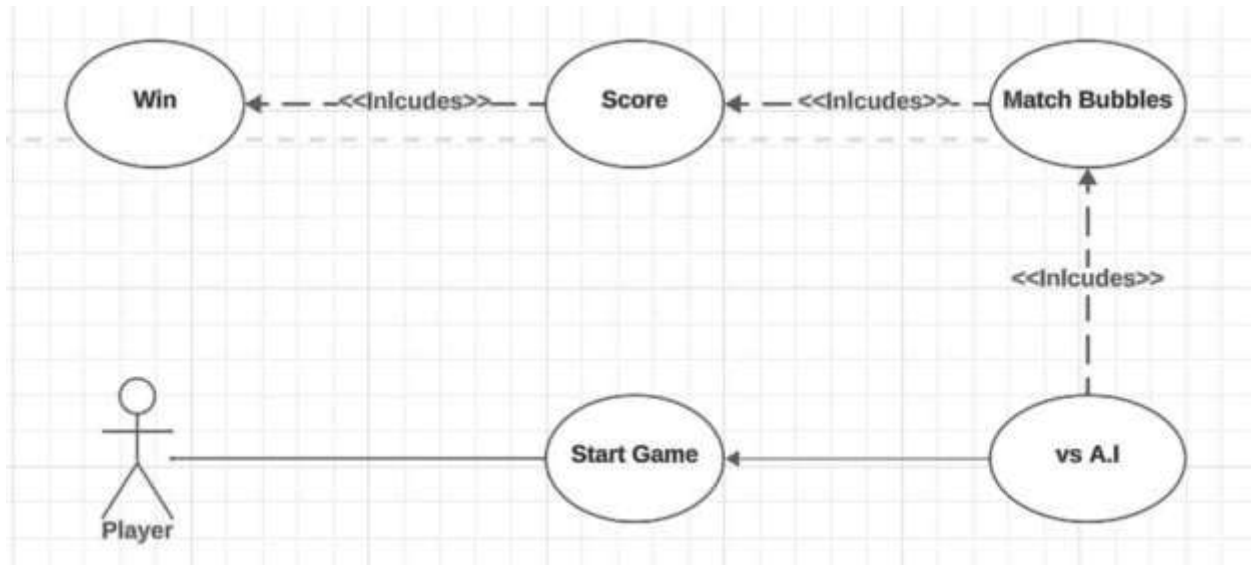


Figure 23 Use Case Diagram vs A.I Wins

Use Case ID	UC_22	
Use Case Name	Win	
Reference	See FR 22	
Description	After scoring more against the A.I the opponent the player will win the game.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Should score more than the A.I opponent player. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring more the player will be sent to the winner screen and the reward point will be added to it's profile.	
Basic Flow	Actor Action	System Action
	The player scores more than the A.I player.	The system will reward the player with experience points.
Alternative Flow	The player can choose to play the next round or exit.	

Table 38 Use Case win (vs A.I)

23. vs A.I Next Match (goes to next match after winning)

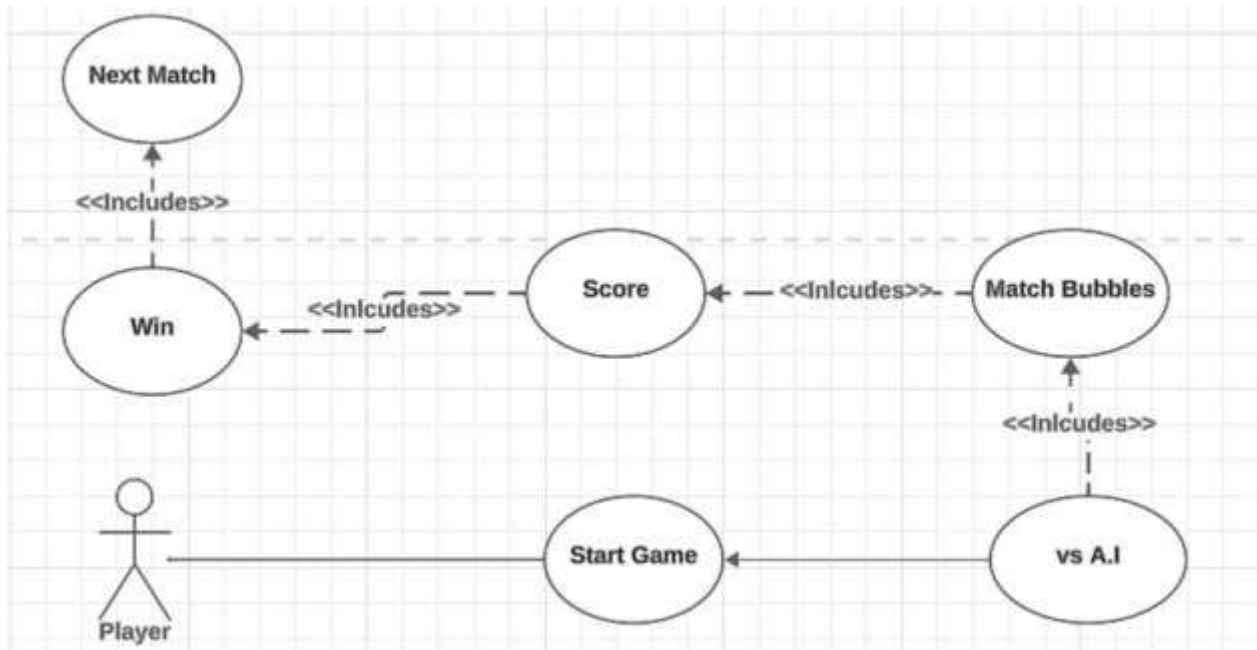


Figure 24 Use Case Diagram vs A.I Next Match

Use Case ID	UC_23	
Use Case Name	Next Match	
Reference	See FR 23	
Description	After Winning the match the player will be transferred to the next more difficult match against the more capable A.I.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Multiplayer Match. 2. Should win the Multiplayer Match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After winning the match the player will be sent to the next better match to challenge his/her new opponent.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the next match button.	The system will then send the player to another match with a better A.I opponent.
Alternative Flow	The player can choose to exit.	

Table 39 Use Case Next Match (vs A.I)

24. vs A.I Win Exit (exit the game after winning)

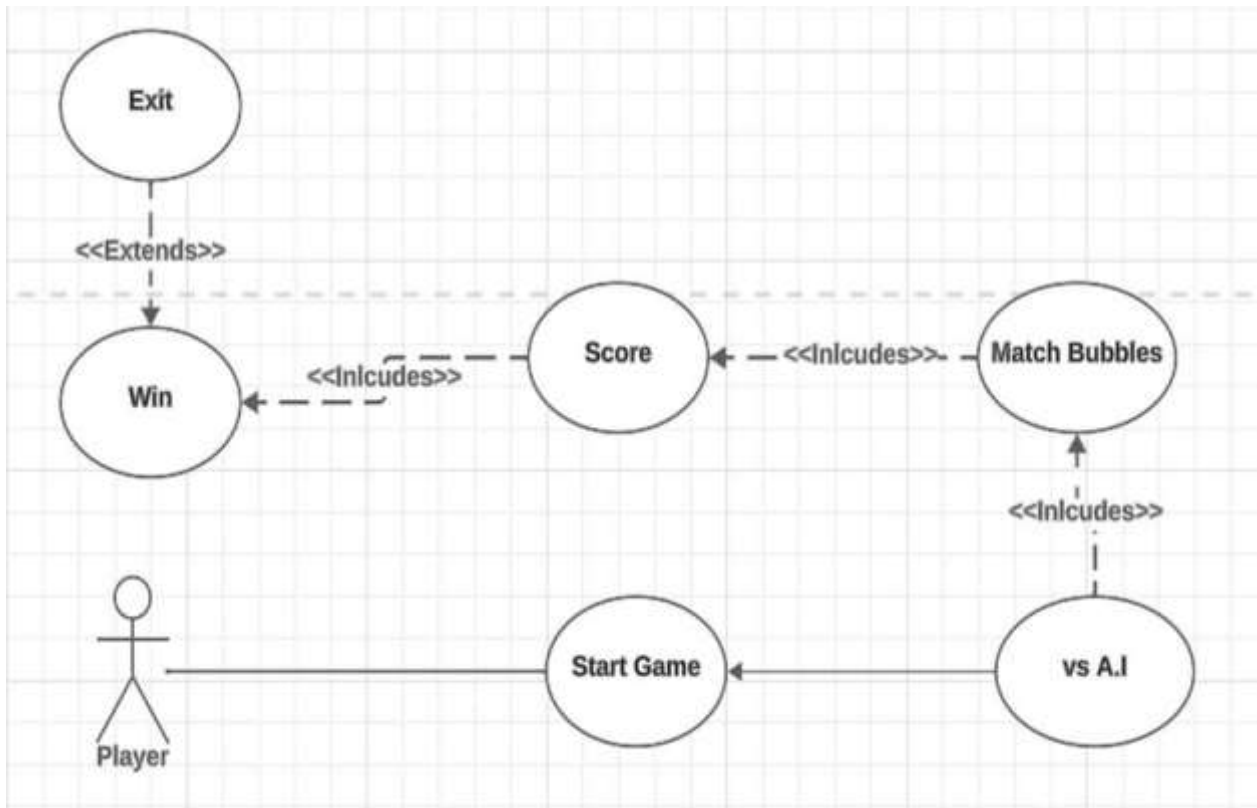


Figure 25 Use Case Diagram vs A.I Win Exit

Use Case ID	UC_24	
Use Case Name	Exit	
Reference	See FR 24	
Description	After Winning the match the player will be exited from the vs A.I mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Should win against an A.I opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	The player will be sent to the start game screen where he will be able to choose from other game modes.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the exit button.	The system will then send the player to another start game screen.
Alternative Flow	The player can choose to play the next match.	

Table 40 Use Case Exit (vs A.I Win)

25. vs A.I lose (Scored less)

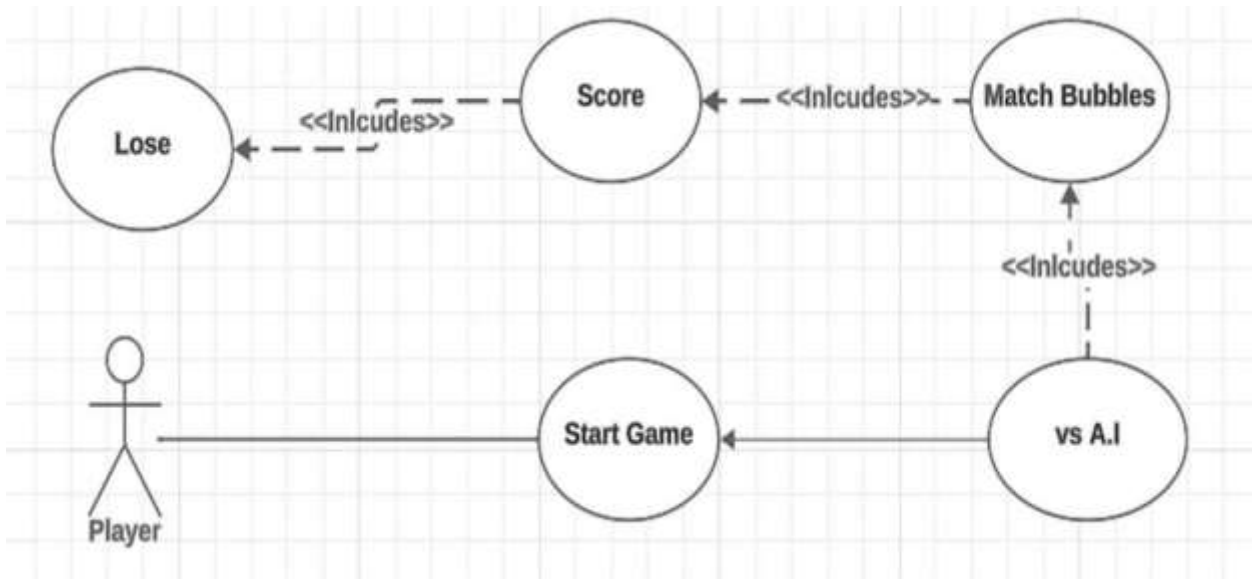


Figure 26 Use Case Diagram vs A.I lose

Use Case ID	UC_25	
Use Case Name	Lose	
Reference	See FR 25	
Description	After entering the vs A.I mode the player lost the game against the A.I opponent.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Scored less than the opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring less than the A.I the opponent player the player will lose the game.	
Basic Flow	Actor Action	System Action
	The player scored less than the A.I opponent he was competing against.	The system will send the player to the losing screen and will collect the point he lost.
Alternative Flow	The player can replay the game if he wants.	

Table 41 Use Case Loose (vs A.I)

26. vs A.I Rematch (Retry)

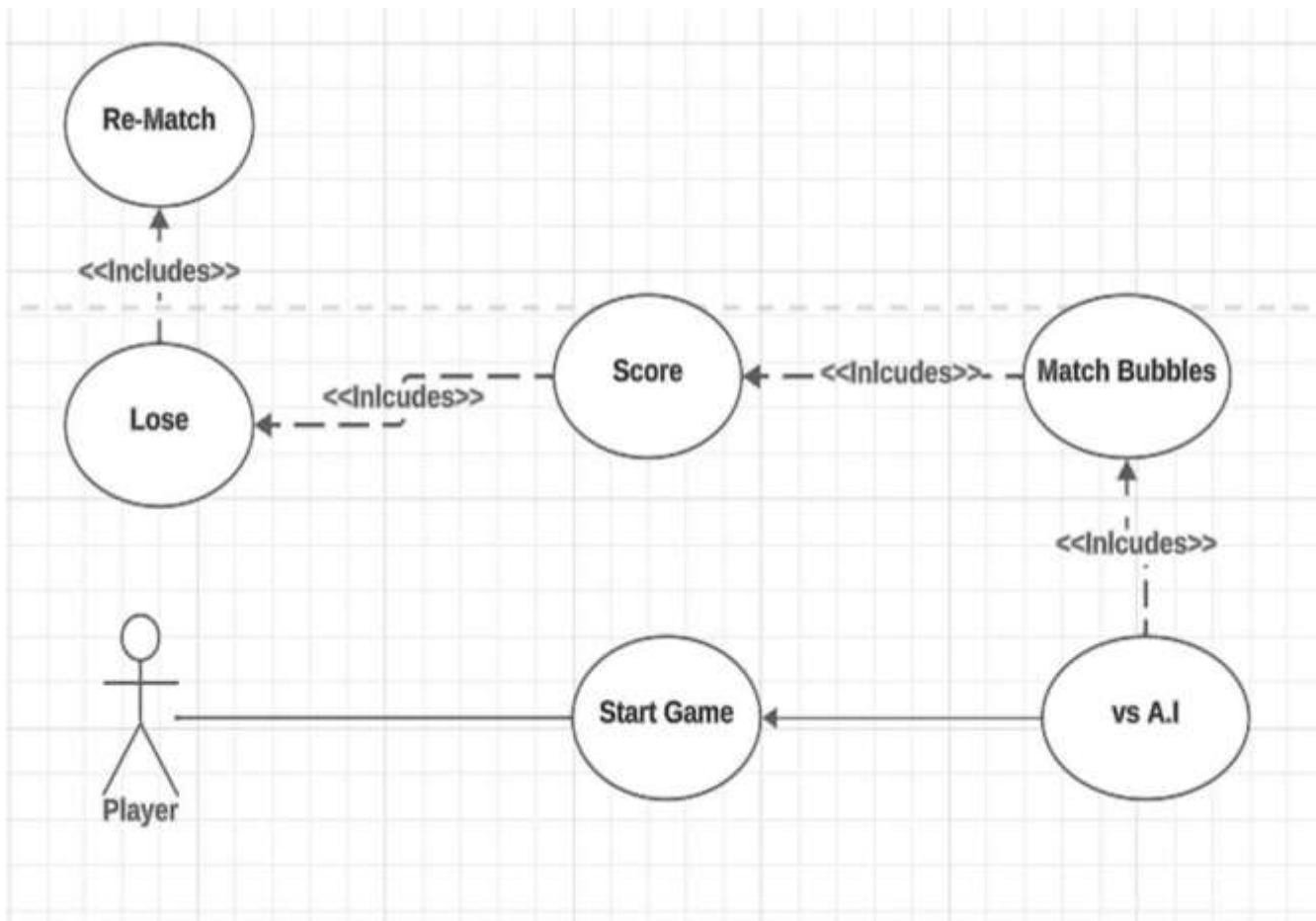


Figure 27 Use Case Diagram vs A.I Rematch

Use Case ID	UC_26	
Use Case Name	Re-Match	
Reference	See FR 26	
Description	After losing the game vs. AI mode the player is asking for a rematch.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Should lose against the A.I opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to play rematch.	
Basic Flow	Actor Action	System Action
	The player lost against the A.I played and clicked on the rematch button.	The system will arrange another match between the player and A.I opponent.
Alternative Flow	The player can Exit the game mode.	

Table 42 Use Case Rematch (vs A.I)

27. vs A.I Exit (Exit the VS A.I mode)

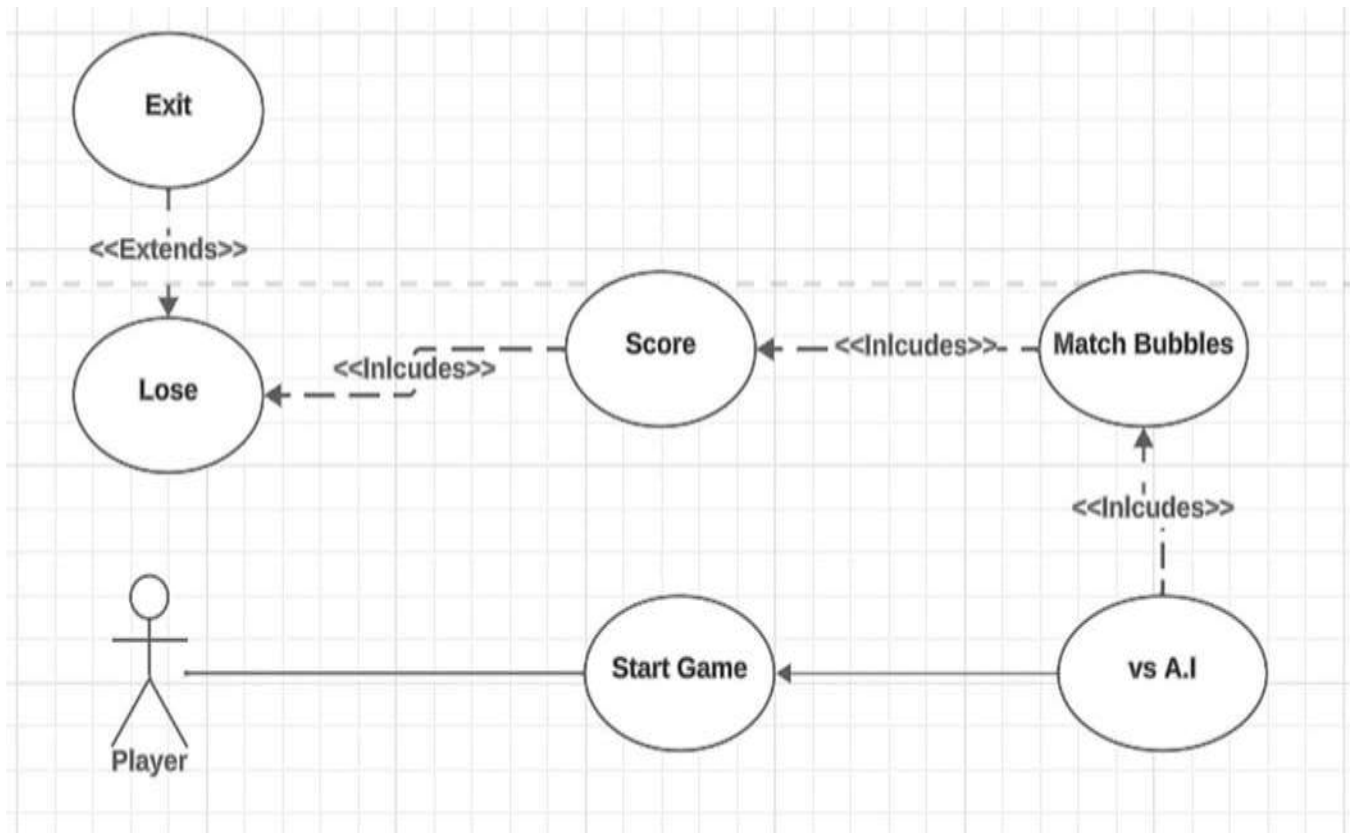


Figure 28 Use Case Diagram vs A.I Exit

Use Case ID	UC_27	
Use Case Name	Exit	
Reference	See FR 27	
Description	After losing the game vs. AI game mode the player is exiting the multiplayer game mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the vs A.I Match. 2. Should lose against the A.I opponent. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to exit the vs A.I game mode and explore other game modes.	
Basic Flow	Actor Action	System Action
	The player lost against the A.I played and clicked on the Exit button.	The system will take the player to the game mode screen to choose from the other game modes.
Alternative Flow	The player can ask for a rematch.	

Table 43 Use Case Exit (vs A.I Loose)

28. Single player (Start Single player)

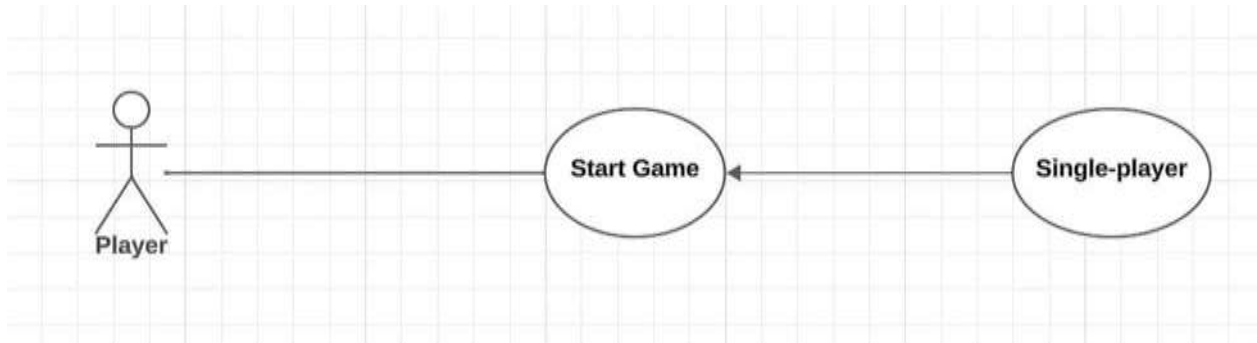


Figure 29 Use Case Diagram Single player

Use Case ID	UC_28	
Use Case Name	Single Player	
Reference	See FR 28	
Description	This use case Is a generalized use case that will allow the player to choose the Single-player mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should be logged in. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After choosing the Single Player option the player will be able against a time limit.	
Basic Flow	Actor Action	System Action
	Players click on the Single Player option.	The system will match the player where the player will have to clear all the bubbles with limited resources.
Alternative Flow	The player will be able to choose other two game modes.	

Table 44 Use Case Single Player

29. Single player Match Bubbles (Match bubbles of the same color)

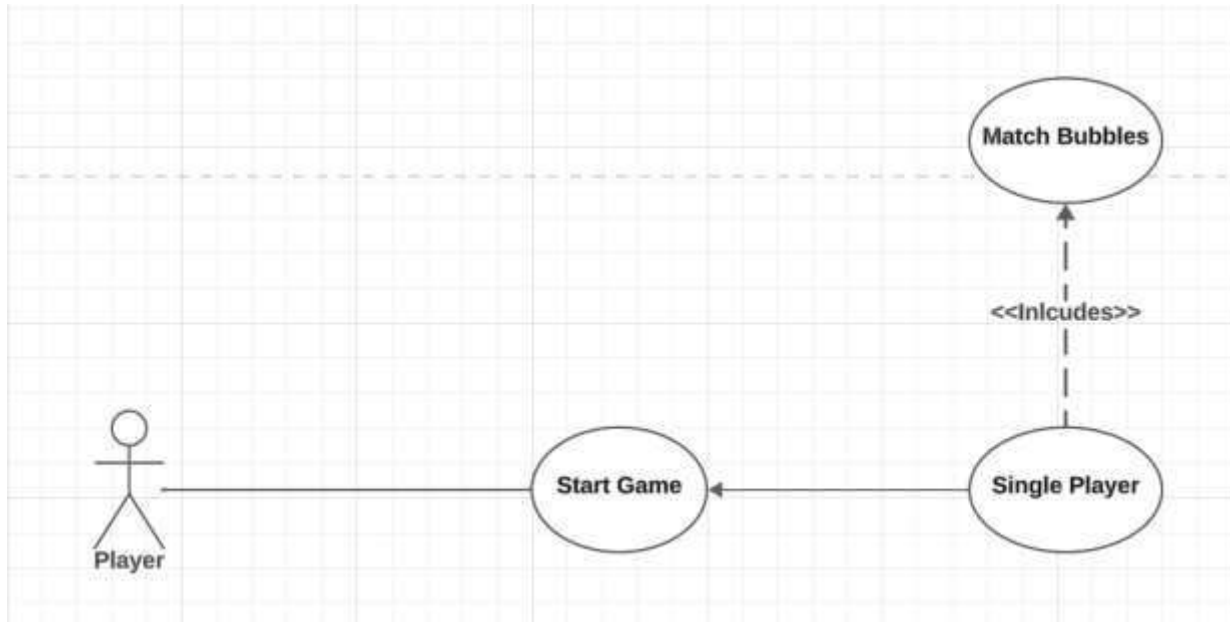


Figure 30 Use Case Diagram Single player Match Bubbles

Use Case ID	UC_29	
Use Case Name	Match Bubbles	
Reference	See FR 29	
Description	After entering the single-player mode the player has to match the match with at least 3 bubbles of the same color.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player mode. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player will be able to clear the remaining bubbles with the resources given.	
Basic Flow	Actor Action	System Action
	The player tries to match at least 3 bubbles of the same color.	The system will reward the player on how many matches he/she has achieved.
Alternative Flow	The player can exit the game if he wants.	

Table 45 Use Case vs Match Bubbles (Single Player)

30. Single player Score (Score determines outcome)

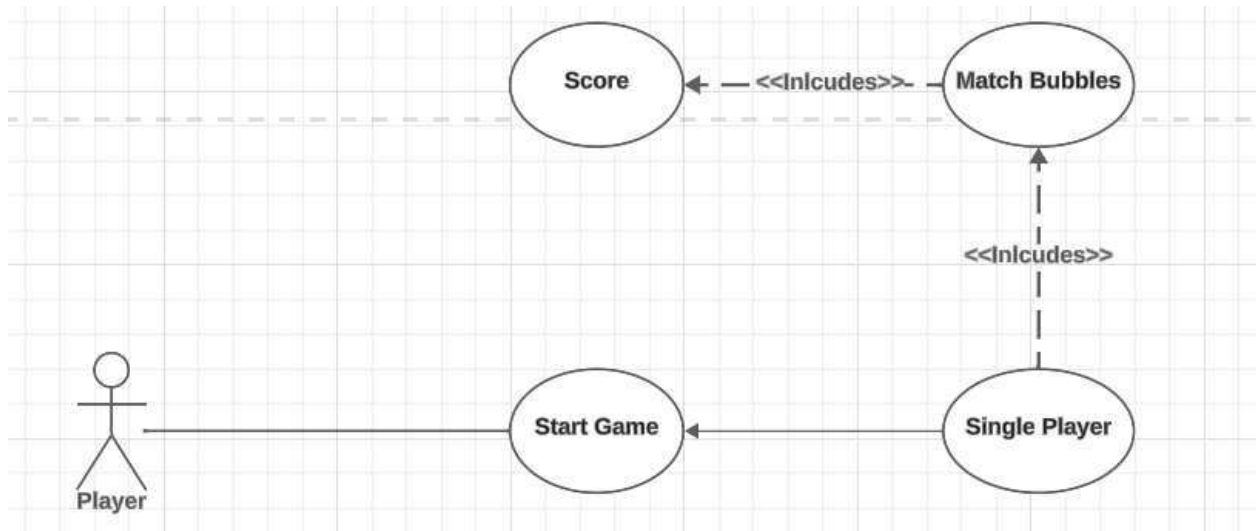


Figure 31 Use Case Diagram Single player Score

Use Case ID	UC_30	
Use Case Name	Score	
Reference	See FR 30	
Description	After entering the Single-player mode the player will be awarded with score on how well he performs.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player match. 2. Match the same color bubbles to increase the score. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After matching the same color bubbles the player's score will increase.	
Basic Flow	Actor Action	System Action
	The player will try to score more by matching the same bubbles.	The system will reward the player with an increment in score increasing chances of winning.
Alternative Flow	The player can exit the game if he wants.	

Table 46 Use Case Score (Single Player)

31. Single Player Wins (Score more to win)

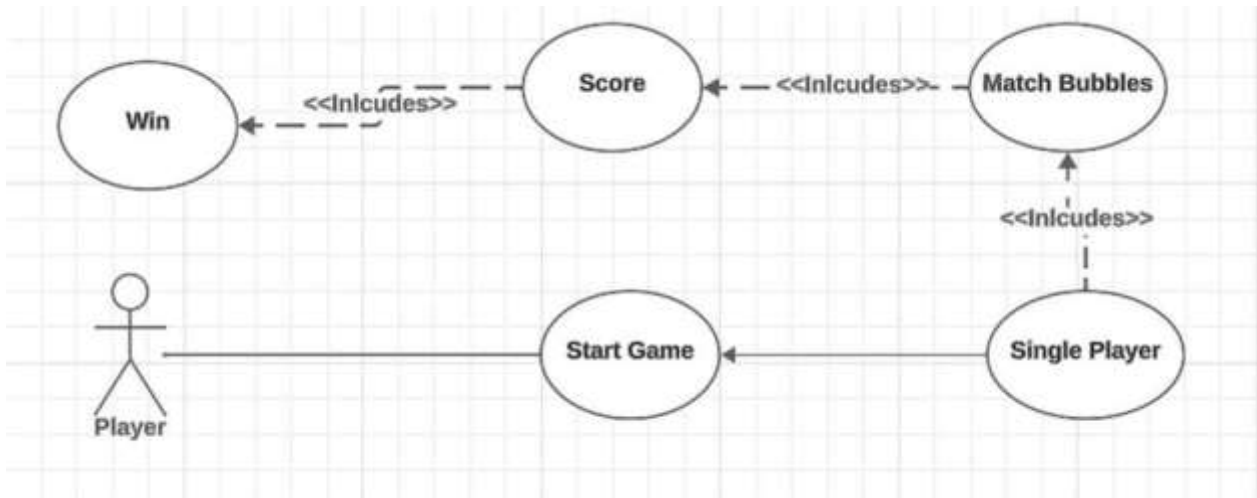


Figure 32 Use Case Diagram Single Player Wins

Use Case ID	UC_31	
Use Case Name	Win	
Reference	See FR 31	
Description	After scoring the required or more the player will win the match.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start a single-player match. 2. Should match the required score using limited resources. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring more than the challenged score within limited time or resources the player will win.	
Basic Flow	Actor Action	System Action
	The player scores more than required to pass the level.	The system will reward the player with experience points.
Alternative Flow	The player can choose to play the next round or exit.	

Table 47 Use Case win (Single player)

32. Single player Next Match (goes to next level after winning)

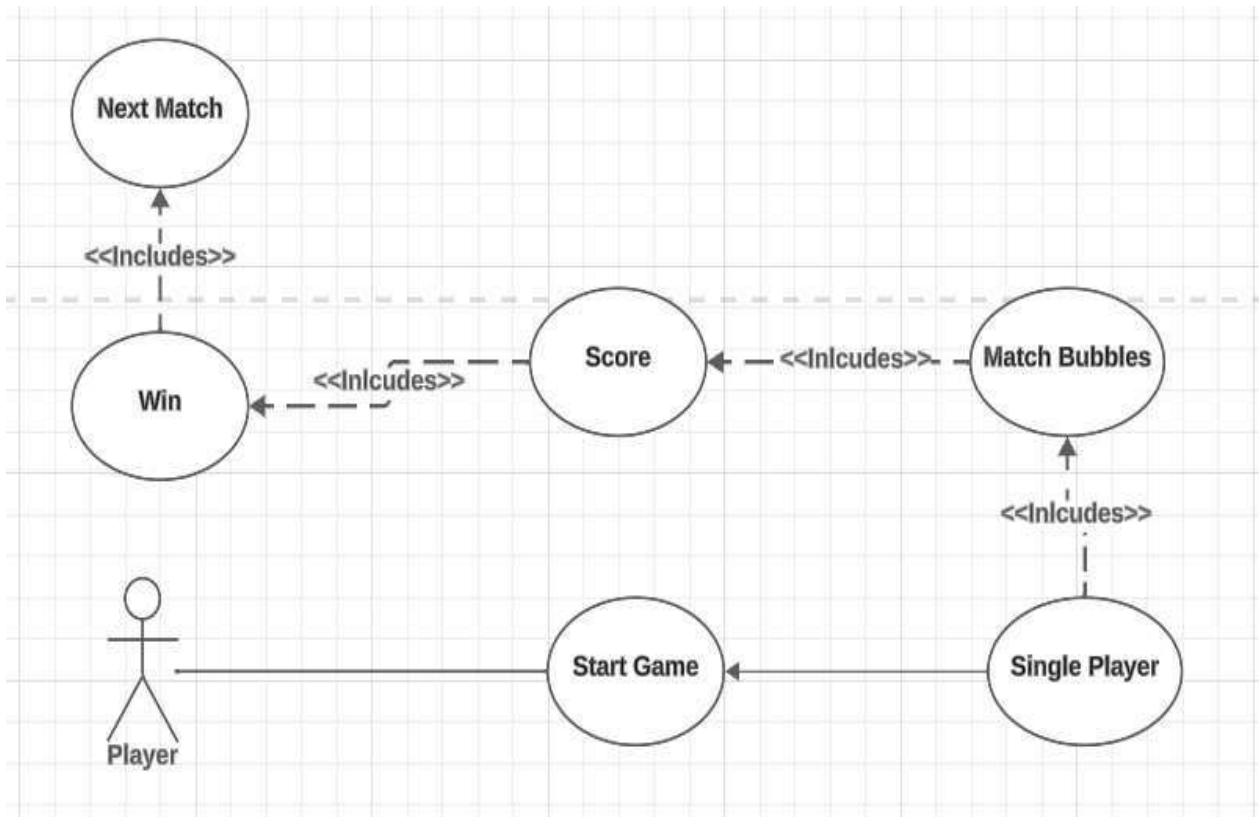


Figure 33 Use Case Diagram Single player Next Match

Use Case ID	UC_32	
Use Case Name	Next Match	
Reference	See FR 32	
Description	After Winning the match the player will be transferred to the next more difficult level with more tough challenges.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player match. 2. Should win the Single-player match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After winning the match the player will be sent to the next more challenging level.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the next match button.	The system will then send the player to a more difficult level.
Alternative Flow	The player can choose to exit.	

Table 48 Use Case Next Match (Single player Win)

33. Single player Exit (exit the game after winning)

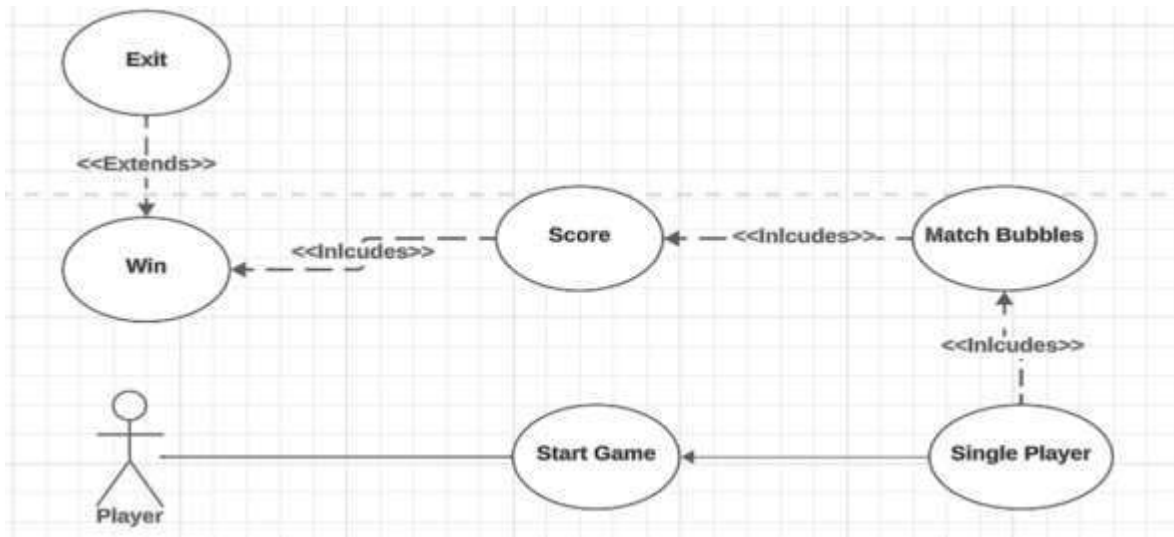


Figure 34 Use Case Diagram Single player Exit

Use Case ID	UC_33	
Use Case Name	Exit	
Reference	See FR 33	
Description	After Winning the match the player will be exited from the vs A.I mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player Match. 2. Should win a Single player match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	The player will be sent to the start game screen where he will be able to choose from other game modes.	
Basic Flow	Actor Action	System Action
	The player has won the game and clicked on the exit button.	The system will then send the player to another start game screen.
Alternative Flow	The player can choose to play the next match.	

Table 49 Use Case vs Exit (Single player Win)

34. Single player lose (Scored less)

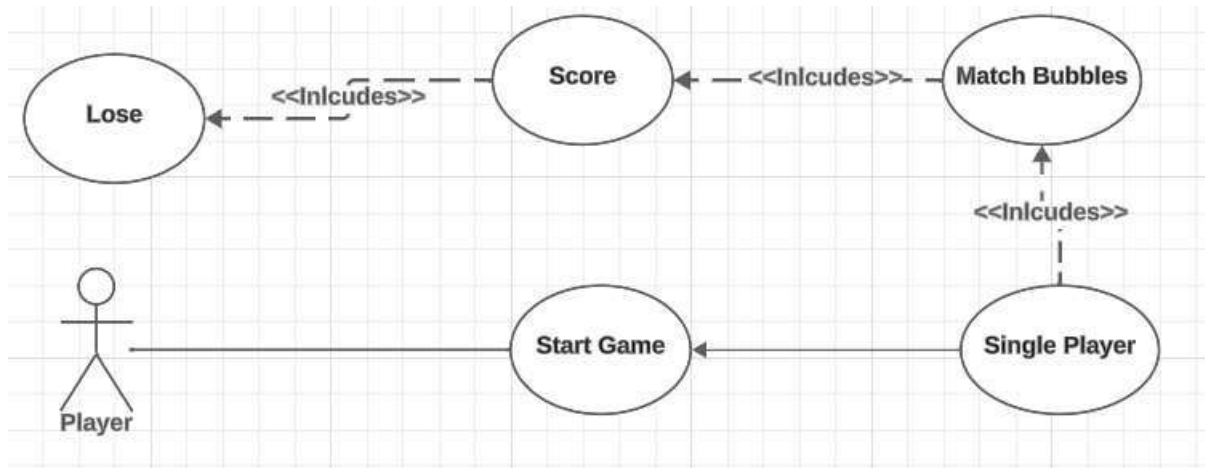


Figure 35 Use Case Diagram Single player lose

Use Case ID	UC_34	
Use Case Name	Lose	
Reference	See FR 34	
Description	After entering the Single player mode the player lost the game.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player match. 2. Score less than the challenged scored. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After scoring less than what was expected of the player the player will lose the match and lose points.	
Basic Flow	Actor Action	System Action
	The player scored less than the minimum required score.	The system will send the player to the losing screen and will collect the point he lost.
Alternative Flow	The player can replay the game if he wants.	

Table 50 Use Case Lose (Single player)

35. Single player Rematch (Retry)

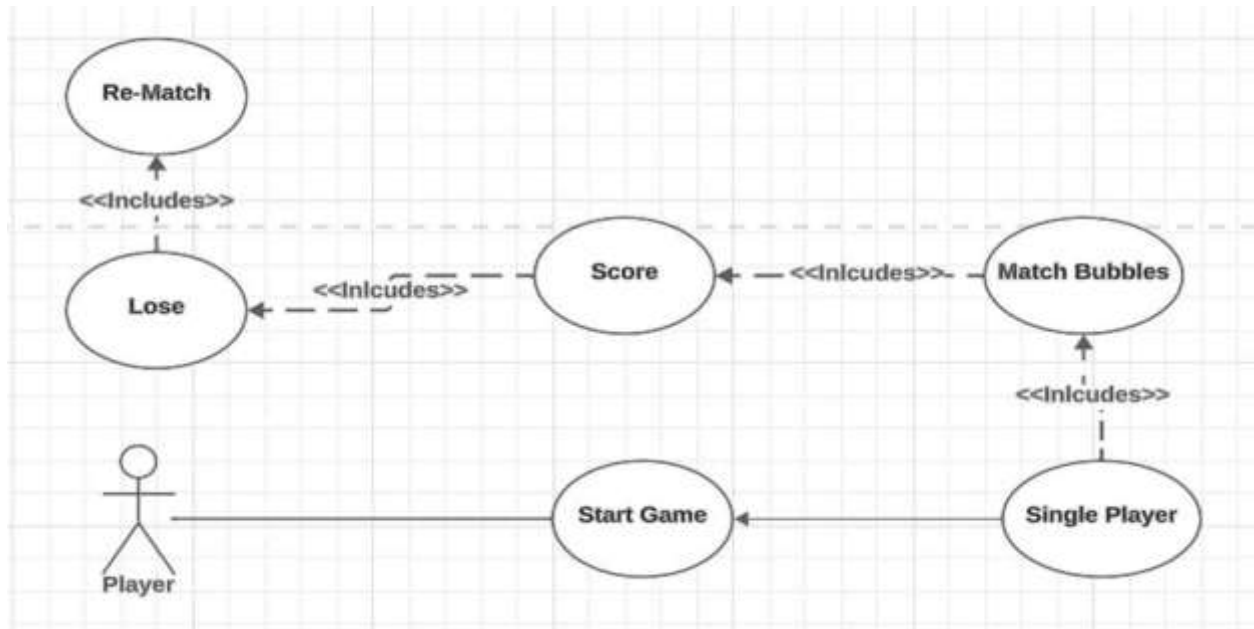


Figure 36 Use Case Diagram Single player Rematch

Use Case ID	UC_35	
Use Case Name	Re-Match	
Reference	See FR 35	
Description	After losing the game in vs Single player mode the player is asking for a rematch.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player match. 2. Should lose the match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to play rematch.	
Basic Flow	Actor Action	System Action
	The player lost and clicked on the rematch button.	The system will arrange another match with the same condition for the player.
Alternative Flow	The player can Exit the game mode.	

Table 51 Use Case Rematch (Single player Lose)

36. Single player Exit (Exit the Single player mode)

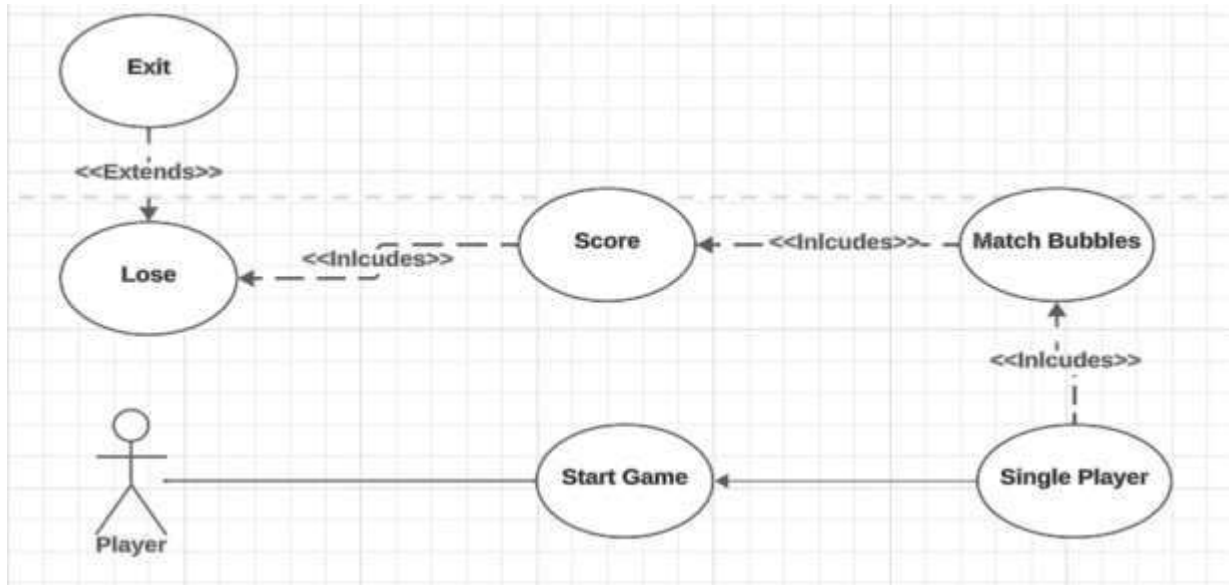


Figure 37 Use Case Diagram Single player Exit

Use Case ID	UC_36	
Use Case Name	Exit	
Reference	See FR 36	
Description	After losing the game in Single-player game mode the player is exiting the multiplayer game mode.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the Single-player match. 2. Should lose the match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After losing the game the player will be able to exit the vs Single player game mode and explore other game modes.	
Basic Flow	Actor Action	System Action
	The player lost the match and clicked on the Exit button.	The system will take the player to the game mode screen to choose from the other game modes.
Alternative Flow	The player can ask for a rematch.	

Table 52 Use Case Exit (Single player Loose)

37. Pause (Pause the game mode)

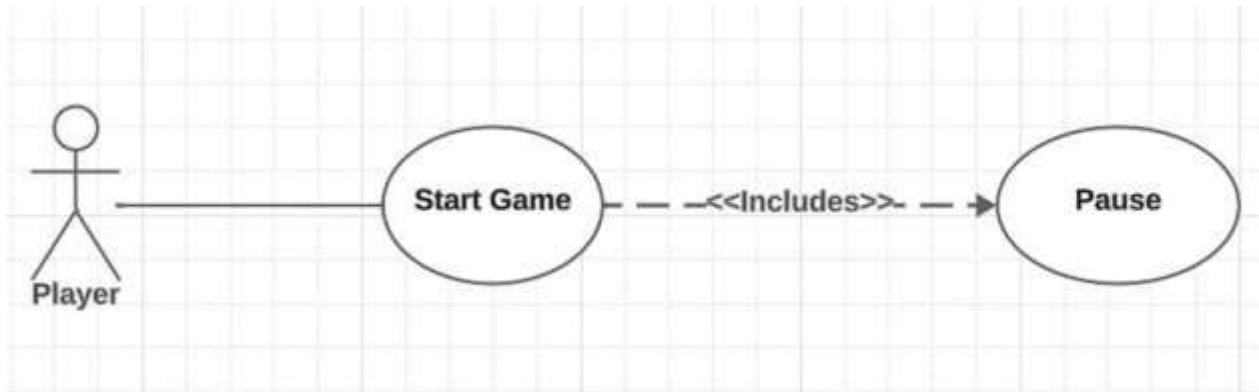


Figure 38 Use Case Diagram Pause

Use Case ID	UC_37	
Use Case Name	Pause	
Reference	See FR 37	
Description	After starting the game if the player wants to pause in the middle of the game he can click on the pause button.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should be in the match. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After pausing the game the player will be able to put a halt to his/her current gaming activities and do something else.	
Basic Flow	Actor Action	System Action
	The player has started the game and has clicked on the pause button.	The system will take the player to the pause screen where his game will stop.
Alternative Flow	The player can exit or resume the game if he wants.	

Table 53 Use Case Pause

38. Pause Exit (Exit the game after pausing)

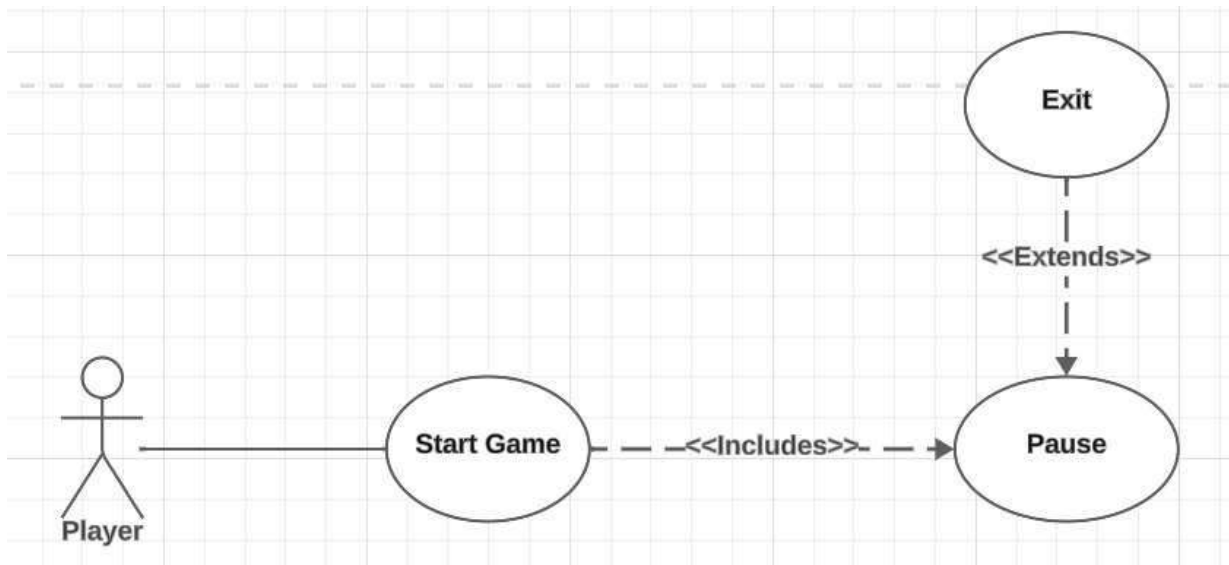


Figure 39 Use Case Diagram Pause Exit

Use Case ID	UC_38	
Use Case Name	Exit	
Reference	See FR 38	
Description	If the player wants to leave the game in the middle of the match, he can do so by pausing the game and clicking on the exit button.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should start the game. 2. Should pause the game. 3. Working internet connection. 4. Opened software in the system.	
Post-Conditions	After pausing and clicking on the exit the person will be sent to the game mode screen where he can choose other modes if he likes to.	
Basic Flow	Actor Action	System Action
	The player has to click on the pause and then on the exit button.	The system will transfer the player to the main menu screen.
Alternative Flow	The player can resume the game.	

Table 54 Use Case Exit (Pause)

39. Setting (Customize the setting)



Figure 40 Use Case Diagram Setting

Use Case ID	UC_39	
Use Case Name	Setting	
Reference	See FR 39	
Description	If the player wants to change some settings to his/her liking then he can do so by clicking the setting button.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should have an open main menu. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player will be able to customize the game setting to his/her personal preferences.	
Basic Flow	Actor Action	System Action
	The player has to click on the setting button.	The system will take the player to the settings screen.
Alternative Flow	The player can explore other options if he likes to.	

Table 55 Use Case Setting

40. Volume (Change the volume)

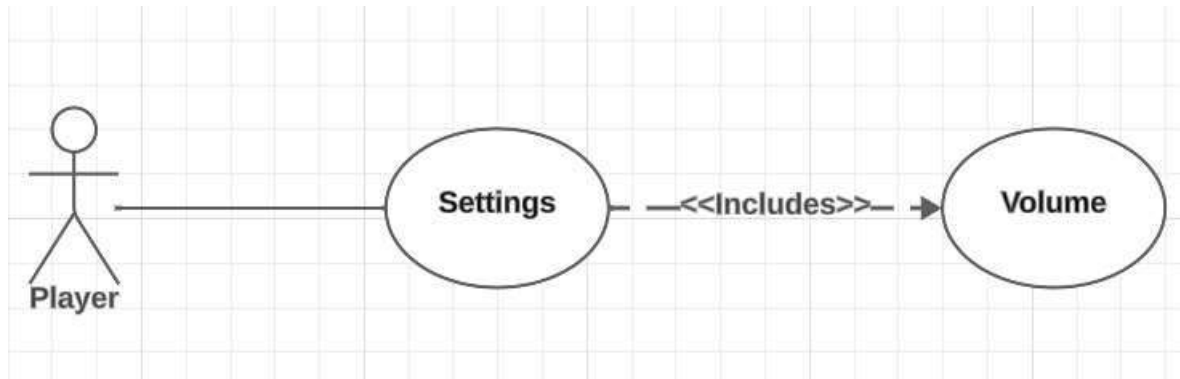


Figure 41 Use Case Diagram Volume

Use Case ID	UC_40	
Use Case Name	Volume	
Reference	See FR 40	
Description	The player can change the volume of the game according to his/her liking.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should be in the settings menu. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player will be able to customize the game volume to his/her personal preferences.	
Basic Flow	Actor Action	System Action
	The player has to click on the setting button and then on the volume button.	The system will take the player to the volume nob screen where he can do as he likes.
Alternative Flow	The player can explore other options and may exit the settings.	

Table 56 Use Case Volume

41. Difficulty (Change the Difficulty)

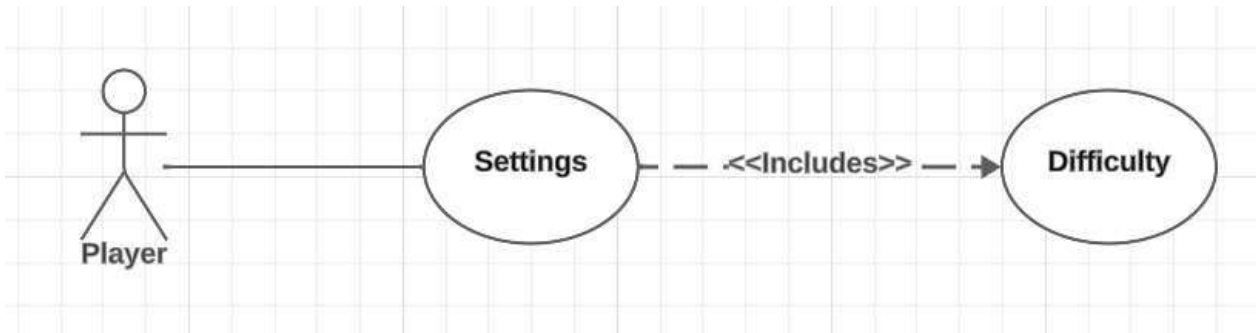


Figure 42 Use Case Diagram Difficulty

Use Case ID	UC_41	
Use Case Name	Difficulty	
Reference	See FR 41	
Description	If the game is too hard or too simple for the player then he can temper the difficulty level as he sees fit.	
Primary actor	Players	
Secondary Actor		
Pre-Conditions	1. Should be in the settings menu. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player will be able to customize the game difficulty level to his/her personal preferences.	
Basic Flow	Actor Action	System Action
	The player has to click on the setting button and then on the difficulty button.	The system will take the player to the difficulty level screen.
Alternative Flow	The player can explore other options and may exit the settings.	

Table 57 Use Case Difficulty

42. Logout (Logout from the game)

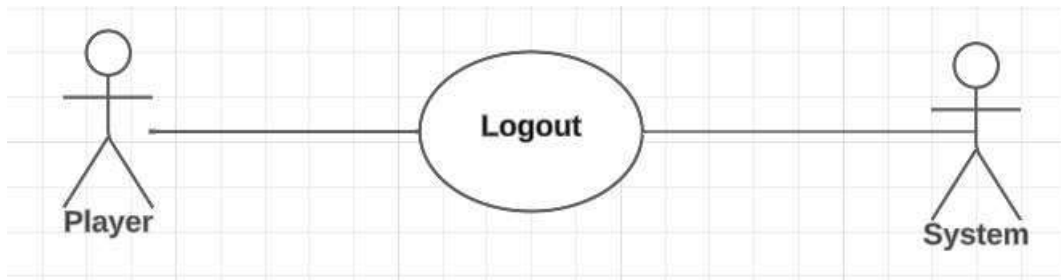


Figure 43 Use Case Diagram Logout

Use Case ID	UC_42	
Use Case Name	Logout	
Reference	See FR 42	
Description	If the player wants to log out or needs to log in from another account he can do so by clicking on the logout button.	
Primary actor	Players	
Secondary Actor	System	
Pre-Conditions	1. Should be in the main menu. 2. Working internet connection. 3. Opened software in the system.	
Post-Conditions	The player then be logged out of the game if he wants to, he can log in again whenever he/she likes or wants to play multiplayer login is a must.	
Basic Flow	Actor Action	System Action
	The player has to click on the logout button.	The system will log the player out of his/her active account.
Alternative Flow	The player can choose to log in again or play some game modes directly.	

Table 58 Use Case Logout

Chapter 4: Design

4.1 Architecture Diagram

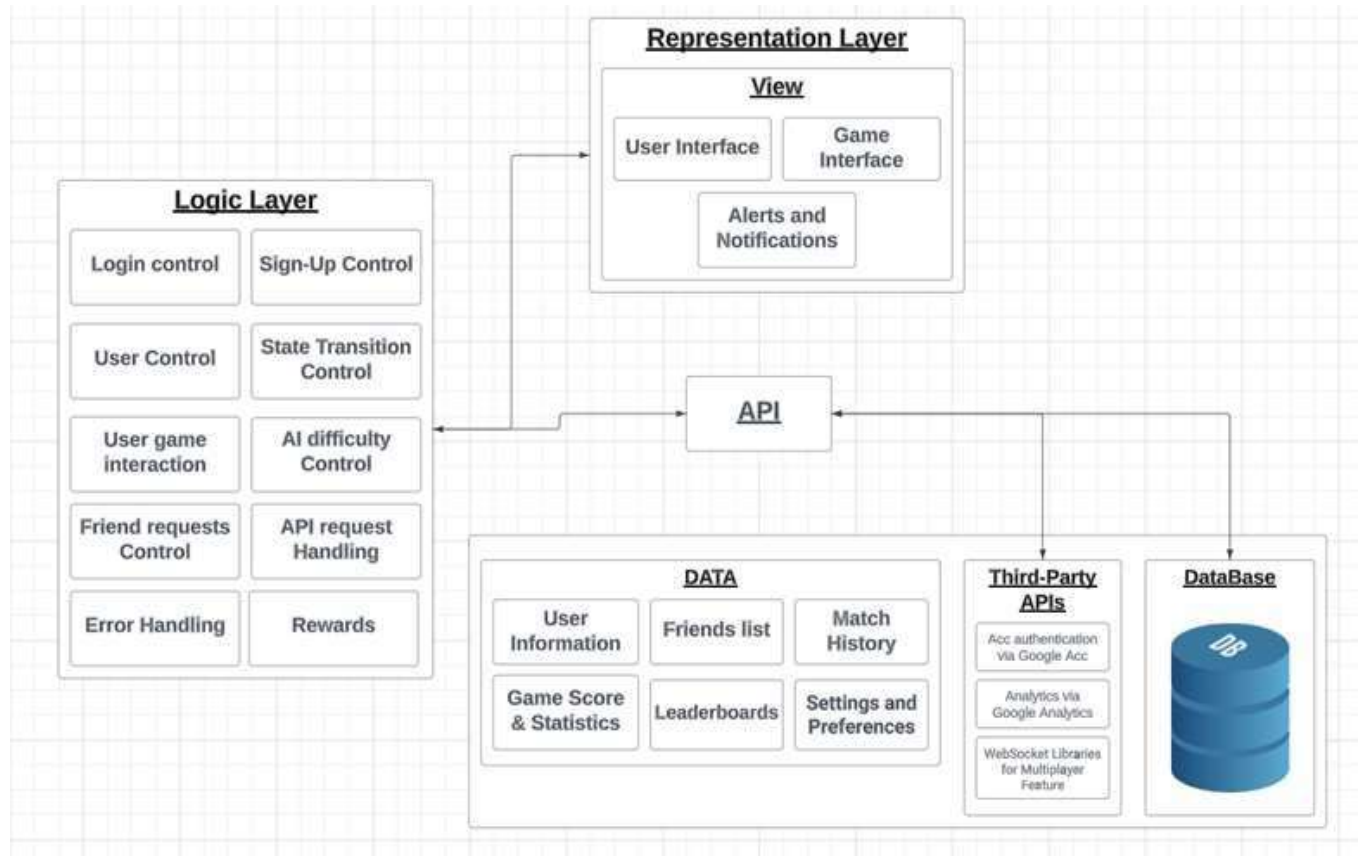


Figure 44 Architecture Diagram

4.2 ERD with data dictionary

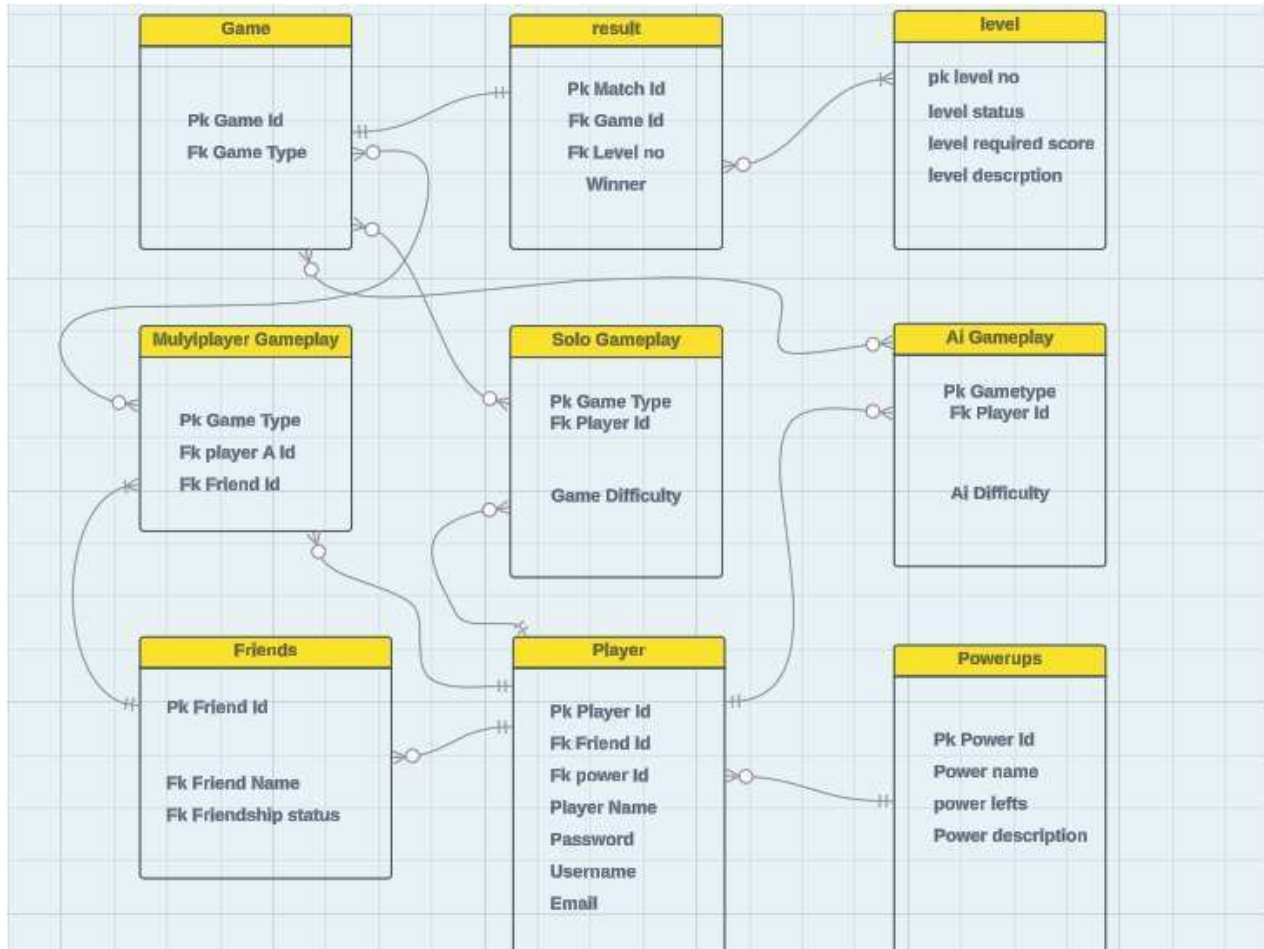


Figure 45 ERD

4.3 Data Flow diagram

4.3.1 The level 0

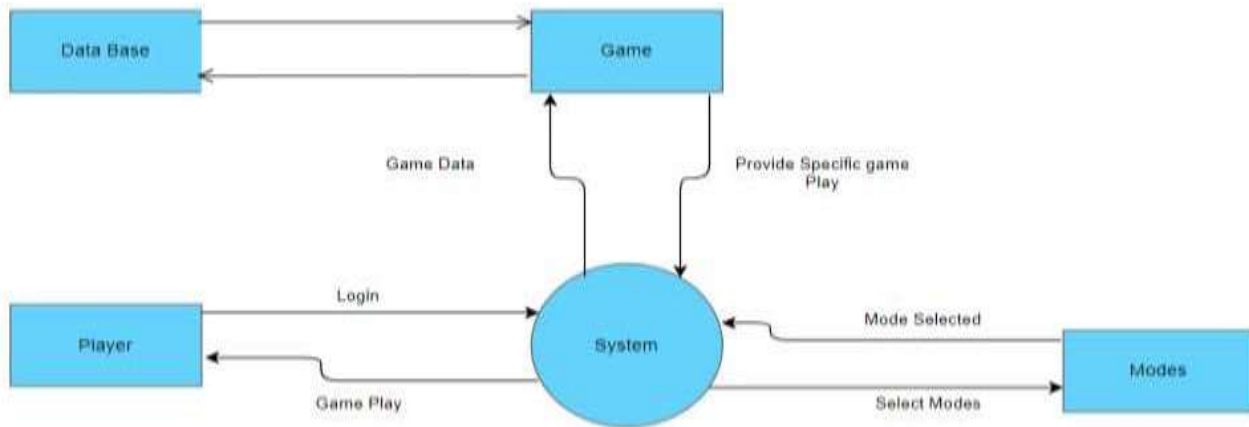


Figure 46 Level 0 DFD

4.3.2 The level 1

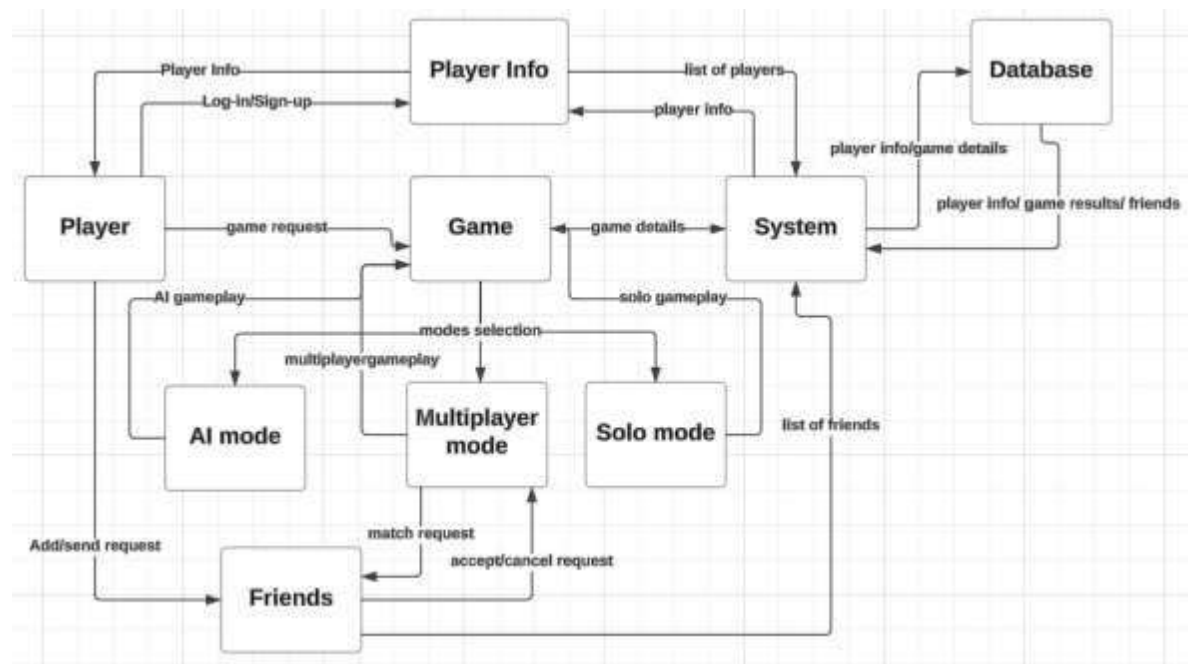


Figure 47 Level 1 DFD

4.4 Class Diagram

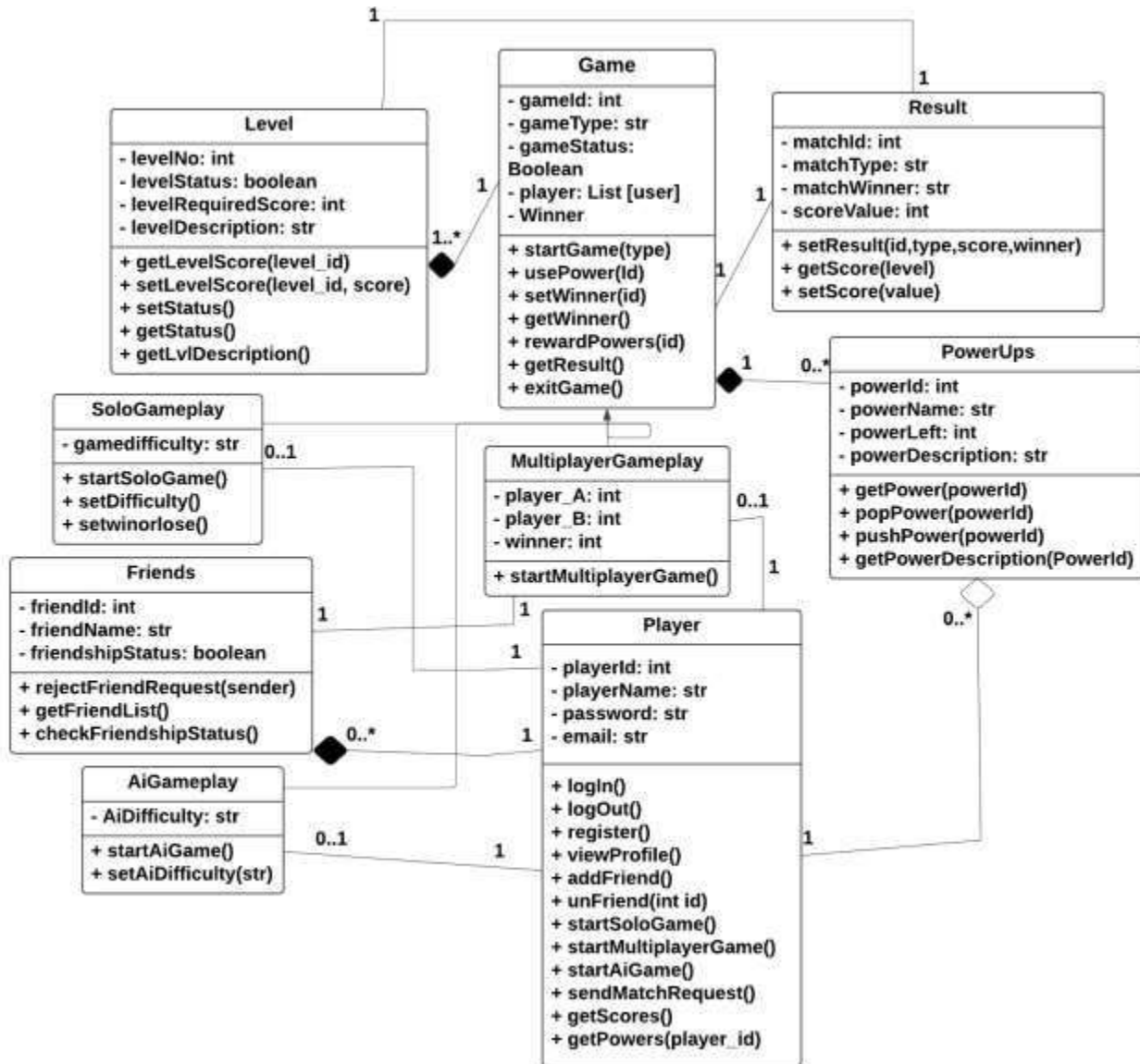


Figure 48 Class Diagram

4.5 Activity Diagram

4.5.1 Login

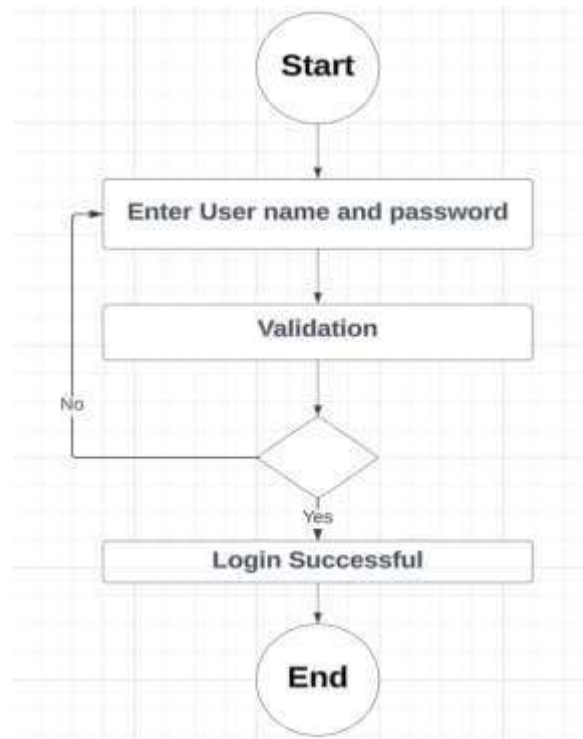


Figure 49 Activity Diagram Login

4.5.2 Reset Password (Password Forgotten?)

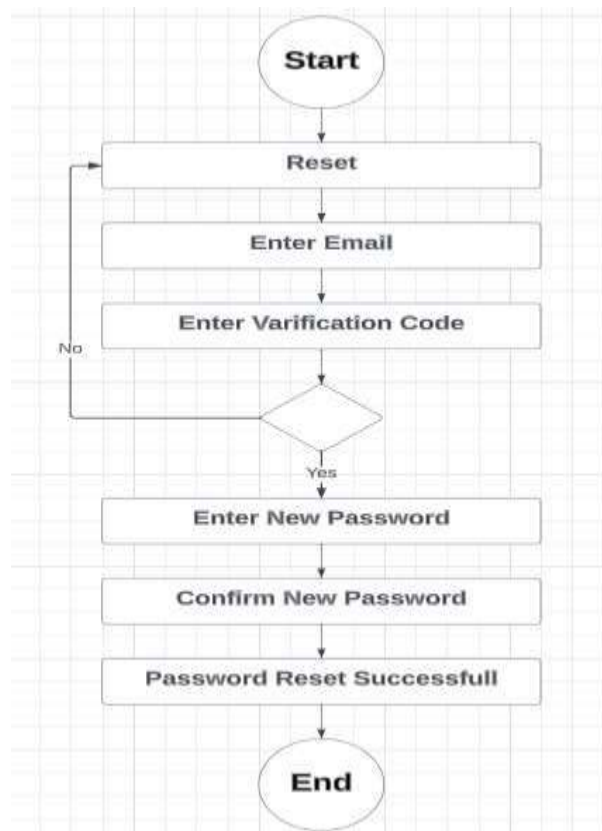


Figure 50 Activity Diagram Reset Password

4.5.3 Logout

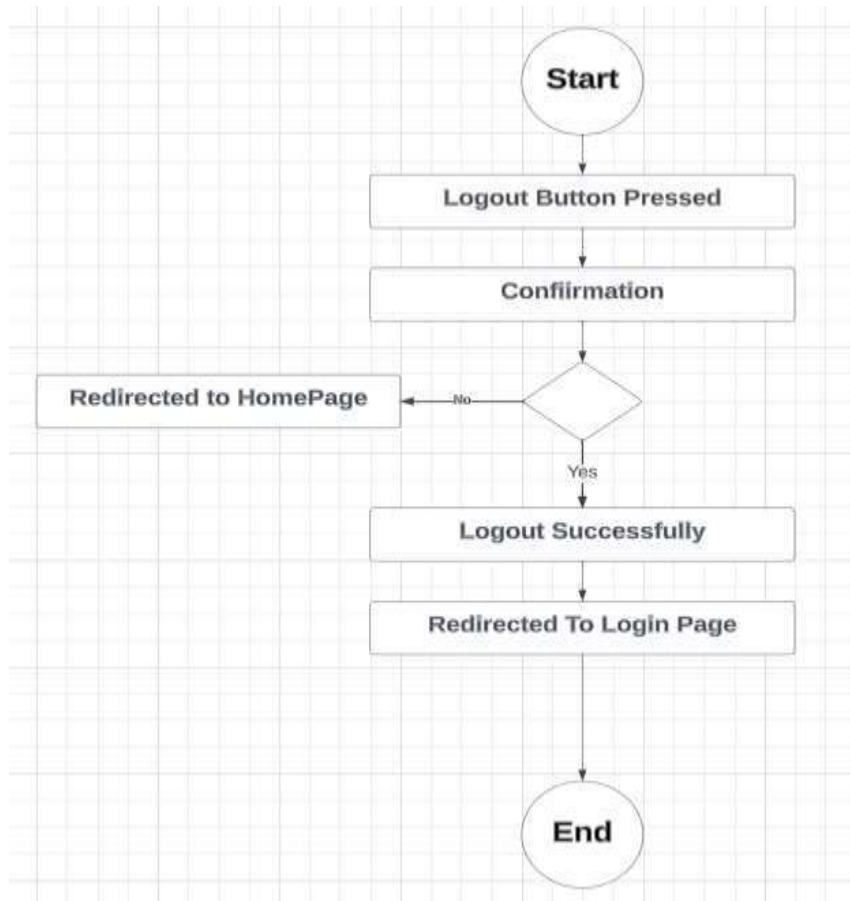


Figure 51 Activity Diagram Logout

4.5.4 Sign-up

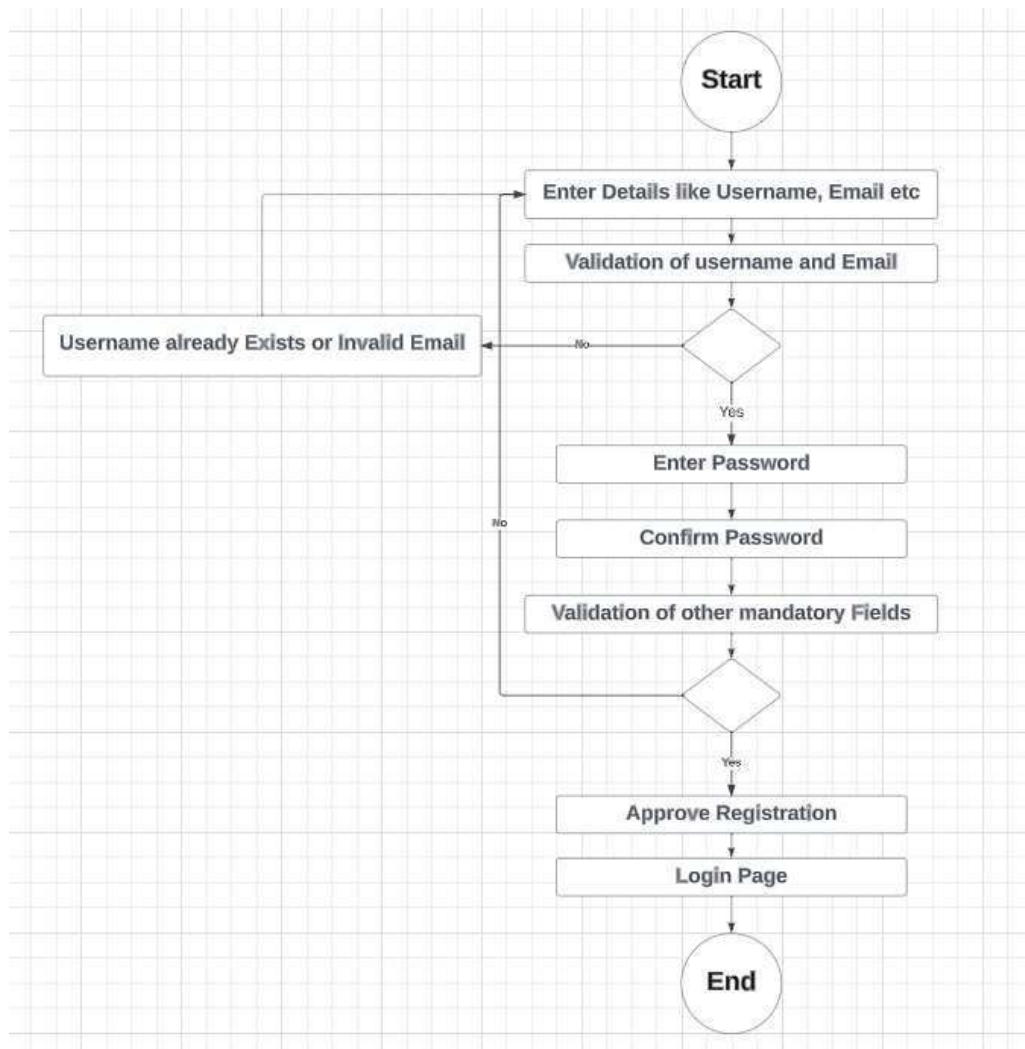


Figure 52 Activity Diagram *Sign-up*

4.5.5 Update Profile

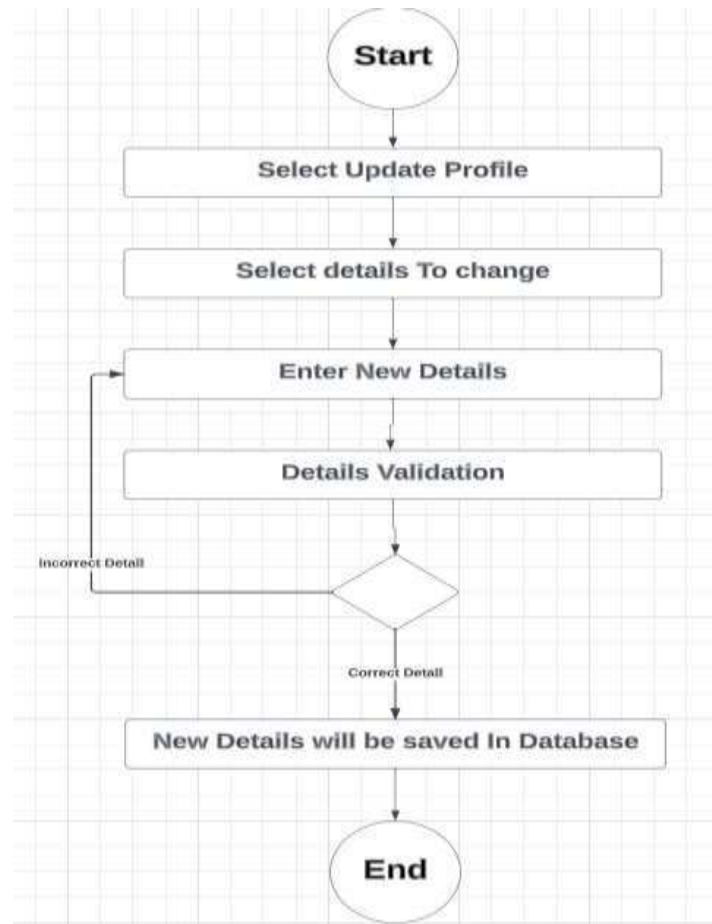


Figure 53 Activity Diagram Update Profile

4.5.6 Starting Game

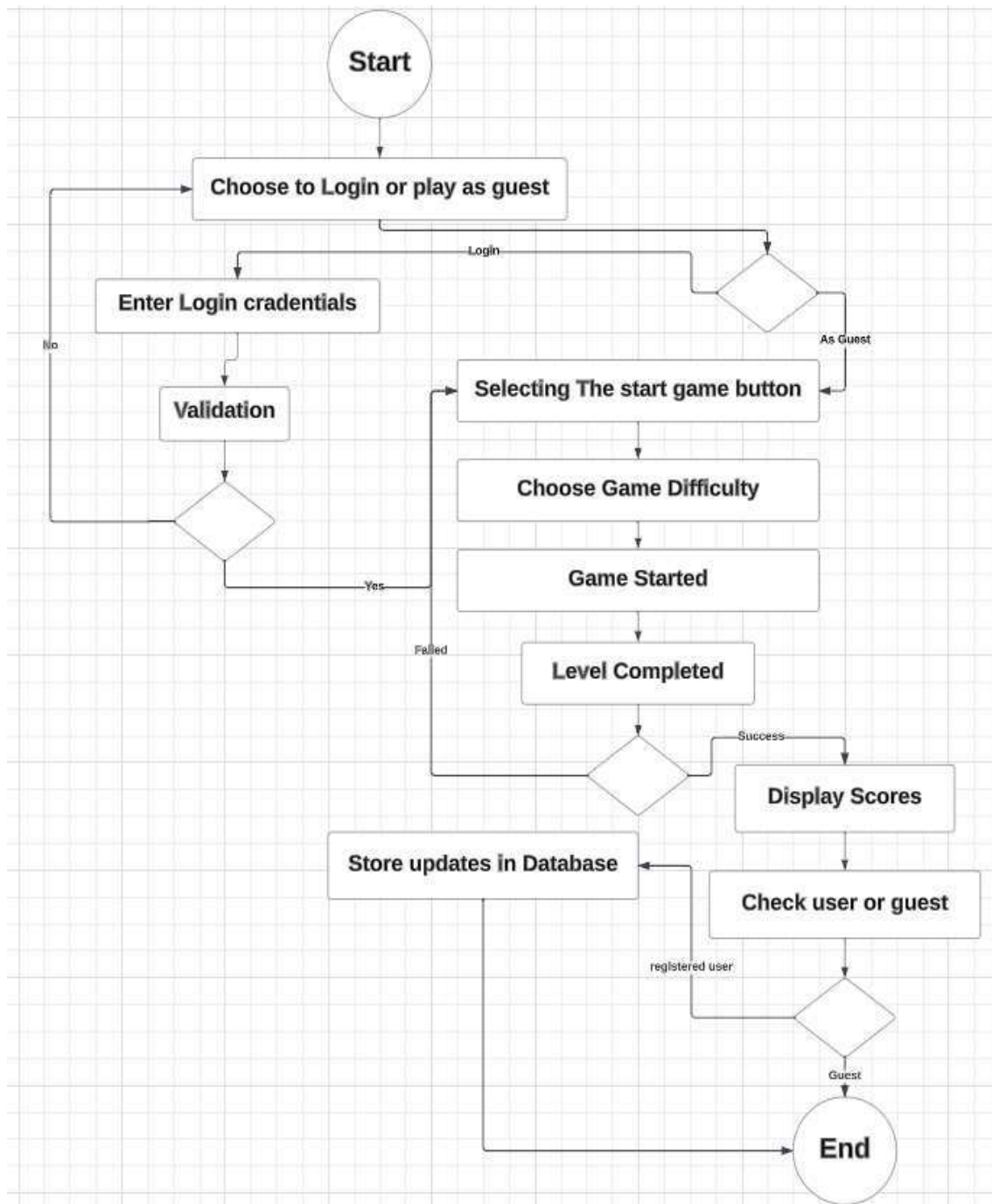


Figure 54 Activity Diagram Starting Game

4.5.7 Level Failed

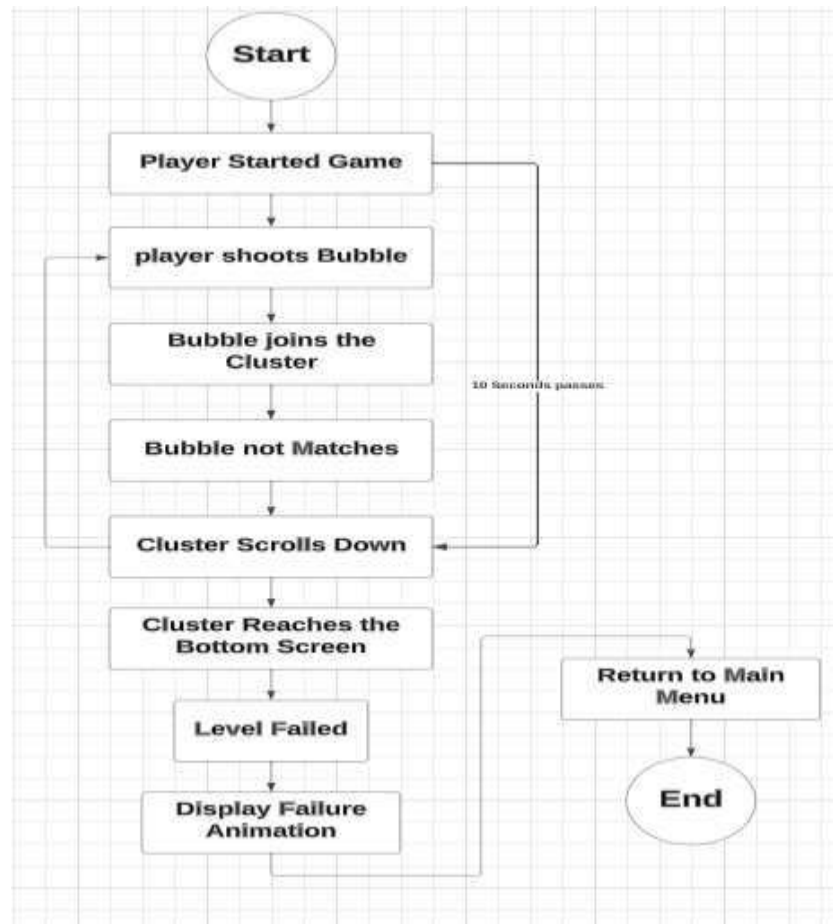


Figure 55 Activity Diagram Level Failed

4.5.8 Restarting or Exiting Game

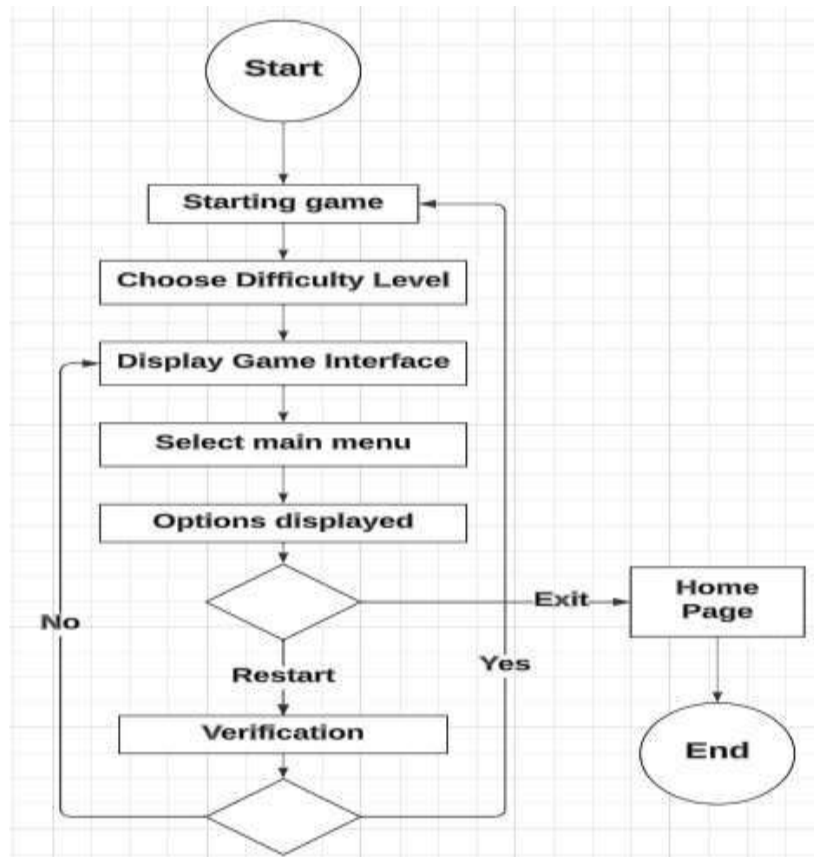


Figure 56 Activity Diagram Restarting or Exiting Game

4.5.9 Sending Friend Request

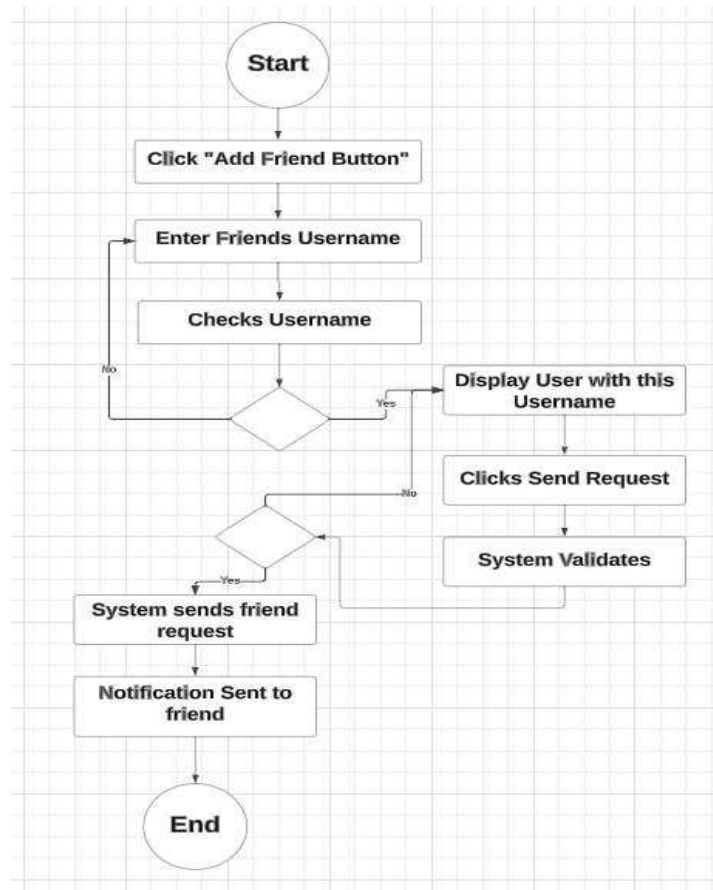


Figure 57 Activity Diagram Sending Friend Request

4.5.10 Accepting or cancelling friend request

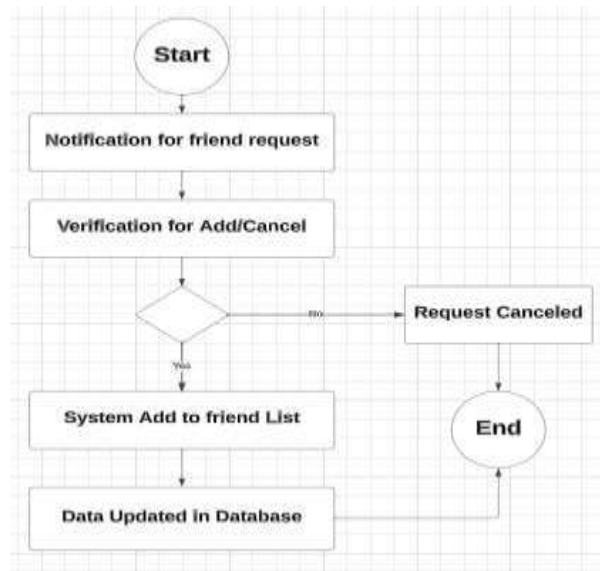


Figure 58 Activity Diagram Accepting or cancelling friend request

4.5.11 Requesting Friend for Match

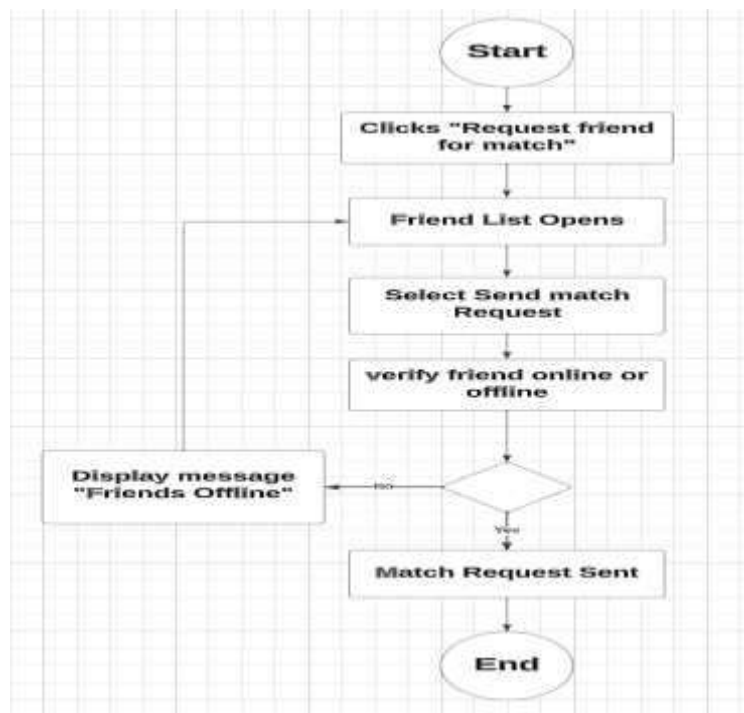


Figure 59 Activity Diagram Requesting Friend for Match

4.5.12 Accepting Friend Request of Match

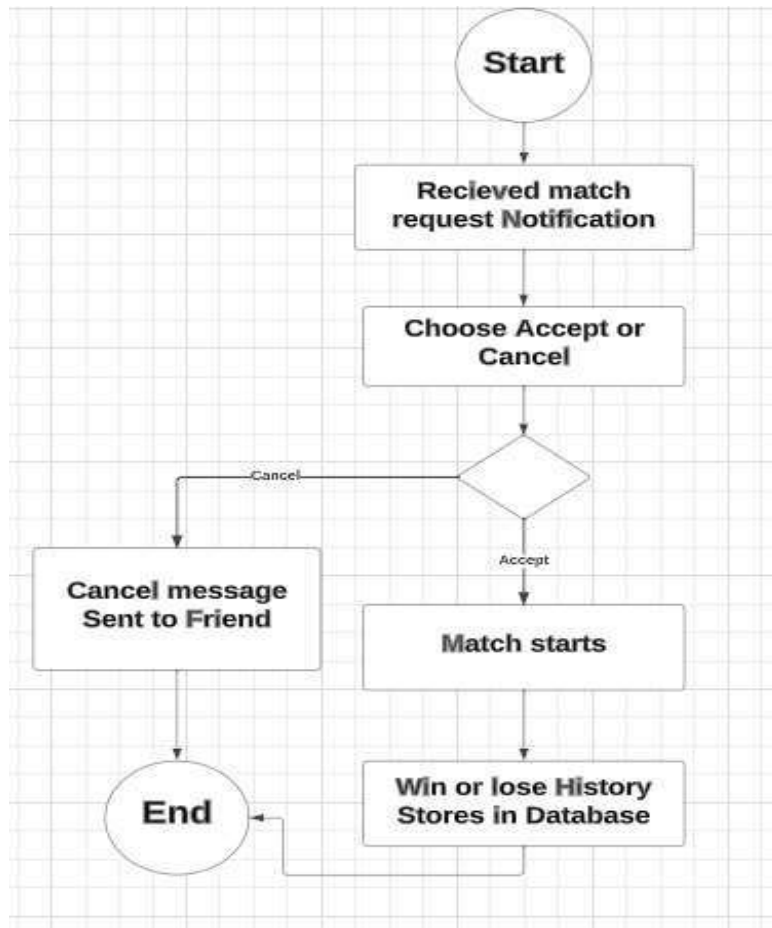


Figure 60 Activity Diagram Accepting Friend Request of Match

4.5.13 Using Power-Ups

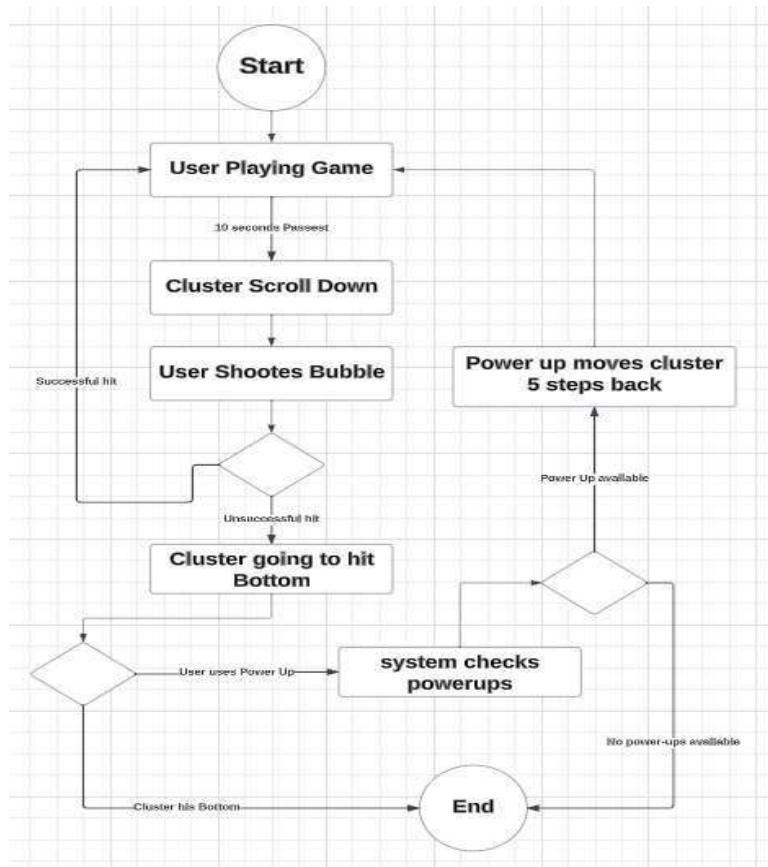


Figure 61 Activity Diagram Using Power-Ups

4.5.14 Selecting AI in multiplayer game

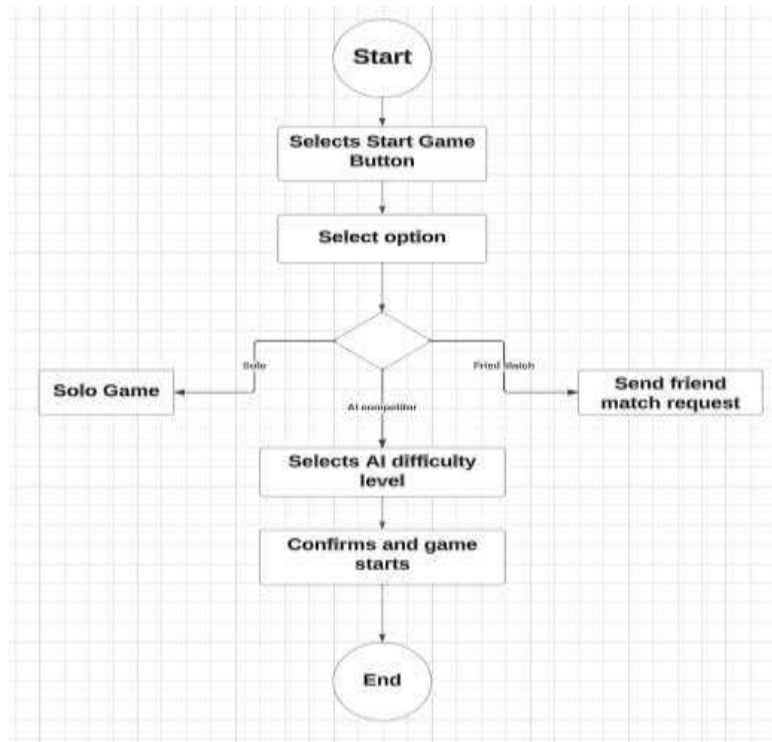


Figure 62 Activity Diagram Selecting AI in multiplayer game

4.5.15 User Ratings and feedback

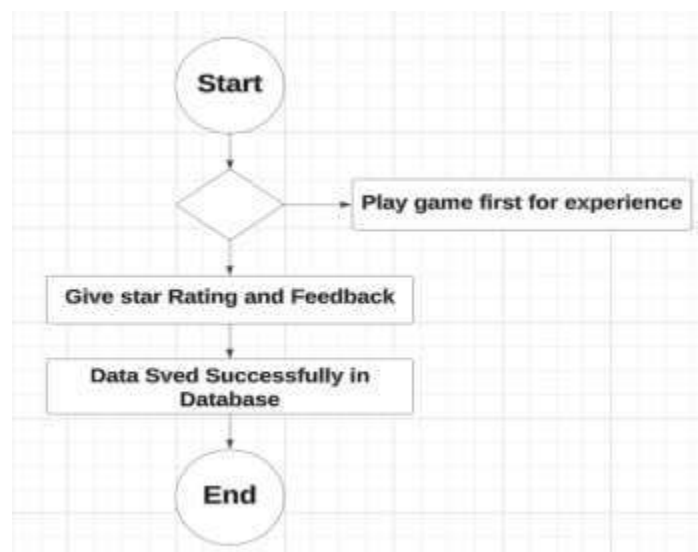


Figure 13 Activity Diagram User Ratings and feedback

4.6 Sequence Diagram

4.5.15 Login

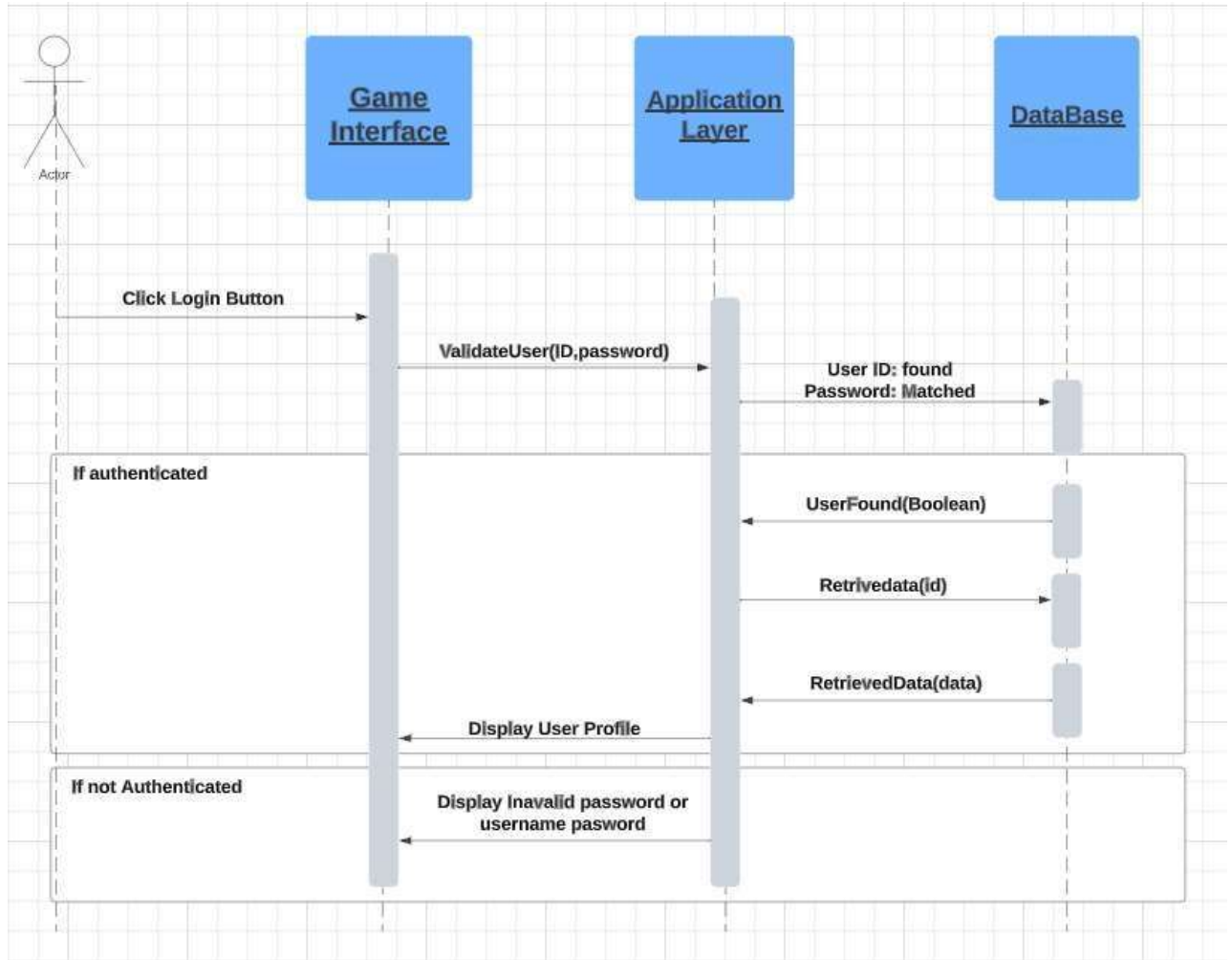


Figure 24 Sequence Diagram login

4.5.15 Reset Password (Password Forgotten?)

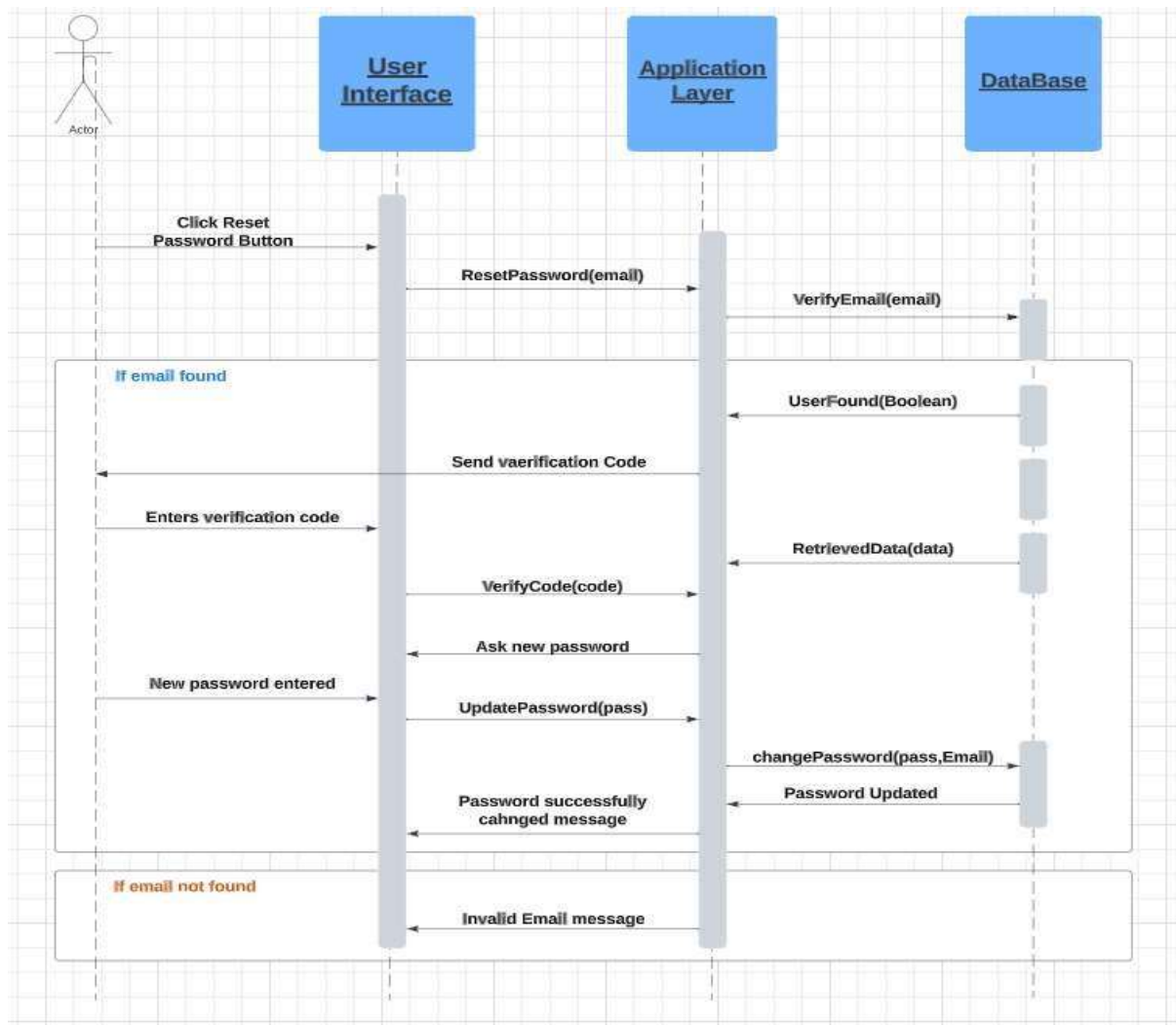


Figure 35 Sequence Diagram Reset Password (Password Forgotten?)

4.5.15 Logout

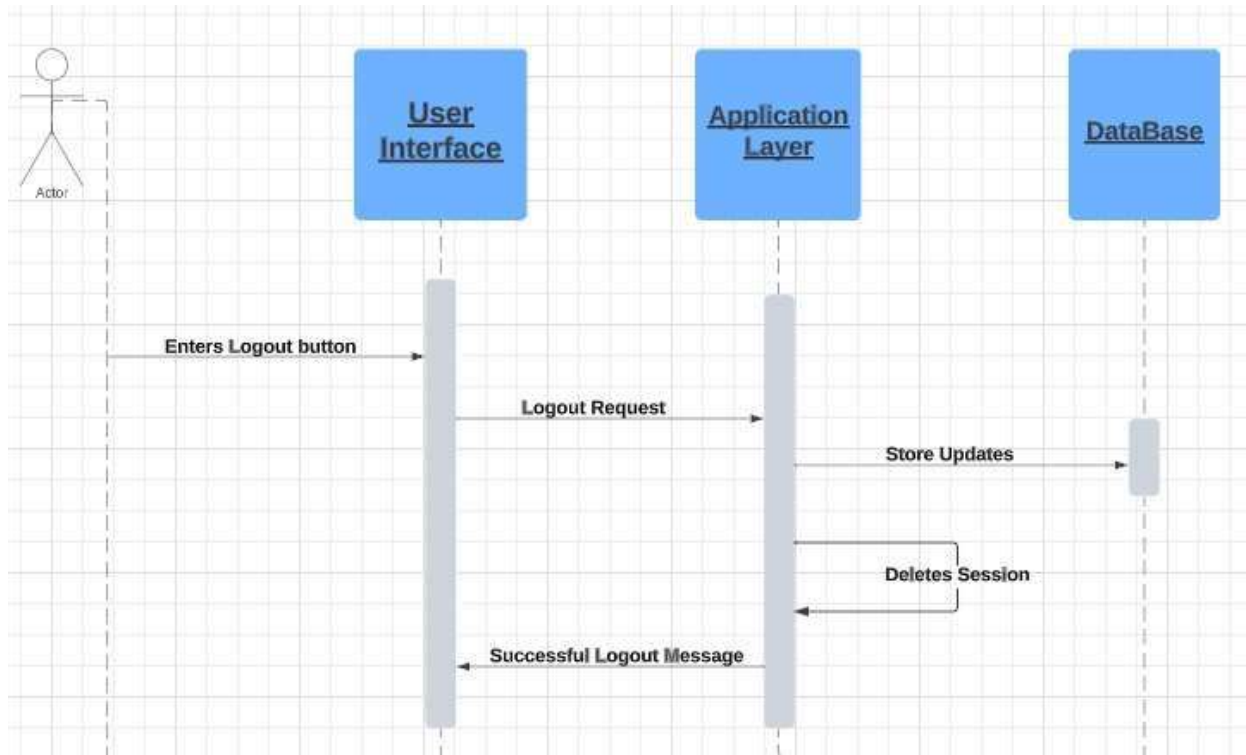


Figure 46 Sequence Diagram Logout

4.5.15 Sign-Up

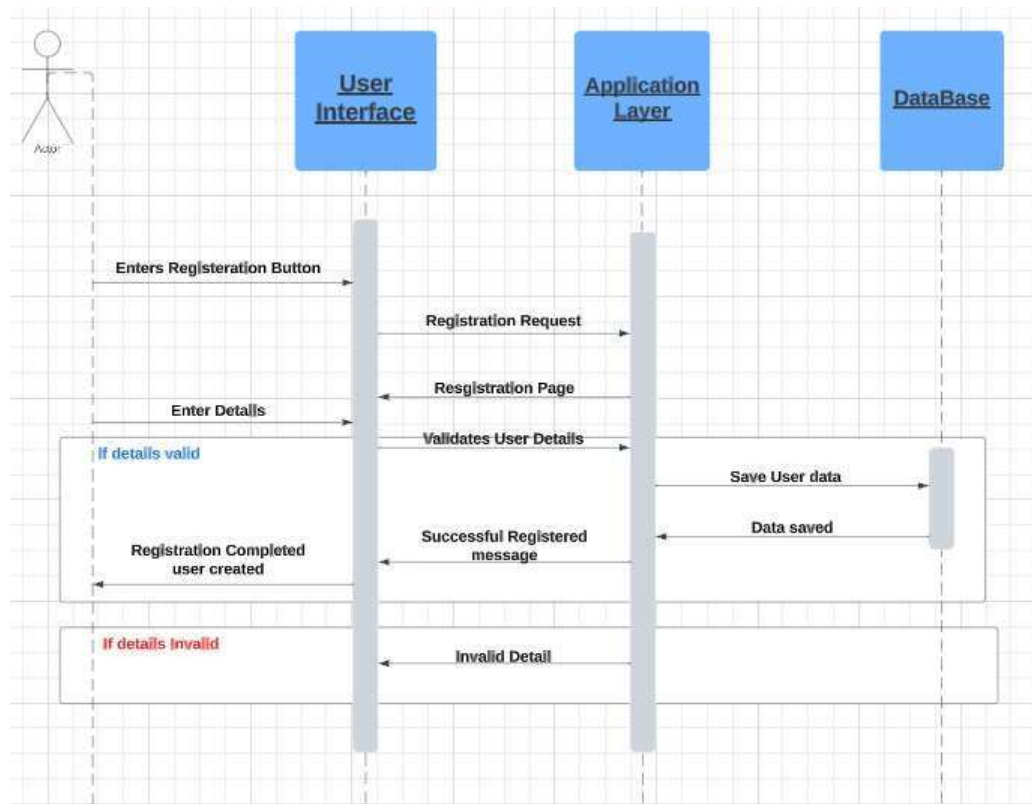


Figure 57 Sequence Diagram Sign-Up

4.5.15 Update profile

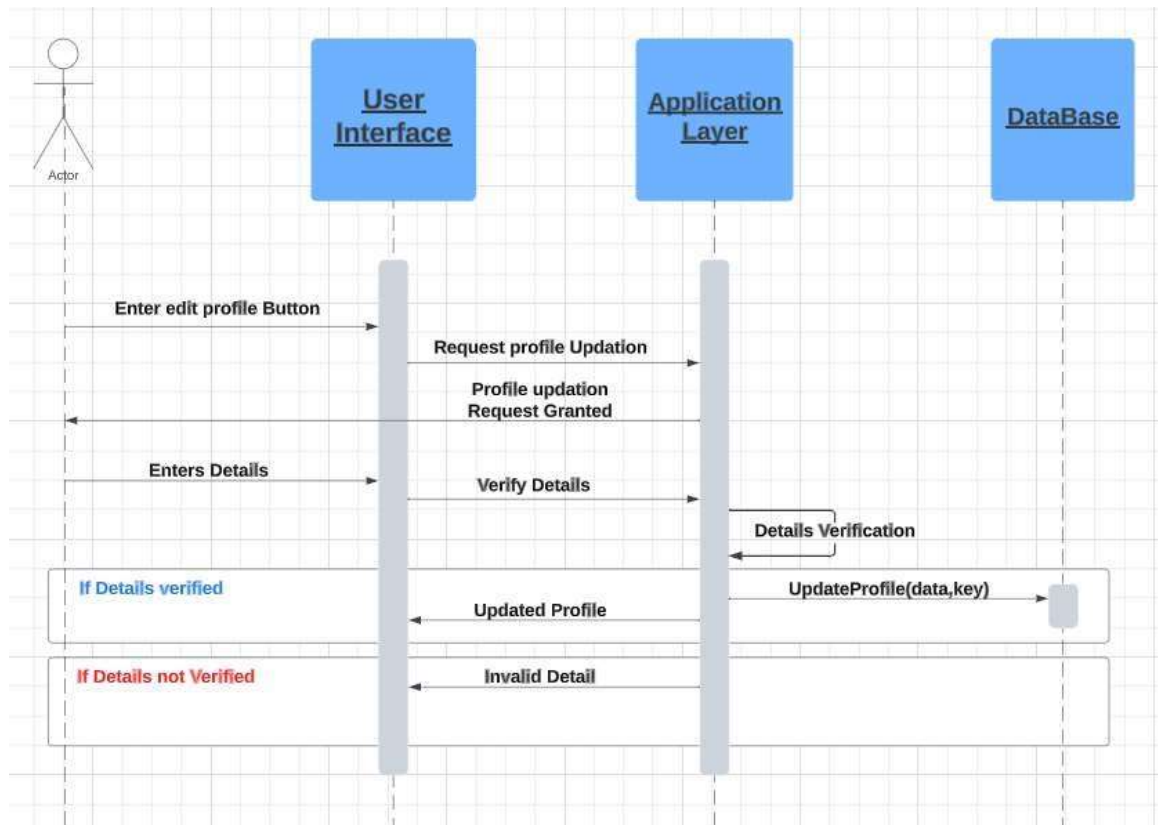


Figure 68 Sequence Diagram Update profile

4.5.15 Starting Game

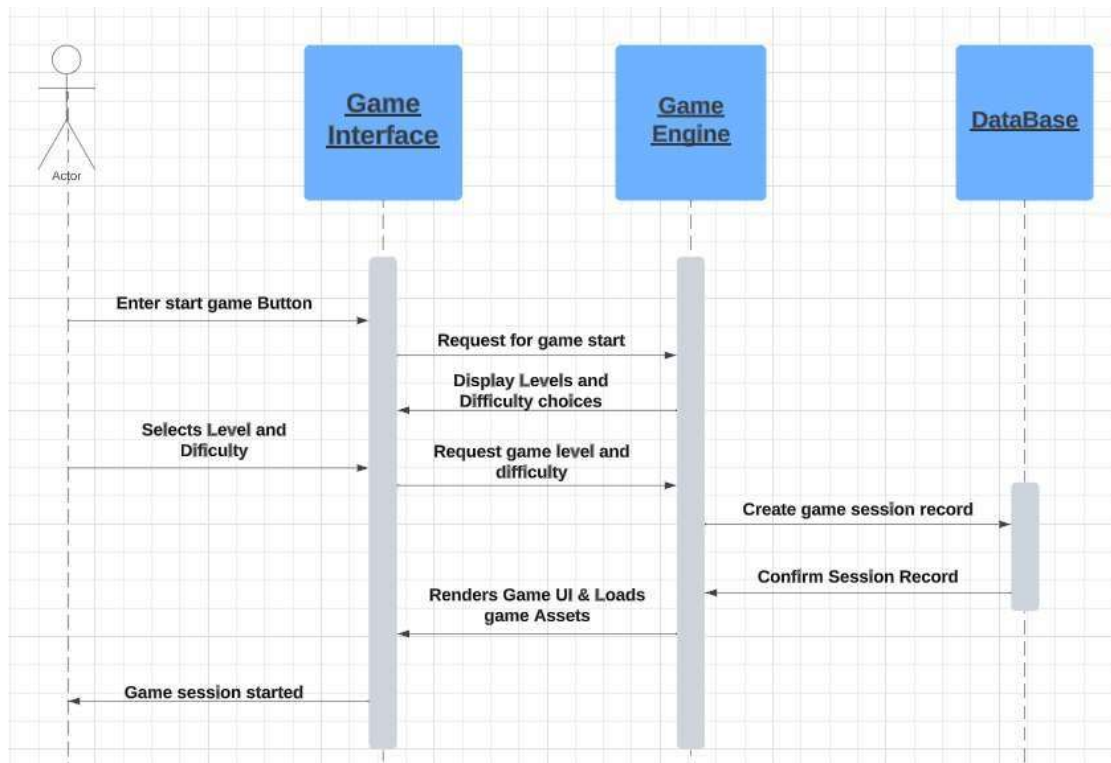


Figure 79 Sequence Diagram Starting Game

4.5.15 Sending Friend Request

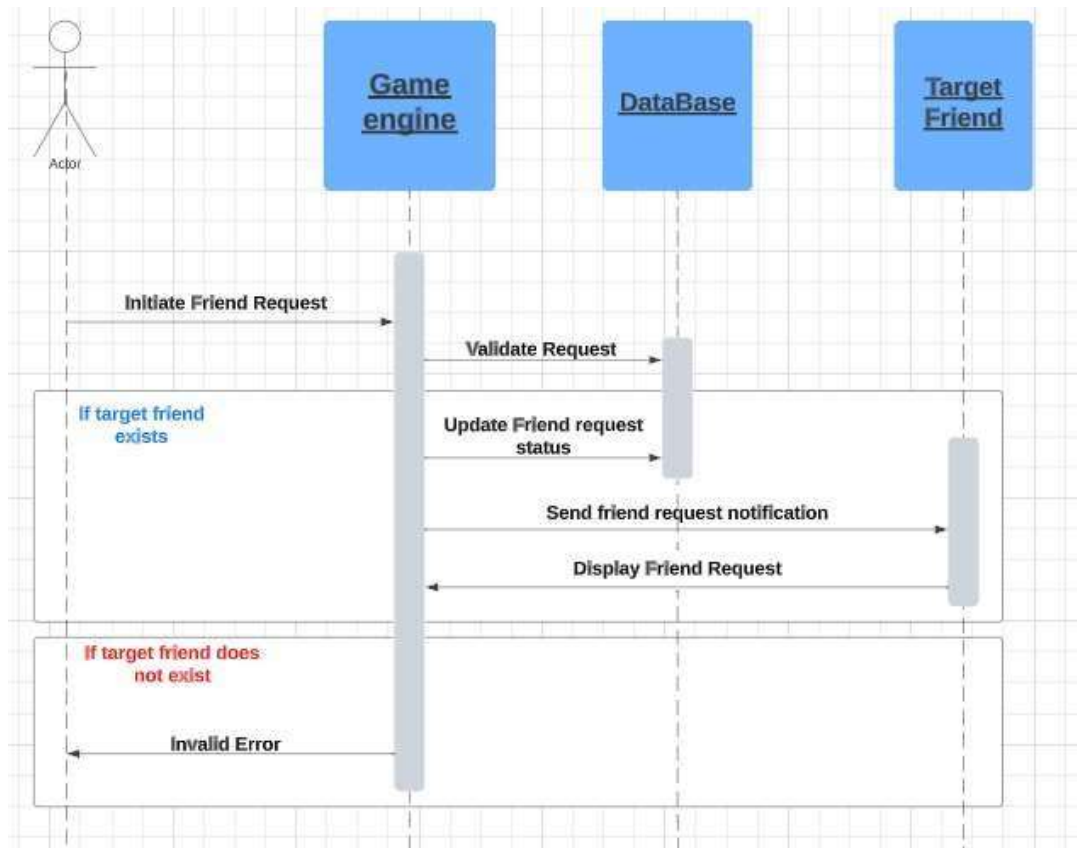


Figure 70 Sequence Diagram Sending Friend Request

4.5.15 Recoding game scores and rewards

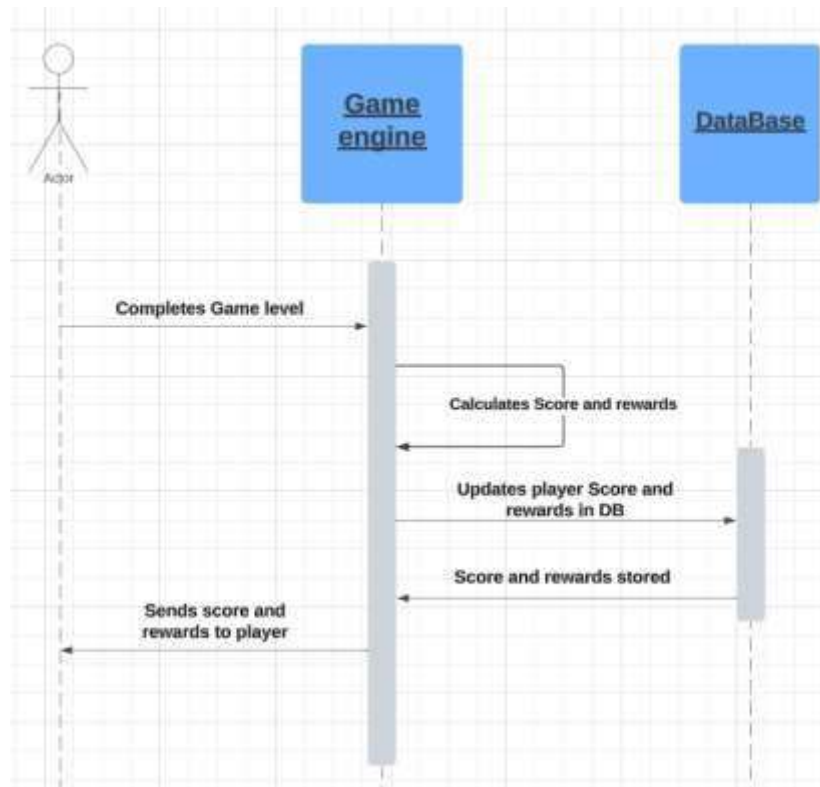


Figure 71 Sequence Diagram Recoding game scores and rewards

4.5.15 Requesting friend for Match

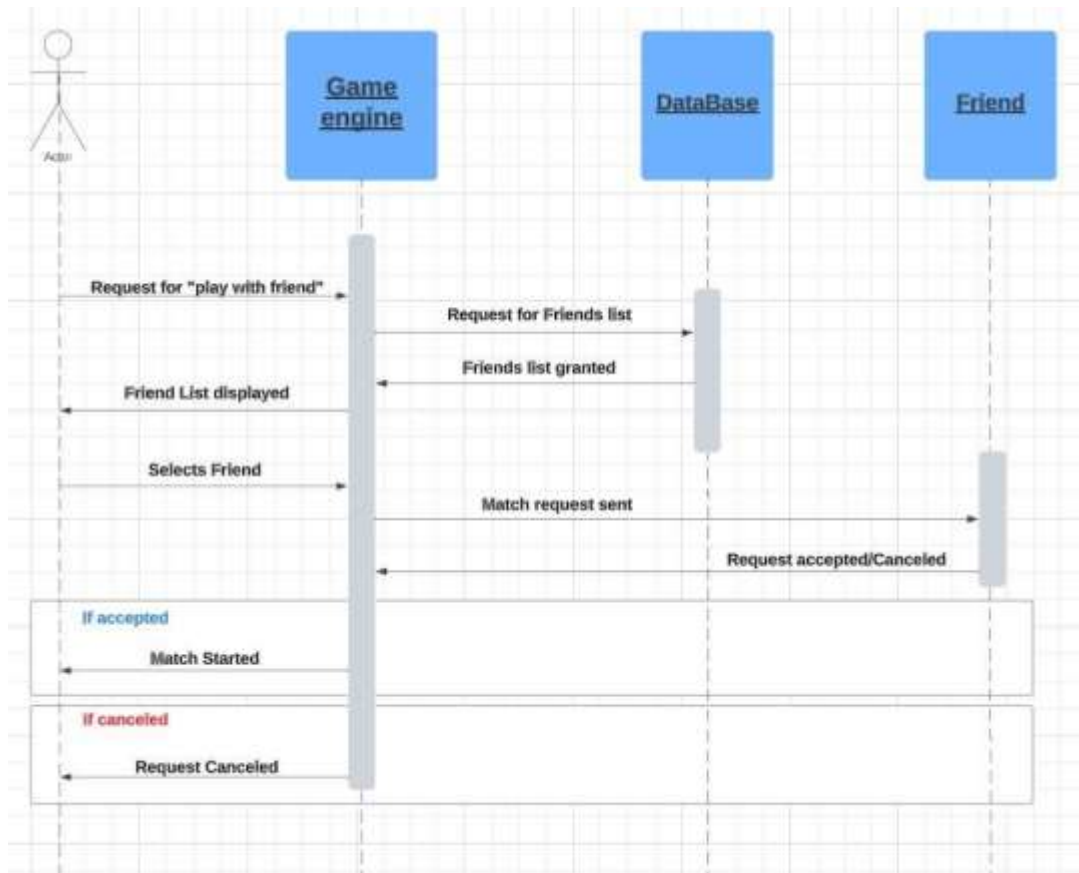


Figure 72 Sequence Diagram Requesting friend for Match

4.5.15 Playing with AI

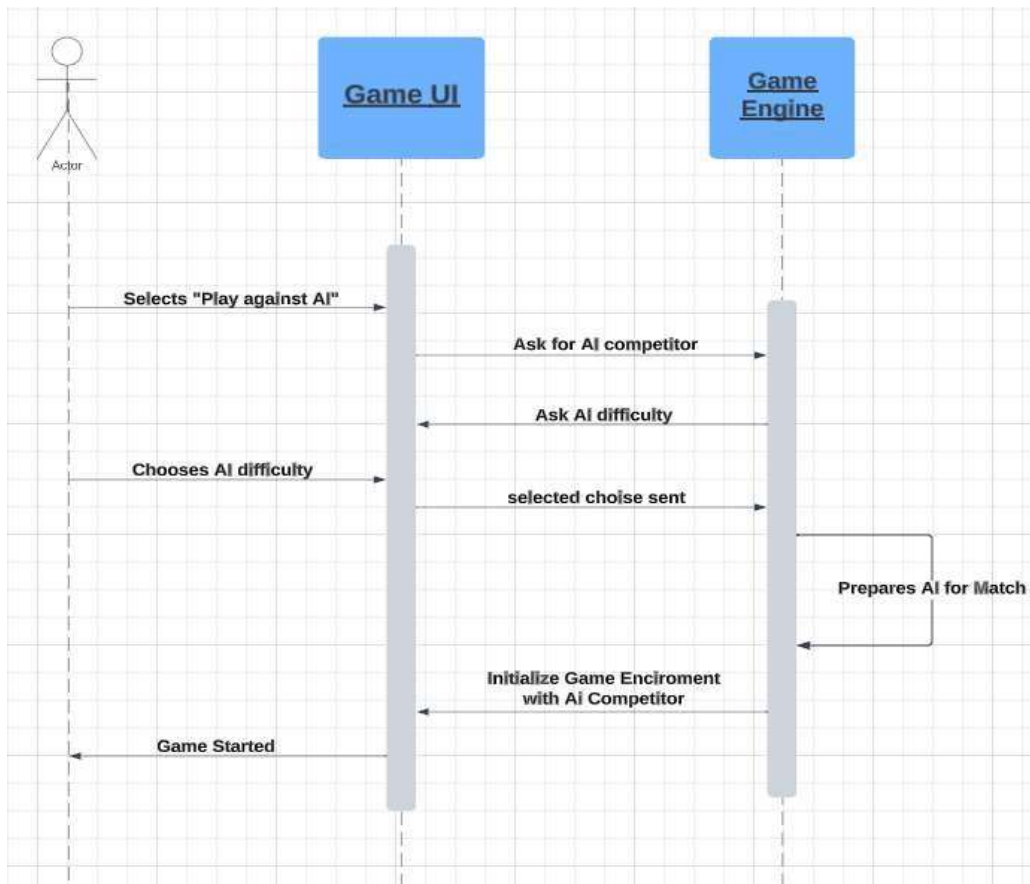


Figure 73 Sequence Diagram Playing with AI

4.7 Collaboration Diagram

4.7.1 Collaboration Diagram Registration/Signup

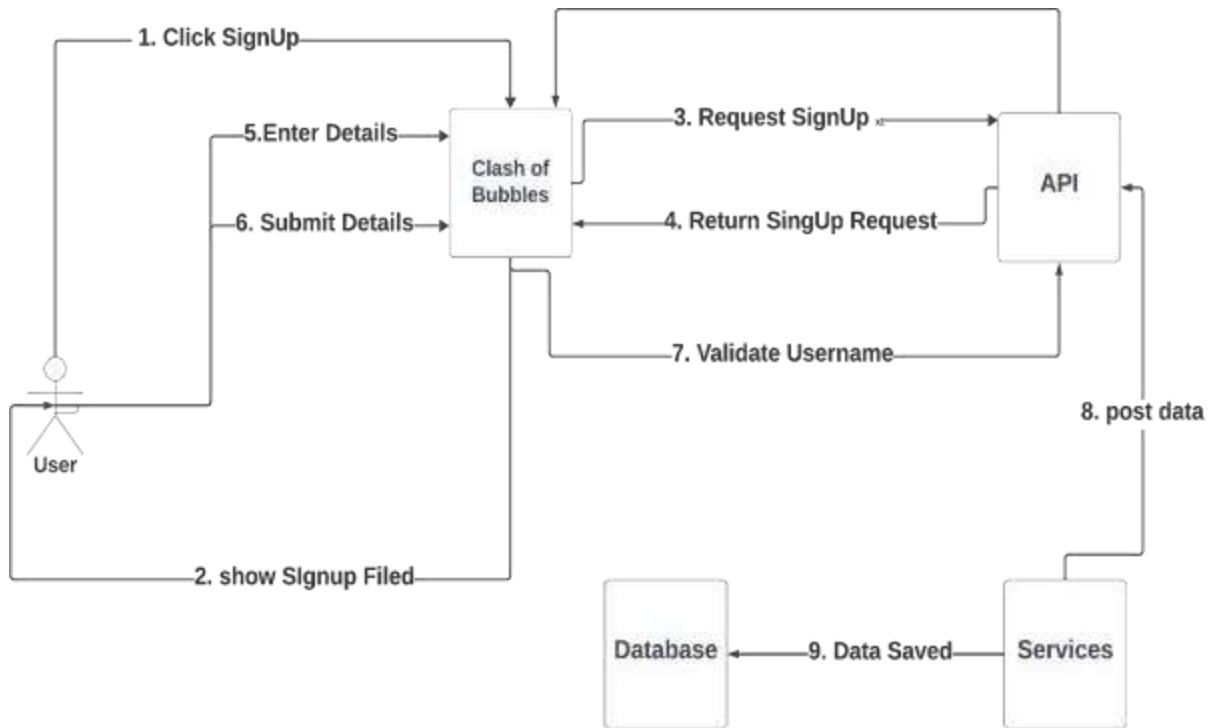


Figure 74 Collaboration Diagram Registration/Signup

4.7.2 Collaboration Diagram Login

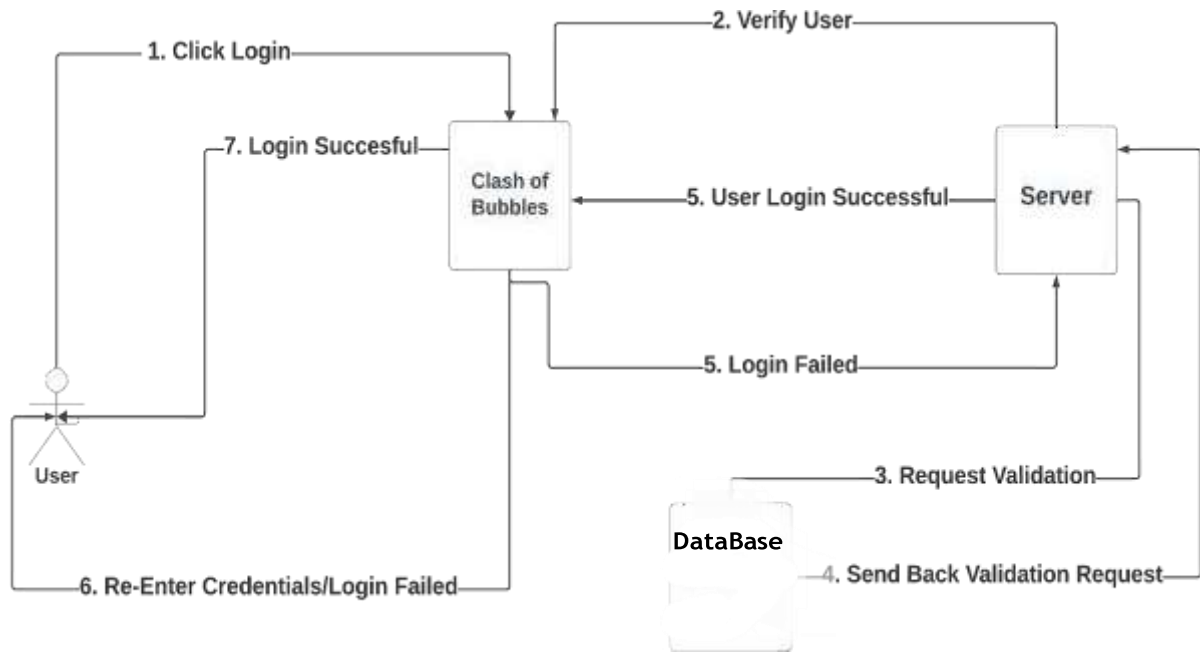


Figure 75 Collaboration Diagram Login

4.7.3 Collaboration Diagram Log-Out

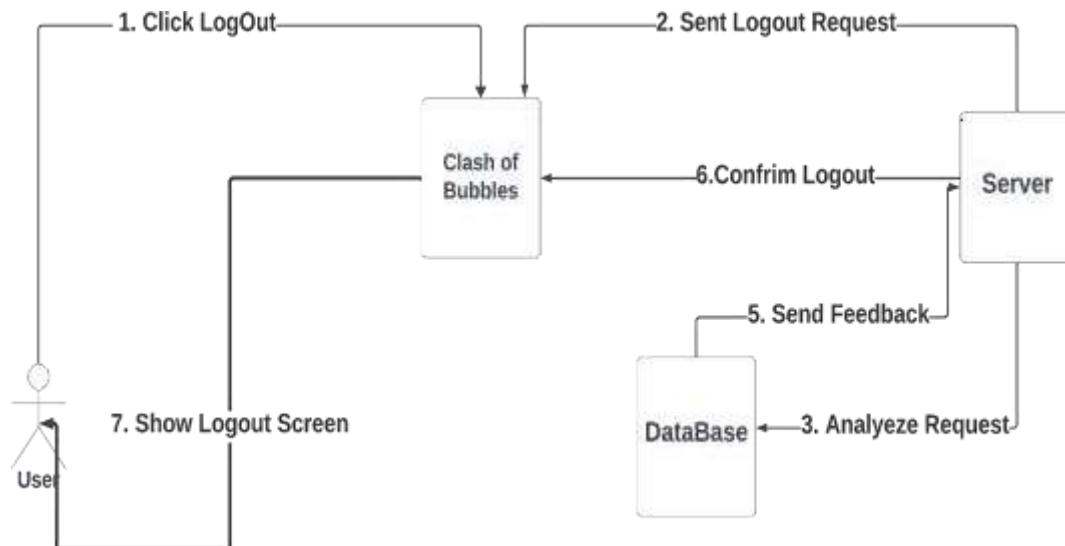


Figure 76 Collaboration Diagram Log-Out

4.7.4 Collaboration Diagram Multiplayer

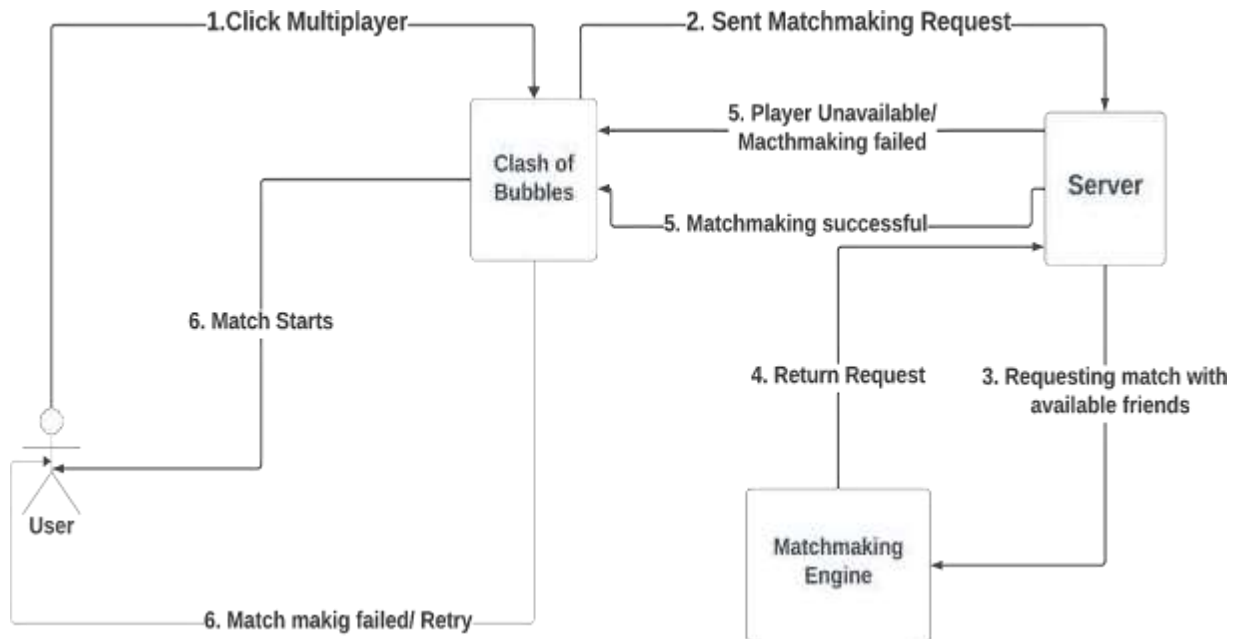


Figure 77 Collaboration Diagram Multiplayer

4.7.5 Collaboration Diagram Edit Profile

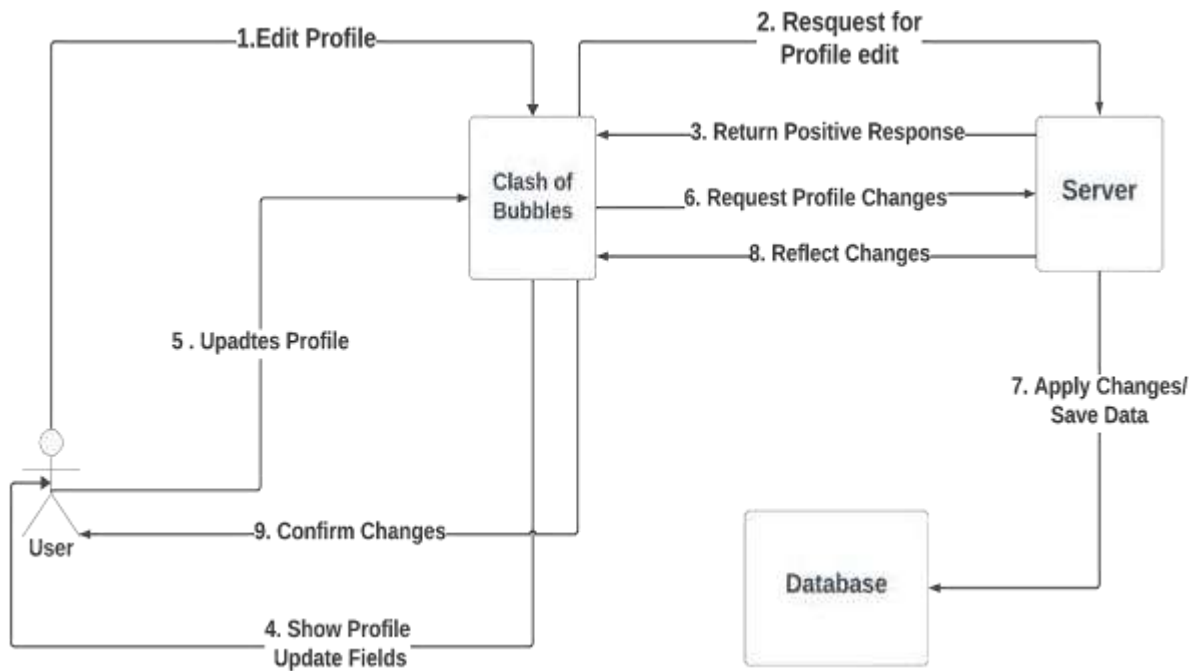


Figure 78 Collaboration Diagram Edit Profile

4.7.6 Collaboration Diagram Friend Request

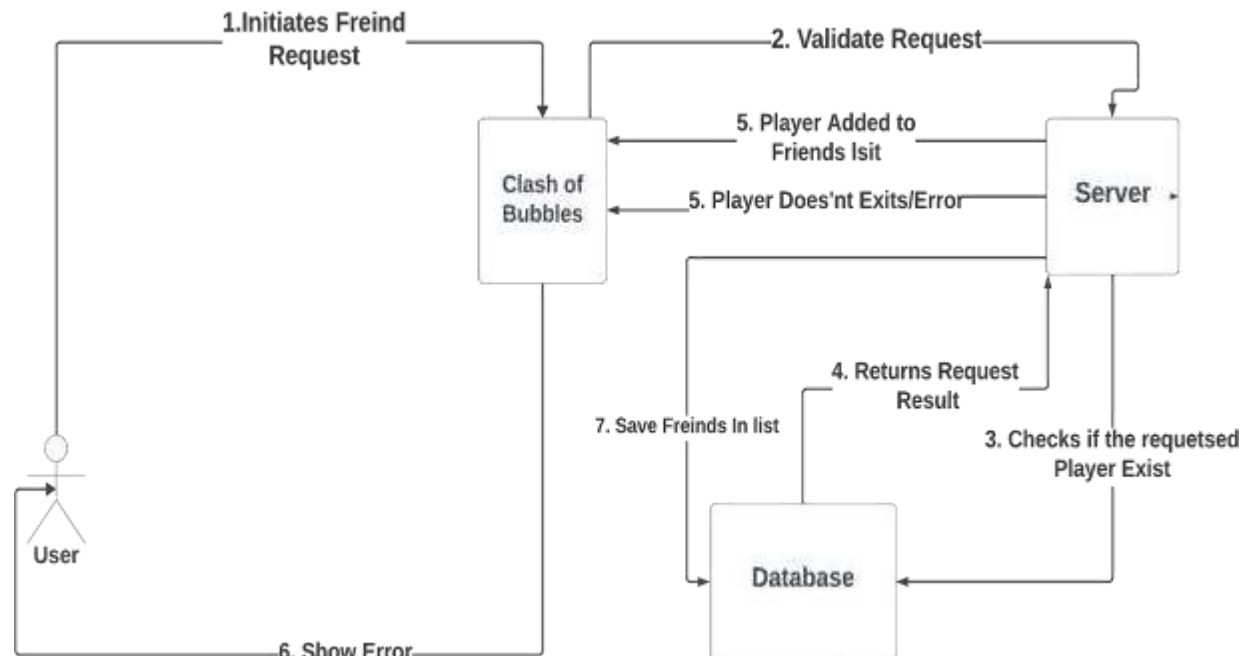


Figure 79 Collaboration Diagram Friend Request

4.7.7 Collaboration Diagram Score Record

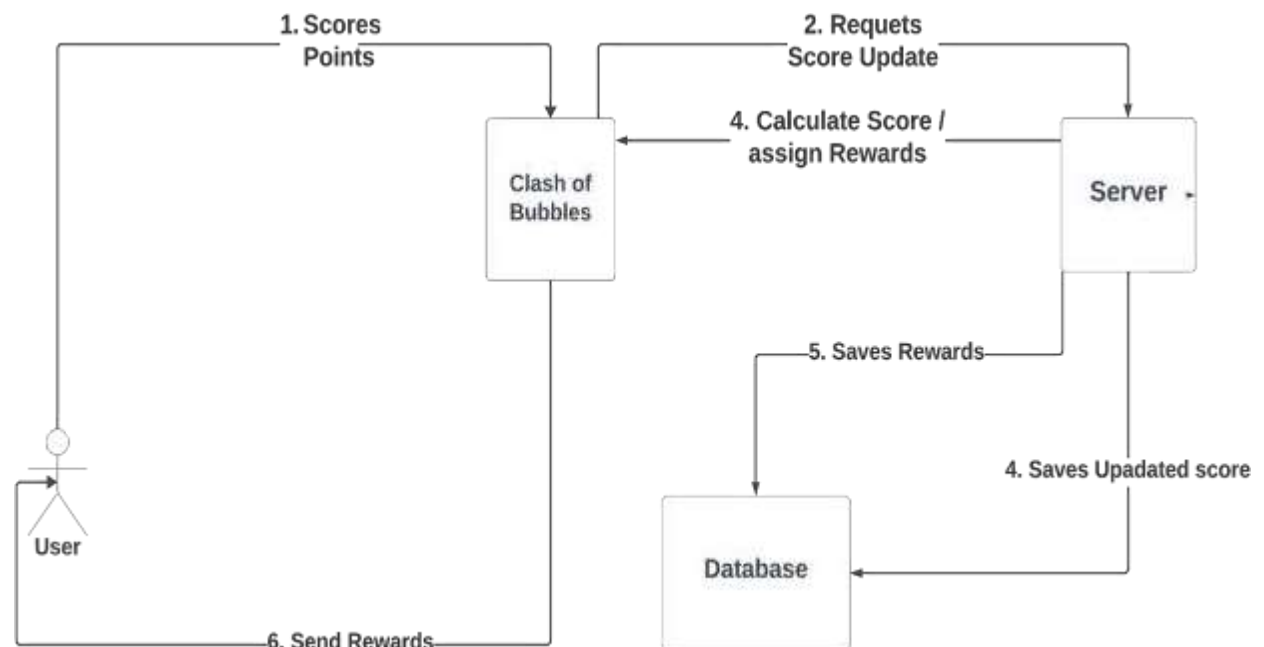


Figure 80 Collaboration Diagram Score Record

4.7.8 Collaboration Diagram vs A.I Match

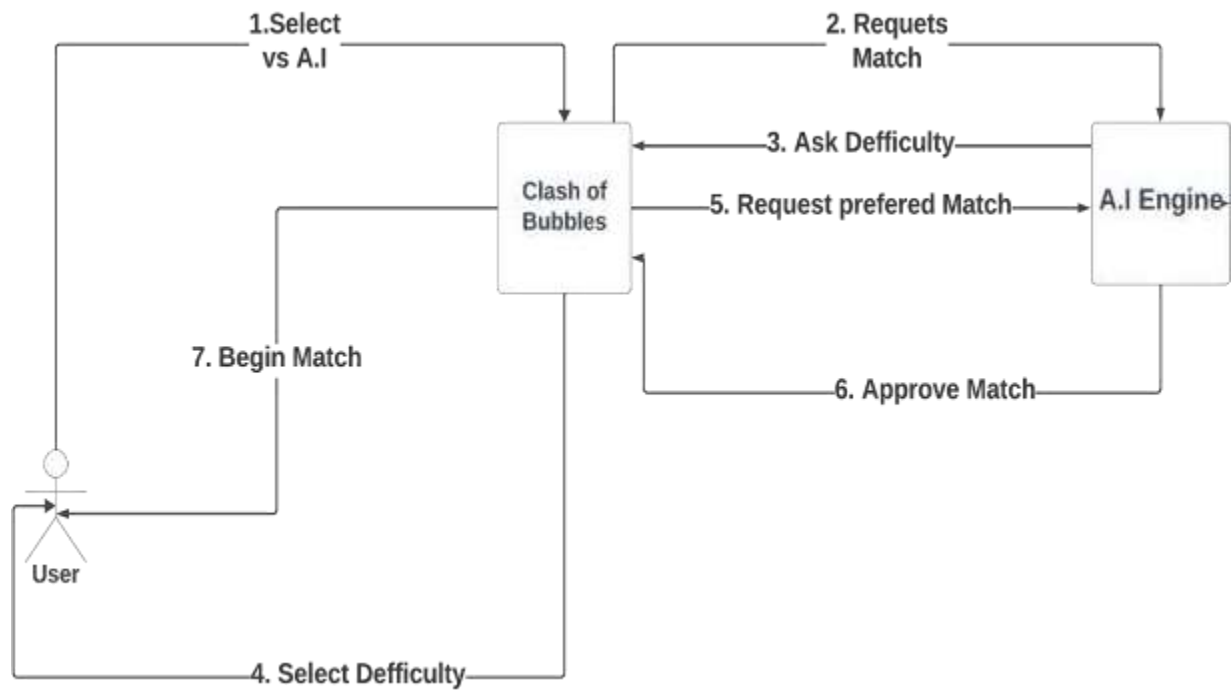


Figure 81 Collaboration Diagram

4.8 State Transition Diagram

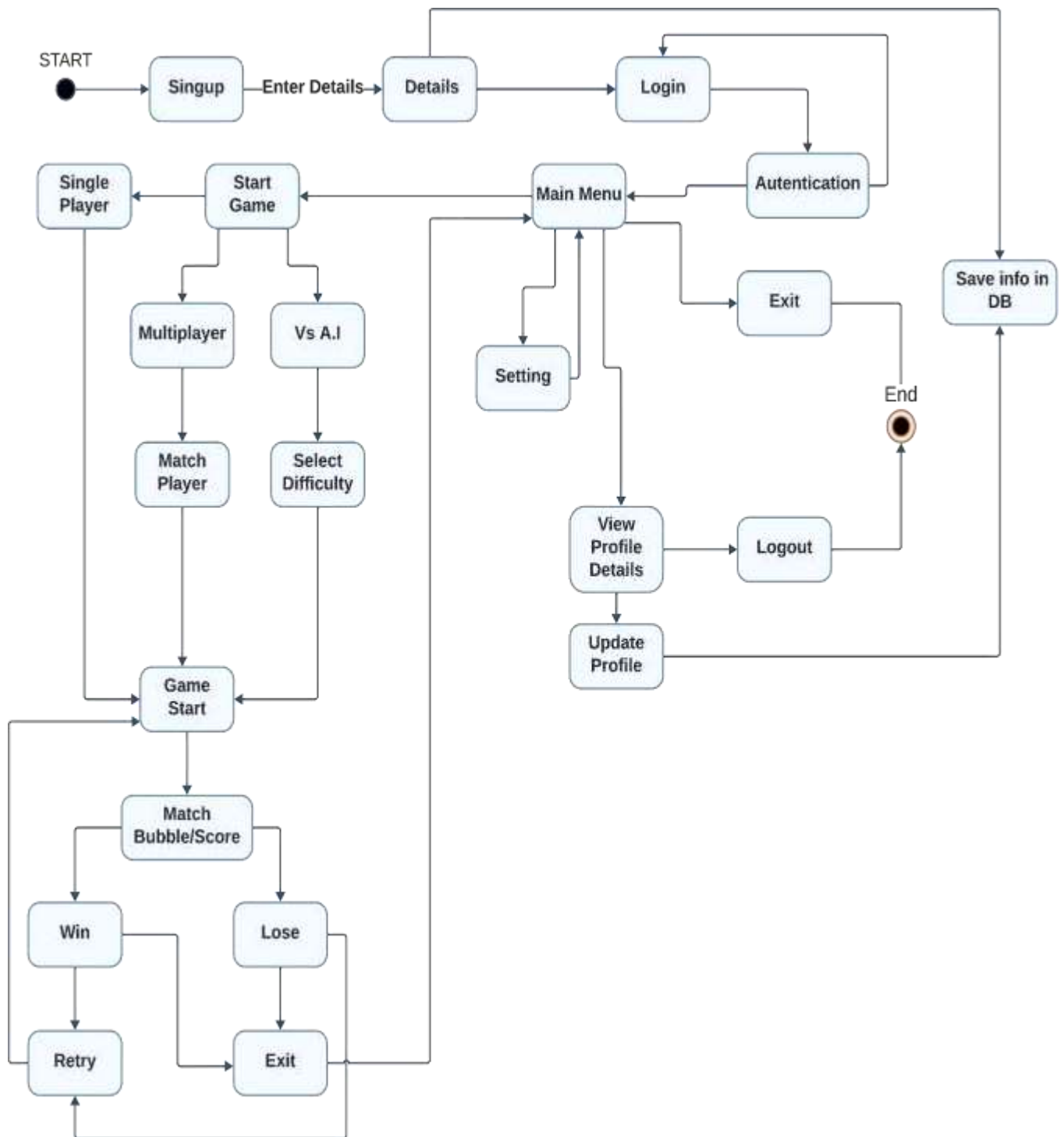


Figure 82 State Transition Diagram

4.9 Component Diagram

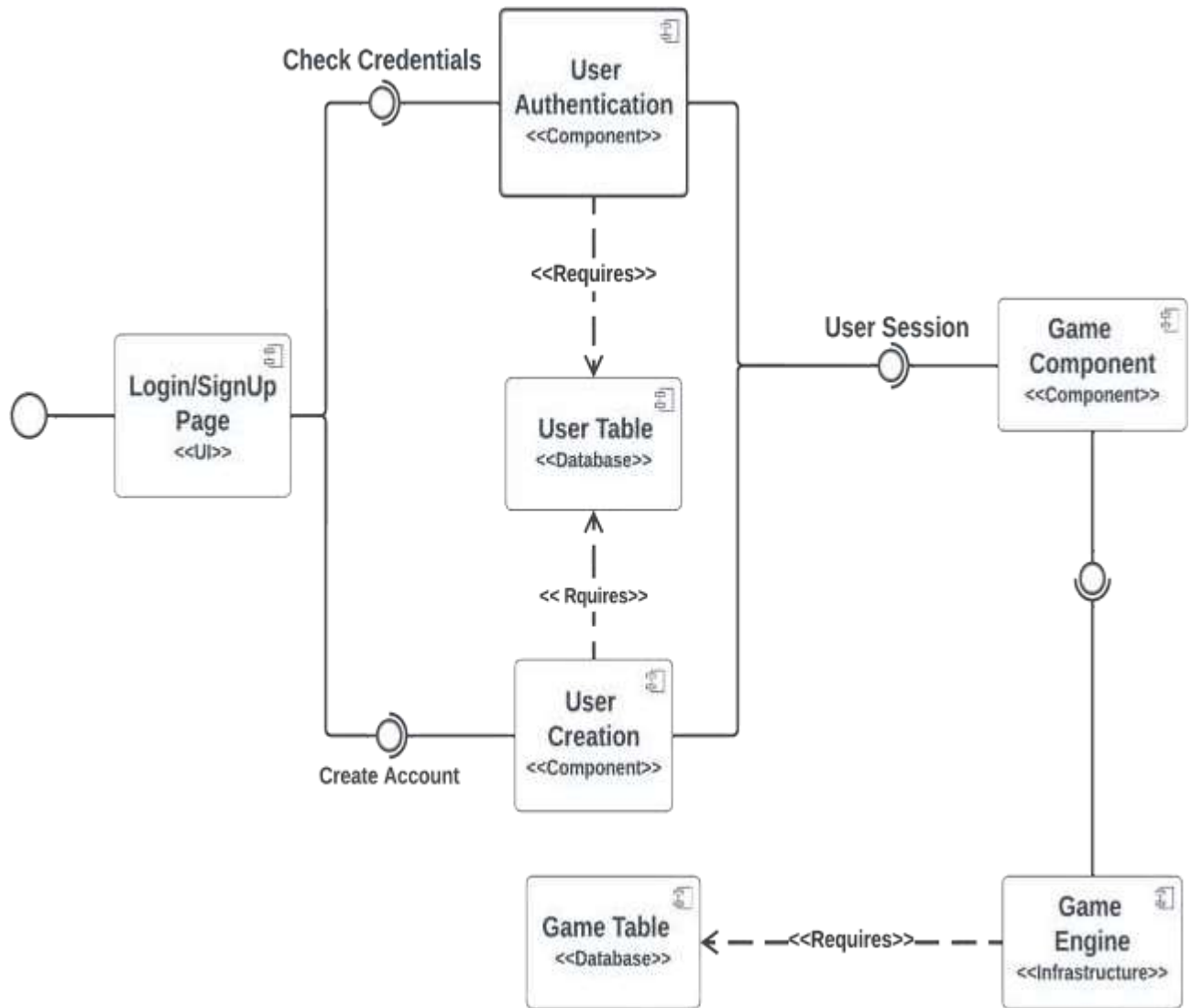


Figure 83 Component Diagram

4.10 Deployment Diagram

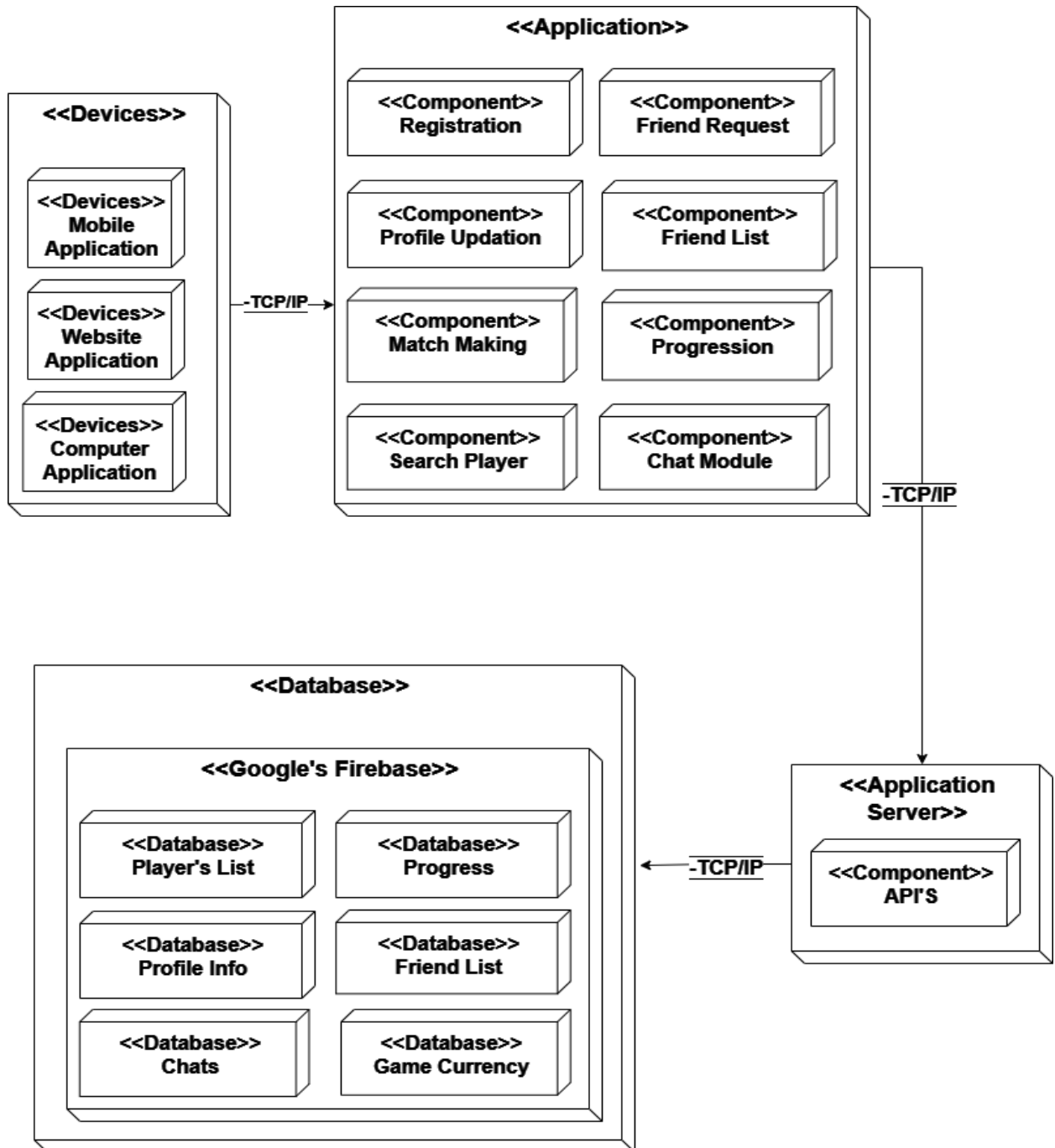


Figure 84 Deployment Diagram

Chapter 5: Testing

5.1 Test Case Specifications

This Testing phase includes all the Test Cases of the functional requirements of project

5.1.1 Test Case for Register Account/SignUp

Positive Test Case	
ID	TC_REGISTER_SUCCESS
Priority	High
Description	This test case is about registering User which will give them access to the application's features.
Reference	Functional Requirement Reference
Users	Playe
Pre-requisites	1. Open the app. 2. Connected to the internet
Steps	1. Open the app. 2. Enter the required details to create account (username, password). 3. Click on Signup.
Input	Name, Phone number, Email, Password, Confirmed Password.
Expected result	Successfully created account on application.
Status	Tested, passed.

Table 59 Positive Test Case For Register Account

Negative Test Case	
ID	TC_REGISTER_FAILURE
Priority	High
Description	This test case is about registering of User which will give them access to the application's features.
Reference	Functional Requirement Reference
Users	Player
Pre-requisites	1. Open the app. 2. Connected to the internet
Steps	1. Open app. 2. Enter the required details to create account (username, password). 3. Click on Sign In.
Input	Invalid Name/ Phone number/ Email/ Password/ unmatched Confirmed Password.
Expected result	Won't create account on application.
Status	Tested, passed.

Table 60 Negative Test Case For Register Account

5.1.2 Test Case for Login Account

	Positive Test Case
ID	TC_LOGIN_SUCCESS
Priority	High
Description	To verify user authentication to system.
Reference	Functional Requirement reference
Users	Player
Pre-requisites	1. System is online. 2. User must have active login credentials provided by system administrator. 3. User has internet access.
Steps	1. Open the app. 2. Enter login email. 3. Enter Password. 4. Press Login.
Input	Login email and password
Expected result	Successfully enters the system and main home page opens.
Status	Tested, passed.

Table 61 Positive Test Case For Login Account

Negative Test Case	
ID	TC_LOGIN_FAILURE
Priority	High
Description	To verify user authentication to system.
Reference	Functional Requirement reference
Users	Player
Pre-requisites	1. System is online. 2. User must have active login credentials provided by system administrator. 3. User has internet access.
Steps	1. Open the app. 2. Enter login email. 3. Enter Password. 4. Press Login.
Input	Incorrect Login email or password or deactivated credentials.
Expected result	Does not allow access to system features and notifies the error.
Status	Tested, passed.

Table 62 Negative Test Case For Login Account

5.1.3 Test Case for Shooting Bubble

	Positive Test Case
ID	TC_BUBBLE_SHOT_SUCCESS
Priority	High
Description	To verify if the system shoot bubbles on user gesture.
Reference	Functional Requirement reference.
Users	Customers/Admin/Vendor
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Enter touch on the screen to select trajectory. 4. Leave the touch screen to shoot bubble.
Input	Screen push and pull gesture on bubble gun.
Expected result	Successfully shoots the bubble to the wanted trajectory.
Status	Tested, passed.

Table 63 Positive Test Case For Shooting Bubble

	Negative Test Case
ID	TC_BUBBLE_SHOT_FAILED
Priority	High
Description	To verify if the system shoot bubbles on user gesture.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Enter touch on the screen to select trajectory. 4. Leave the touch screen to shoot bubble.
Input	Screen push and but not pull gesture on bubble gun. Selecting the wrong trajectory.
Expected result	Bubble remains unchanged not getting shoot.
Status	Tested, passed.

Table 64 Negative Test Case For Shooting Bubble

5.1.4 Test Case for Bubble Match

	Positive Test Case
ID	TC_BUBBLE_MATCH_SUCCESS
Priority	High
Description	To verify if the Bubbles with the same color pop.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. See If they pop.
Input	Launched the bubble at the bubble cluster of the same color.
Expected result	Successfully pops the same color bubble if the cluster's bubbles are more than 2.
Status	Tested, passed.

Table 65 Positive Test Case For Bubble Match

	Negative Test Case
ID	TC_BUBBLE_MATCH_FAILED
Priority	High
Description	To verify if the Bubbles with the same color pop.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. See If they pop.
Input	Launched the bubble at the bubble cluster of the different color.
Expected result	Bubbles of the different color does not pop.
Status	Tested, failed.

Table 66 Negative Test Case For Bubble Match

5.1.5 Test Case for Bubble Match

	Positive Test Case
ID	TC_LEVEL_CLEAR_SUCCESS
Priority	High
Description	To verify if the Player win the game After the winning conditions met.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. Meet the winning requirements.
Input	Meet the requirement by popping the bubbles or by collecting leafs etc.
Expected result	Level cleared and successfully promoted to next level
Status	Tested, passed.

Table 67 Positive Test Case For Clear Level

	Negative Test Case
ID	TC_LEVEL_CLEAR_FAILED
Priority	High
Description	To verify if the Player lose the game After the winning conditions not met.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. Meet the winning requirements.
Input	Winning requirement were not met.
Expected result	Level lost and now the player have to reply it to clear
Status	Tested, passed.

Table 68 Negative Test Case For Clear Level

5.1.6 Test Case for Health Bar

	Positive Test Case
ID	TC_HEALTH_BAR_SUCCESS
Priority	High
Description	To verify if the Player Health bar decreases with each lose.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. Do not reach the minimum Goals to win.
Input	Lose the game.
Expected result	Level lost and health bar decreases by one.
Status	Tested, passed.

Table 69 Positive Test Case For Health Bar

	Positive Test Case
ID	TC_HEALTH_BAR_FAILED
Priority	High
Description	To verify if the Player Health bar does not decrease with each win.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game mode. 3. Launch the bubble at the color cluster. 4. Reach the minimum Goals to win.
Input	Won the game.
Expected result	Level won and health bar does not decreases by one.
Status	Tested, passed.

Table 70 Negative Test Case For Health Bar

5.1.7 Test Case for Coins

	Positive Test Case
ID	TC_COINS_SUCCESS
Priority	High
Description	To verify if the Player lose coins if he buys something.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game. 3. Buy something in exchange for coins.
Input	Buy the something for in-game coins.
Expected result	Coins should be reduced.
Status	Tested, passed.

Table 71 Positive Test Case For Coins

	Negative Test Case
ID	TC_COINS_FAILED
Priority	High
Description	To verify if the Player gain coins if he complete challenges and levels.
Reference	Functional Requirement reference.
Users	Player
Pre-requisites	1. System is online/offline. 2. User must be in game. 3. User has internet access.
Steps	1. Open the app. 2. Enter game. 3. Complete level/Challenges in exchange for coins.
Input	Complete level/Challenges in exchange for coins.
Expected result	Coins should be increased.
Status	Tested, passed.

Table 72 Negative Test Case For Coins

5.1.8 Test Case for logout

Positive Test Case	
ID	TC_LOGOUT_SUCCESS
Priority	High
Description	This use case is about logging out of account from the RO Plant
Reference	Functional Requirement Reference
Users	Player
Pre-requisites	1. App should be open in mobile. 2. Connected to the internet 3. Must be logged in to the App
Steps	1. Click logout button. 2. Selects Yes on the confirmation message.
Input	Click on the “Logout” button to logout account
Expected result	Successfully logged out of account.
Status	Tested, passed.

Table 73 Positive Test Case For logout

Negative Test Case	
ID	TC_LOGOUT_FAILED
Priority	High
Description	This use case is about logging out of account from the RO Plant
Reference	Functional Requirement Reference
Users	Player
Pre-requisites	4. App should be open in mobile. 5. Connected to the internet 6. Must be logged in to the App
Steps	3. Click logout button. 4. Selects No on the confirmation message.
Input	Click logout button and Select No on the confirmation message.
Expected result	Doesn't logout of application.
Status	Tested, passed.

Table 74 Negative Test Case For logout

5.2 Black Box Test Cases

Black box testing also known as Behavioral Testing, is a software testing method in which the internal structure/ design/ implementation of the item being tested is not known to the tester. These tests can be functional or non-functional, though usually functional.

This method is named so because the software program, in the eyes of the tester, is like a black box; inside which one cannot see. This method attempts to find errors in the following categories:

- Incorrect or missing functions
- Interface errors
- Errors in data structures or external database access
- Behavior or performance errors
- Initialization and termination errors

5.2.1 Equivalence Partitions (EP)

Equivalence class partitioning (EP) is a very widely used method to decrease the number of possible test cases that are required to test a system.

5.2.1.1 Create Account

Variables	Valid Classes	Invalid Classes
Email	1. Compulsory field.	1. Symbols other than [-, _, @, .]. 2. Empty Field.
Password	1. Length should be greater than 7 characters. May contain symbols, alphabets [a-z A-Z] and digits [0-9].	1. Length less than 8 characters. 2. Empty field. 3. Special characters/symbols.
Name	1. Compulsory field. 2. May contain alphabets [a-z A-Z].	1. Empty field. Special characters/symbols digits.
Phone	1. Compulsory field. 2. May contain digits [0-9].	1. Empty field. 2. Special characters/symbols and alphabets [a-z A-Z].

Table 75 Black Box Create Account

5.2.1.2 Login Account

Variables	Valid Classes	Invalid Classes
Email	1. Compulsory field. 2. May contain symbols [-, _, @,], alphabets [a-z A-Z] and digits[0-9].	1. Symbols other than [-, _, @, .]. 2. Empty Field.
Password	1. Length should be greater than 7characters. 2. May contain symbols, alphabets[a-z A-Z] and digits [0-9].	1. Length less than 8 characters. 2. Empty field. 3. Special characters/symbols.

Table 76 Black Box Login Account

5.2.1.3 Bubble

Variables	Valid Classes	Invalid Classes
Bubbles	1. Must be a cluster of 3 or more to pop. 2. Must be of same color. 3. The popping bubble must be launched.	1. Must be less than 3. 2. Must be of different color. 3. Not in (abc@gmail.com) format.

Table 77 Black Box Bubbles

5.2.1.4 Level

Variables	Valid Classes	Invalid Classes
Level	1. Must complete the given goals. 2. Still last shooting bubble left. 3. Must not be out of lives.	1. Must not complete the given challenges. 2. Run out of shooting bubbles. 3. Run out of lives

Table 78 Black Box Level

5.2.1.5 Logout

Variables	Valid Classes	Invalid Classes
Email	1. Must be in (abc@gmail.com) format. 2. Email same as given while. 3. registering Compulsory field.	1. Email other than registered account. 2. Empty Field. 3. Not in (abc@gmail.com) format.

Table 79 Black Box Logout

5.2.2 Boundary Value Analysis

A boundary value is an input or output value on the border of an equivalence partition, includes minimum and maximum values at inside and outside boundaries. Normally Boundary value analysis is part of stress and negative testing.

5.2.2.1 Login:

Test Case	Invalid (min-1)	Valid (min)	Valid (min+1)	Valid (max-1)	Valid (max)	Invalid (max+1)
Email Length: (7-254)	6	7	8	253	254	255
Password Length: (8-20)	7	8	9	19	20	21

Table 80 Login Boundary Value Analysis

5.2.2.2 Register/SignUp:

Test Case	Invalid (min-1)	Valid (min)	Valid (min+1)	Valid (max-1)	Valid (max)	Valid (max+1)
Username Length: (3-8)	2	3	4	7	8	9
Email Length: (7-254)	6	7	8	253	254	255
Password Length: (8-20)	7	8	9	19	20	21

*Table 81 Register Boundary Value Analysis***5.2.2.3 Bubbles Color:**

Test Case	Invalid (min-1)	Valid (min)	Valid (min+1)	Valid (max-1)	Valid (max)	Invalid (max+1)
Bubbles Color: (1-10)	0	1	2	9	10	11

*Table 82 Boundary Value Analysis Bubble Color***5.2.2.3 Bubbles cluster:**

Test Case	Invalid (min-1)	Valid (min)	Valid (min+1)	Valid (max-1)	Valid (max)	Invalid (max+1)
Bubbles number Need to pop: (3-50)	2	3	4	49	50	51

Table 83 Boundary Value Analysis Bubble Cluster

5.2.2.4 Forget Password:

Test Case	Invalid (min-1)	Valid (min)	Valid (min+1)	Valid (max-1)	Valid (max)	Invalid (max+1)
Email Length: (7-254)	6	7	8	253	254	255
Password Length: (8-20)	7	8	9	19	20	21

*Table 84 Boundary Value Analysis Forgot Password***5.2.3 Decision Table Testing**

Decision Table is a testing method, which aims to ensure that each one of the possible branch from each decision point is executed at least once and thereby ensuring that all reachable code is executed.

5.2.3.1 Login

Condition 1: The user should enter a valid email.

Condition 2: User should enter valid password

Condition 3: User should click the login button

T = True, F = False, E = Error

Condition	1	2	3	4	5	6	7	8
Condition 1	T	T	T	T	F	F	F	F
Condition 2	T	T	F	F	T	T	F	F
Condition 3	T	F	T	F	T	F	T	F
Result								
Successfully move to home page	T	E	E	E	E	E	E	E

Table 85 Decision Table Testing Login

5.2.3.2 Register/Signup

Condition 1: The user should enter unique username & detail.

Condition 2: User should enter valid Email

Condition 3: User should click the register button

T = True, F = False, E = Error

Condition	1	2	3	4	5	6	7	8
Condition 1	T	T	T	T	F	F	F	F
Condition 2	T	T	F	F	T	T	F	F
Condition 3	T	F	T	F	T	F	T	F
Result								
Successfully Create Account	T	E	E	E	E	E	E	E

Table 86 Decision Table Testing Register

5.2.3.3 Forget Password

Condition 1: The user should click forget password button

Condition 2: User should enter valid/registered Email

Condition 3: User should click the submit button

T = True, F = False, E = Error

Condition	1	2	3	4	5	6	7	8
Condition 1	T	T	T	T	F	F	F	F
Condition 2	T	T	F	F	T	T	F	F
Condition 3	T	F	T	F	T	F	T	F
Result								
Successfully Change password	T	E	E	E	E	E	E	E

Table 87 Decision Table Testing Forgot Password

5.2.3.4 Level Up

Condition 1: Complete the challenge or collect fish/leafs.

Condition 2: Don't run out of shooting bubbles.

T = True, F = False, W = Win, L = Lose

Condition	1	2	3
Condition 1	T	T	F
Condition 2	T	F	F
Result			
Successfully View the profile	W	L	L

Table 88 Decision Table Testing Level Up

5.2.3.5 Bubble Pop

Condition 1: Bubbles should be of same color user.

Condition 2: More than 2 Bubbles should match.

Condition 3: Popping should trigger by shooting bubble.

T = True, F = False, P = Pop , N = Not Pop

Condition 1	T	T	T	T	F	F	F	F
Condition 2	T	T	F	F	T	T	F	F
Condition 3	T	F	T	F	T	F	T	F
Result								
Successfully Change profile	P	N	N	N	N	N	N	N

Table 89 Decision Table Testing Bubble pop

5.2.3.6 Logout:

Condition 1: The user should Logged In

Condition 2: User should click on Logout button

T = True, F = False, E = Error

Condition	1	2	3	4
Condition 1	T	T	F	F
Condition 2	T	F	T	F
Result				
Successfully Logout	T	E	E	E

Table 90 Decision Table Testing Logout

5.2.4 State transition Testing

State Transition testing, a black box testing technique, in which outputs are triggered by changes to the input conditions or changes to 'state' of the system. In other words, tests are designed to execute valid and invalid state transitions.

5.2.4.1 Account Creation/SignUp

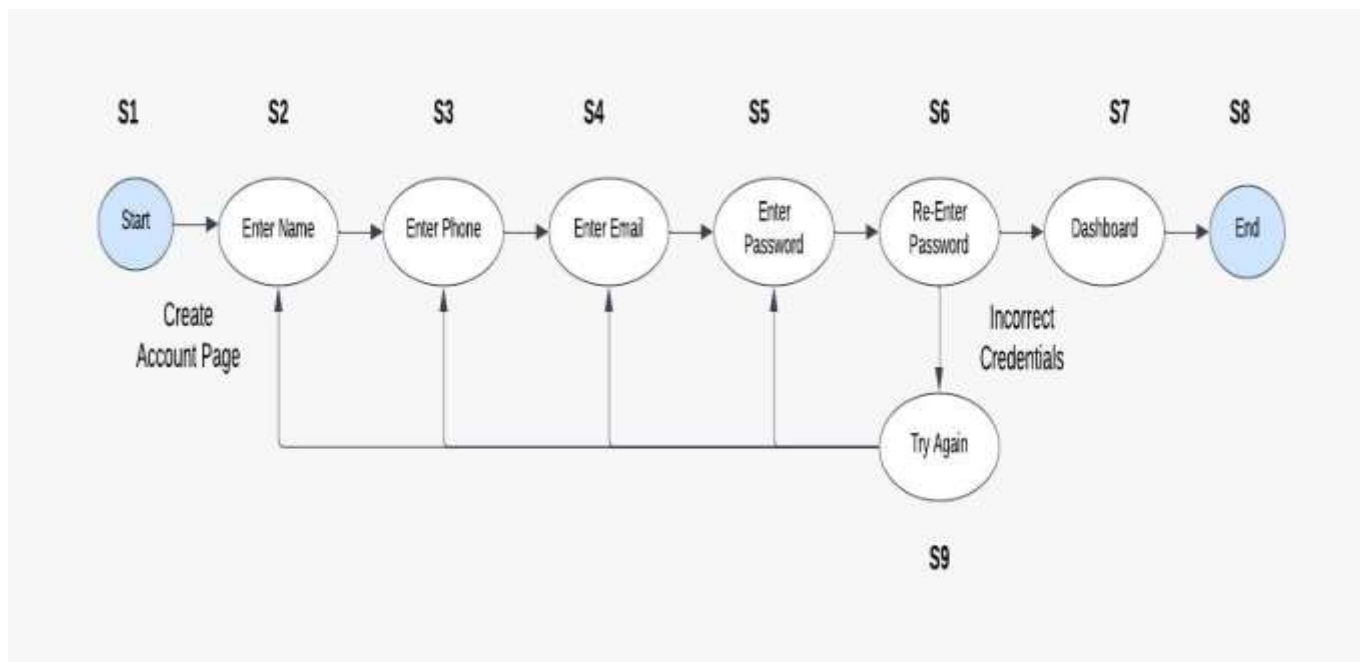


Figure 85 State transition Testing Create Account

Current State	Correct	Incorrect
S1 Start	S2	-
S2 Enter Name	S3	-
S3 Enter Phone no	S4	-
S4 Enter Email	S5	-
S5 Enter Password	S6	-
S6 Re-Enter Password	S7	-
S7 Dashboard	S8	S9
S8 End	-	-
S9 Try Again	S2,S3,S4,S5,S6	-

Table 91 State transition Testing Create Account

5.2.4.2 Login

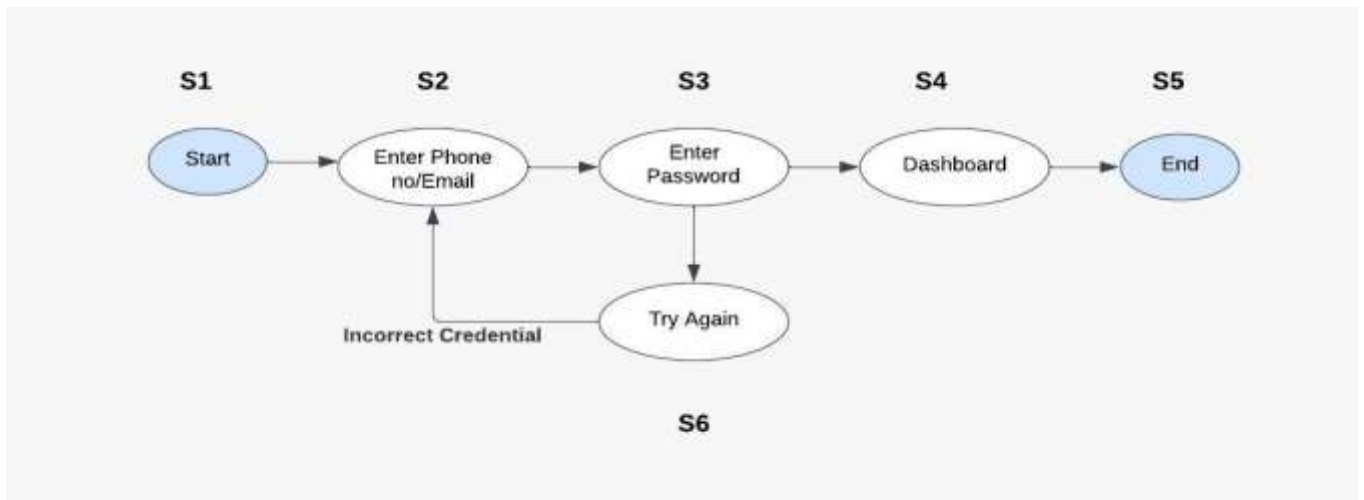


Figure 86 State transition Testing Login Account

Current State	Correct	Incorrect
S1 Start	S2	-
S2 Enter Phone no	S3	-
S3 Enter Password	S3	S6
S4 Dashboard	S5	-
S5 End	-	-
S6 Try Again	S2	-

Table 92 State transition Testing Login Account

5.2.4.3 Bubble shooter game.

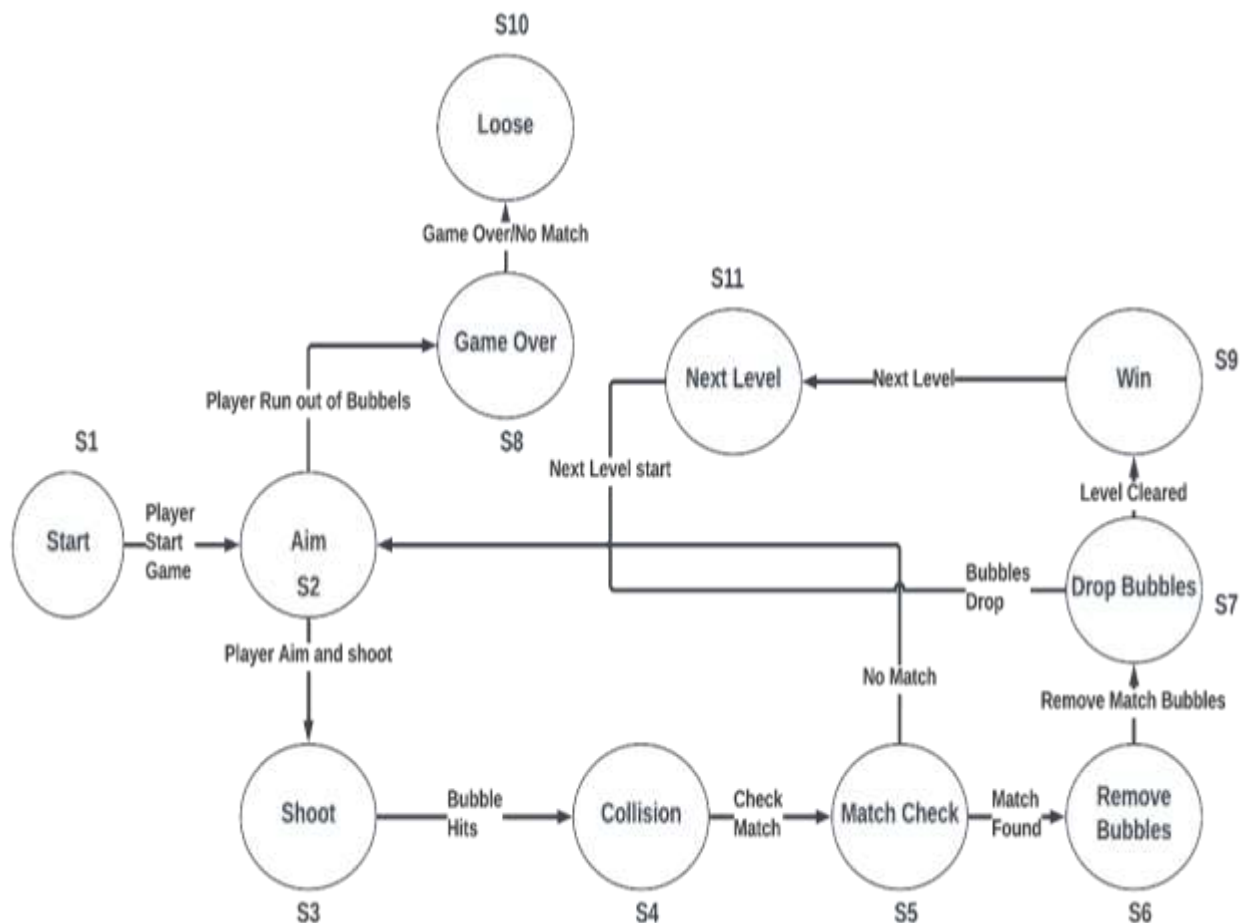


Figure 87 State transition Testing Gameplay

Current State	Event/Action	Next State
S1 Start	Player starts game	S2
S2 Aim	Player aims and shoots bubble	S3
S3 Shoot	Bubble hits another bubble or wall	S4
S4 Collision	Check for matches	S5
S5 Match Check	Match found	S6
S5 Match Check	No match	S2
S6 Remove Bubble	Remove matched bubbles	S7
S7 Drop Bubbles	All bubbles cleared	S11
S7 Drop Bubbles	Bubbles dropped	S2
S11 Next Level	All levels cleared	S9
S11 Next Level	Next level starts	S2
S2 Aim	Player runs out of bubbles	S8
S8 Game Over	Display Game Over	S10

Table 93 State transition Testing Gameplay

5.2.4.4 Logout

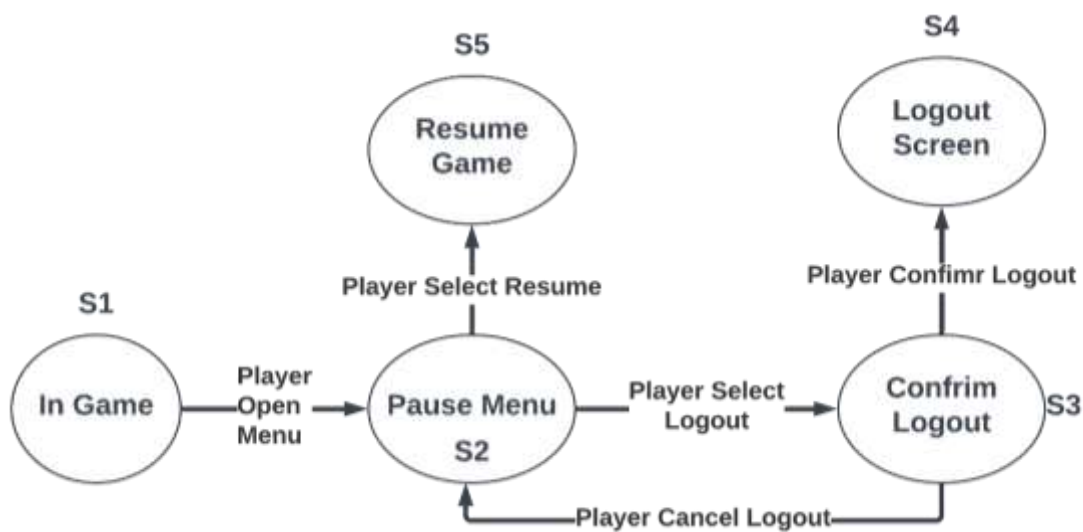


Figure 88 State transition Testing Logout Account

Current State	Event/Action	Next State
S1 In Game	Player opens menu	S2
S2 Pause Menu	Player selects logout	S3
S2 Pause Menu	Player selects resume	S1
S3 Confirm Logout	Player confirms logout	S4
S3 Confirm Logout	Player cancels logout	S2

Table 94 State transition Testing Logout

5.2.5 Use Case Testing

Use Case Testing is a functional black box testing technique that helps testers to identify test scenarios that exercise the whole system on each transaction basis from start to finish.

5.2.5.1 Login

Main Success Scenarios A-Actor	Step	Description
	1	Open login screen.
	2	Enter email and password.
	3	Click the login button.
Extensions	1a	Internet connection failed.
	3a	Internet connection failed.
	3b	Invalid email or password.

Table 95 Use Case Testing Login

5.2.5.2 Register/Signup

Main Success	Step	Description
Scenarios A-Actor	1	Open Register screen.
	2	Insert correct information.
	3	Click the Register button.
Extensions	1a	Internet connection failed.
	3a	Internet connection failed.
	3b	Invalid information.

Table 96 Use Case Testing Signup

5.2.5.3 Start Game

Main Success	Step	Description
Scenarios A-Actor	1	Open Menu Screen.
	2	Select Game mode.
	3	Choose between the mode.
Extensions	3a	Chose vs PC.
	3b	Multiplayer.
	3c	Vs A.I.

Table 97 Use Case Testing Start Game

5.2.5.4 Score

Main Success	Step	Description
Scenarios A-Actor	1	Match Bubbles.
	2	Score.
	3	Win.
Extensions	1a	Collect leaf/fish.
	2a	Score less
	2b	Lose

Table 98 Use Case Score

5.2.5.5 Logout

Main Success	Step	Description
Scenarios	1	Click on Logout button
A-Actor	2	Press Confirm logout
Extensions	1a	Internet connection failed.
	2a	Internet connection failed.

Table 99 Use Case Testing Logout

5.3 White Box Testing

White box testing is a testing technique that examines the program structure and derives test data from the program logic/code. The other names of glass box testing are clear box testing, open box testing, logic-driven testing or path driven testing or structural testing.

5.3.1 Cyclomatic Complexity

Cyclomatic complexity is a software metric that provides a quantitative measure of the logical complexity of a program. When used in the context of the basis path testing method, the value computed for cyclomatic complexity defines the number of independent paths in the basis set of a program and provides you with an upper bound for the number of tests that must be conducted to ensure that all statements have been executed at least once.

The general rule to find Complexity is defined as:

$$M = E - N + 2P$$

E = Number of edges of the graph

N = Number of nodes of the graph

P = Number of connected components

5.3.1.1 Sign Up

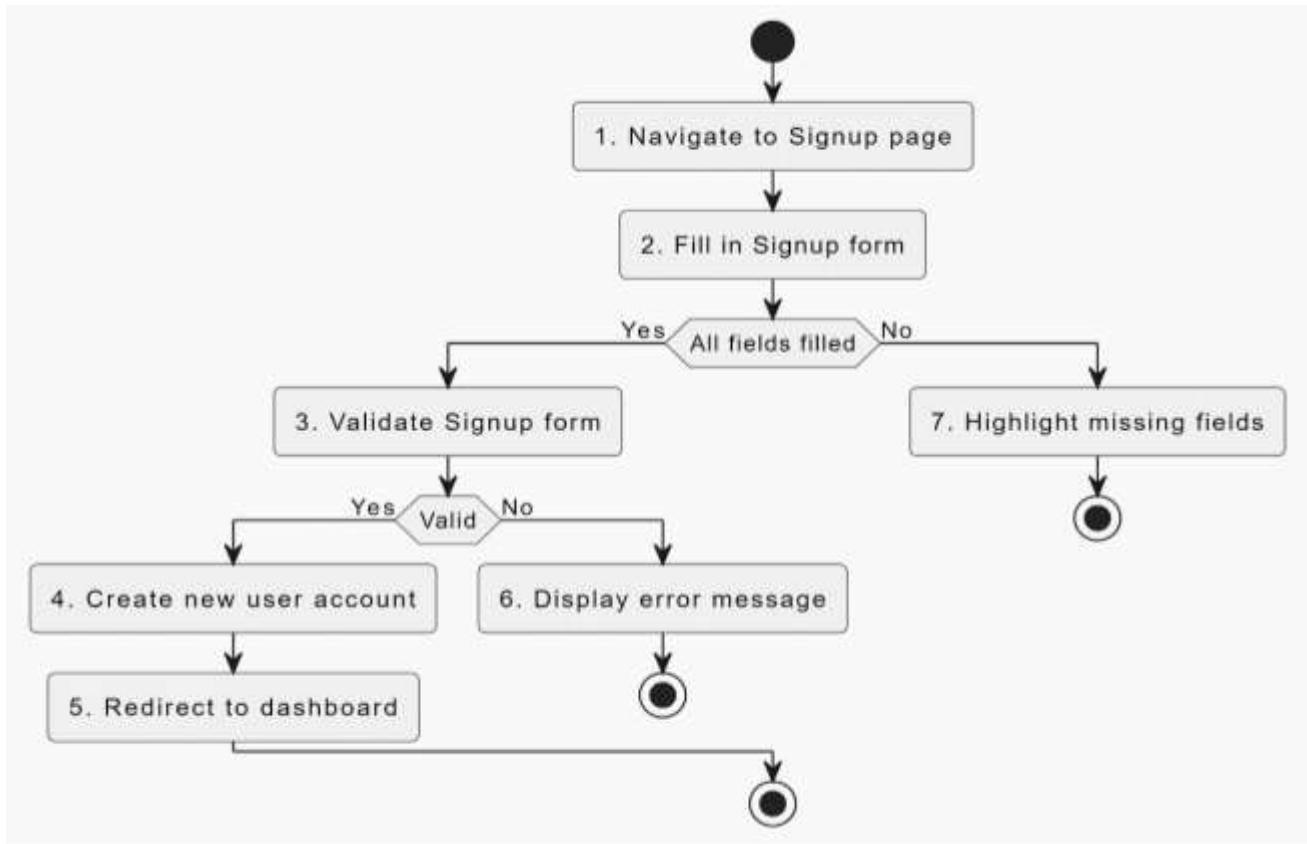


Figure 89 Control flow Testing Signup

Cyclomatic complexity, $V(G)$

$$V(G) = E - N + 2P$$

E = Number of edges of the graph

N = Number of nodes of the graph

P = Number of connected components

Number of nodes = 9

Number of edges = 9

$$V(G) = 9 - 9 + 2 = 2$$

5.3.1.2 Login

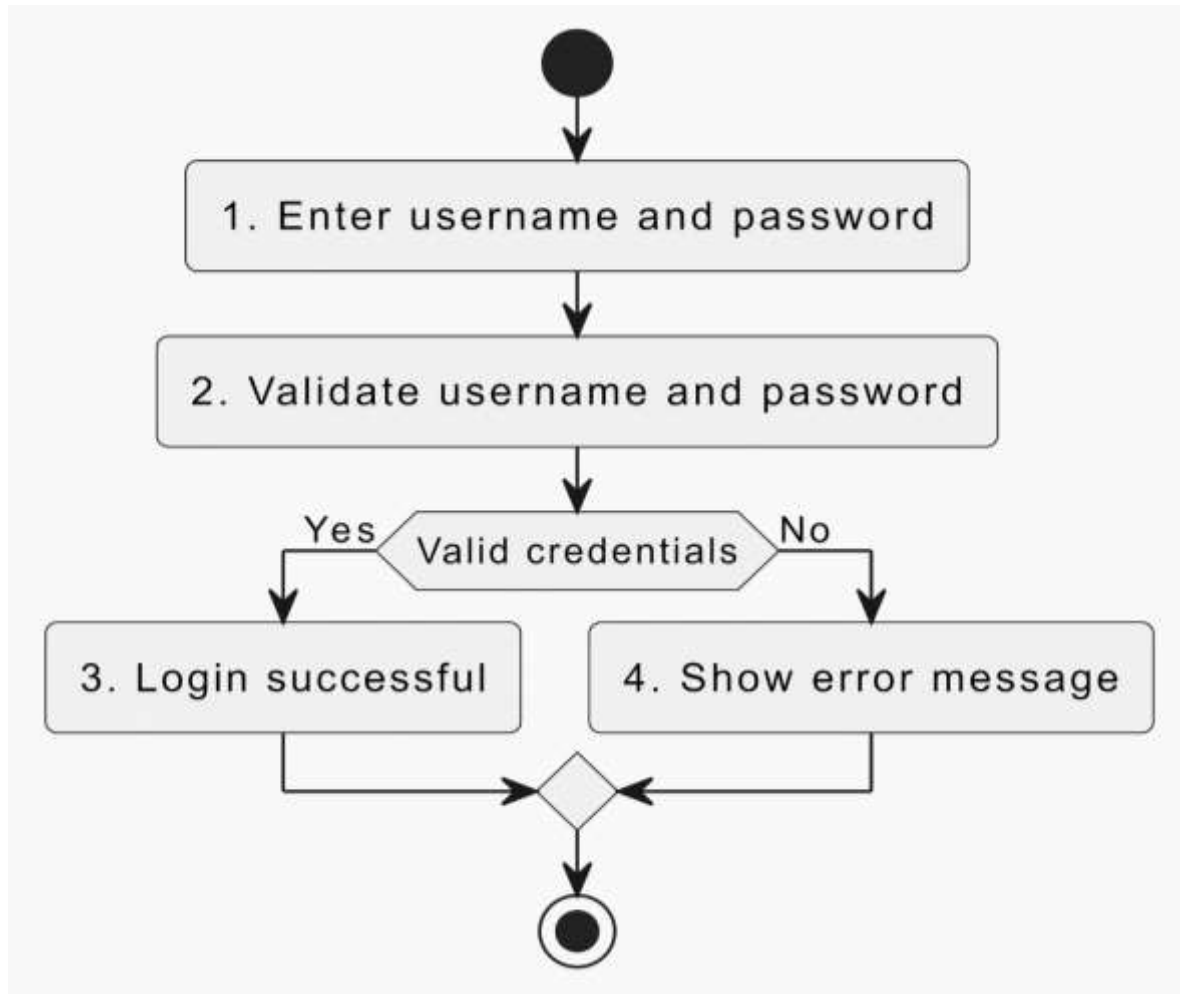


Figure 90 Control Flow Testing Login

Cyclomatic complexity, $V(G)$

$$V(G) = E - N + 2P$$

E = Number of edges of the graph

N = Number of nodes of the graph

P = Number of connected components

Number of nodes = 5

Number of edges = 5

$$V(G) = 5 - 5 + 2 = 2$$

5.3.1.3 Gameplay

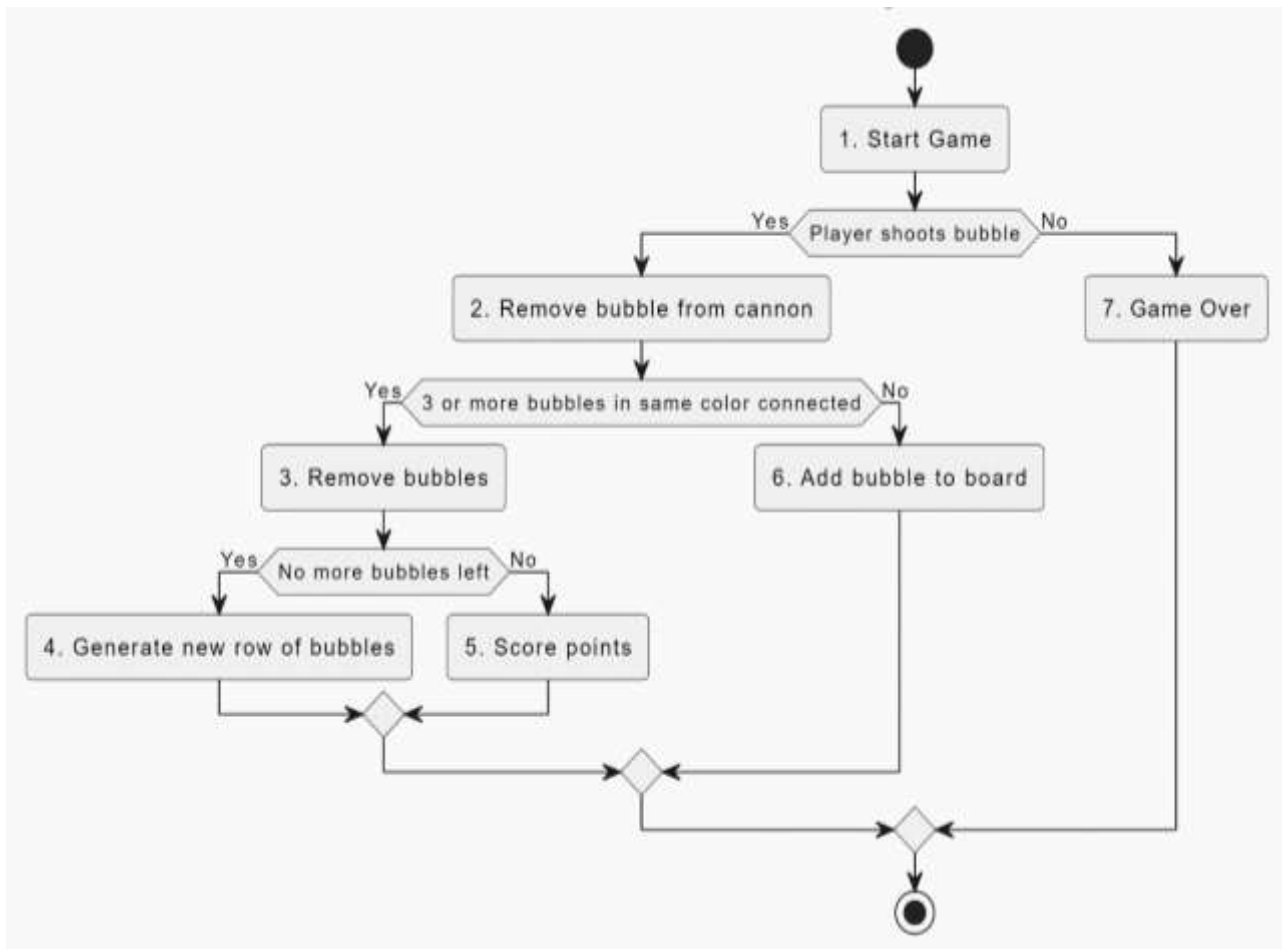


Figure 91 Control Flow Testing Gameplay

Cyclomatic complexity, $V(G)$

$$V(G) = E - N + 2P$$

E = Number of edges of the graph

N = Number of nodes of the graph

P = Number of connected components

Number of nodes = 10

Number of edges = 10

$$V(G) = 10 - 10 + 2 = 2$$

5.3.1.4 Logout

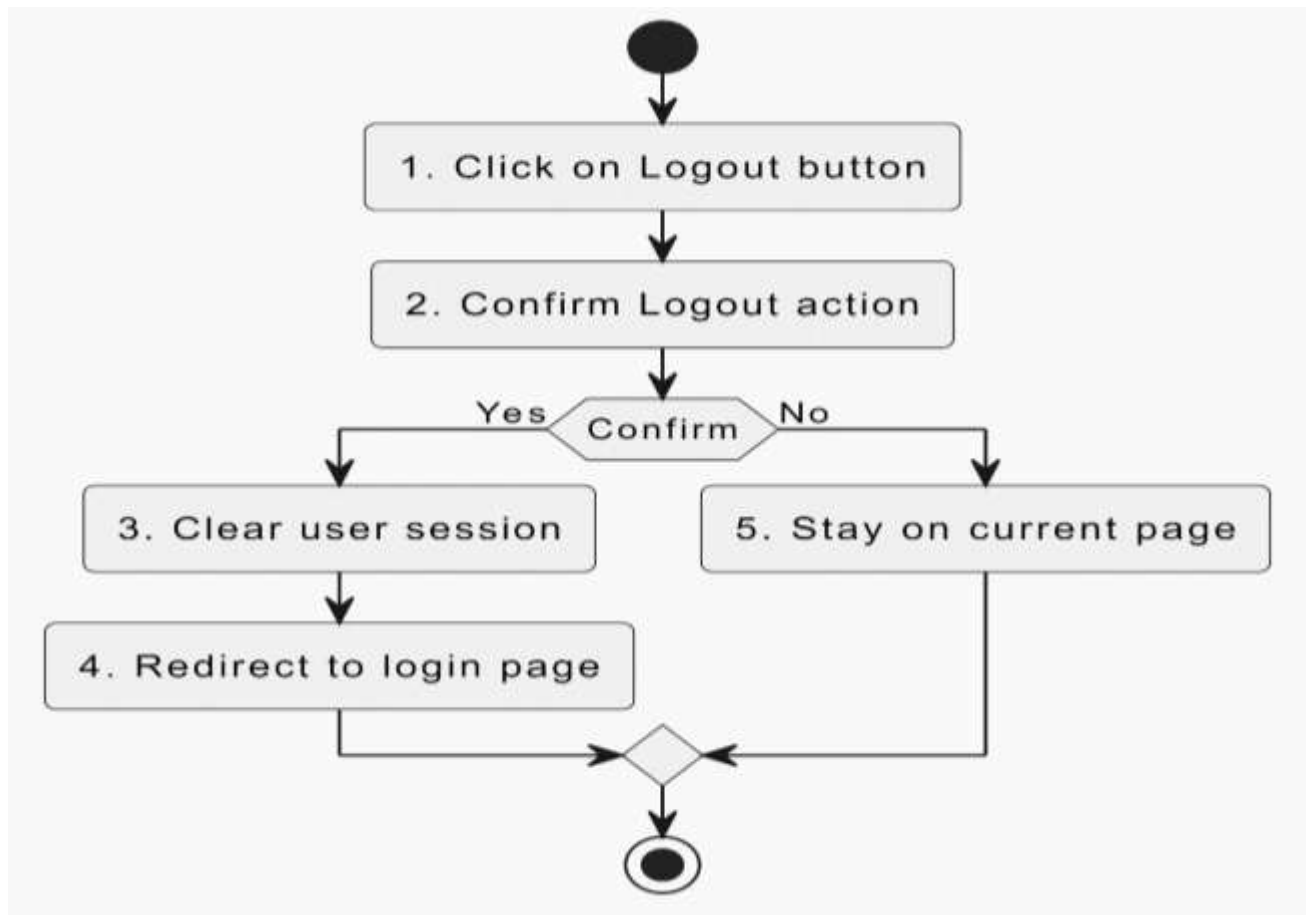


Figure 92 Control Flow Testing Logout

Cyclomatic complexity, $V(G)$

$$V(G) = E - N + 2P$$

E = Number of edges of the graph

N = Number of nodes of the graph

P = Number of connected components

Number of nodes = 10

Number of edges = 10

$$V(G) = 6 - 6 + 2 = 2$$

5.4 Performance testing

Performance testing, a non-functional testing technique performed to determine the RO PLANT Web-App parameters in terms of responsiveness and stability under various workload. Performance testing measures the quality attributes of the system, such as scalability, reliability and resource usage.

- Prepared a dedicated testing environment on a robust server configuration featuring 6 CPU cores and 16GB of RAM. Employed an advanced load testing tool to replicate realistic user interactions and traffic patterns.
- Ensured that accessing the game's main menu had a swift response time of less than 3 seconds, guaranteeing players quick access to gameplay without frustrating delays.
- Thoroughly verified that the game maintained optimal response times throughout various stages, including level loading and active gameplay, to deliver a seamless and immersive experience for players.
- Methodically determined the game's maximum player capacity, carefully monitoring performance metrics to identify any signs of strain or instability before reaching critical thresholds.
- Conducted comprehensive performance tests under different load scenarios, ranging from light to heavy player activity, to assess the game's resilience and responsiveness across varying levels of demand.

5.5 Stress testing

The stress testing of the Clash of Bubbles game demonstrated that it performs well under moderate to high loads but faces performance degradation and stability issues under extreme conditions. The game maintained stable performance with a substantial number of concurrent users, but beyond a certain point, server response times increased and minor lag appeared, with significant issues at peak user loads. High-frequency actions were handled smoothly up to a certain threshold, beyond which frame rate drops became pronounced. Resource utilization showed efficient use of CPU and GPU, with memory usage increasing proportionally with the load. Network tests revealed minimal impact with moderate latency, but higher packet loss caused noticeable delays and synchronization issues. The game effectively recovered from brief network interruptions. To improve, enhancements in server scalability, client optimization for intense actions, better handling of severe network instability, and further resource management optimizations are recommended.

5.6 System Testing

System Testing (ST) is a black box testing technique performed to evaluate the complete system the system's compliance against specified requirements. In System testing, the functionalities of the system are tested from an end-to-end perspective.

System Testing is usually carried out by a team that is independent of the development team in order to measure the quality of the system unbiased. It includes both Functional and Non-Functional testing.

- **Functional Testing:** We systematically tested each function of the game, including aiming, shooting, bubble collision detection, scoring, level progression, and game over conditions. This ensured that all gameplay mechanics worked as intended and provided an enjoyable gaming experience for players.
- **User Interface Testing:** We thoroughly examined the user interface elements such as menus, buttons, icons, and visual effects to ensure they were intuitive, visually appealing, and responsive across different devices and screen resolutions. This included verifying that all interface elements were correctly positioned, sized, and aligned.
- **Compatibility Testing:** We tested the game on various devices, operating systems, and web browsers to ensure compatibility and functionality across different platforms. This included testing on desktop computers, laptops, tablets, and smartphones to ensure a consistent experience for all users.
- **Performance Testing:** As discussed earlier, we conducted performance testing to evaluate the game's responsiveness, stability, and scalability under various load conditions. This included assessing response times, server resource usage, and error rates to identify any performance bottlenecks and optimize the game's performance.
- **Localization Testing:** If the game was intended for a global audience, we conducted localization testing to ensure that all text, graphics, and audio elements were correctly translated and culturally appropriate for different regions and languages. This ensured that the game was accessible and engaging to players worldwide.

5.7 Regression Testing

Regression testing a black box testing technique that consists of re-executing those tests that are

impacted by the code changes. These tests should be executed as often as possible throughout the software development life cycle. Types of Regression Tests:

Final Regression Tests

A "final regression testing" is performed to validate the build that hasn't changed for a period of time. This build is deployed or shipped to customers.

We thoroughly checked the bubble shooter game to ensure recent updates hadn't introduced new issues. We selected critical test cases covering affected areas, executed them to confirm functionality, and addressed any problems promptly. After fixes, we validated changes and documented results, ensuring the game remained stable and enjoyable for players.

Regression Tests

A normal regression testing is performed to verify if the build has NOT broken any other parts of the application by the recent code changes for defect fixing or for enhancement.

- We went back and retested all the features and functions that we had already checked before and made sure that everything still worked as expected, even after we made changes or updates to the game.
- If we found any issues, we fixed them right away to make sure the game stayed in good shape. By doing this, we made sure that new updates didn't accidentally break anything that was already working fine.

5.7.1 Selecting Regression Tests

- Requires knowledge about the system and how it affects by the existing functionalities.
- Tests are selected based on the area of frequent defects.
- Tests are selected to include the area, which has undergone code changes many a times.
- Tests are selected based on the criticality of the features.

This method of game testing can be called a retest of the functionality of the system feature. By regression testing it is checked if after the new modification of the system bugs or errors appeared. In this way, the quality of the system is maintained at high standards.

5.7.2 Regression Testing Steps

Regression tests are the ideal cases of automation which results in better **Return on Investment**

(ROI).

- Select the Tests for Regression.
- Choose the apt tool and automate the Regression Tests
- Verify applications with Checkpoints
- Manage Regression Tests/update when required
- Schedule the tests
- Integrate with the builds
- Analyze the results

Chapter 6: Tools and Techniques

6.1 Introduction

"Clash of Bubble" is developed using Unity, a powerful and versatile game engine known for its capabilities in creating engaging and interactive games. This chapter provides detailed insights into the tools and techniques employed in the development process, showcasing how Unity's extensive features are utilized to achieve the game's objectives.

6.2 Unity Game Engine

Unity was selected due to its robustness in 2D game development and its cross-platform compatibility. Key features of Unity utilized in the game development include:

- **Unity Editor:** The main workspace for developers, the Unity Editor provides tools for designing scenes, importing assets, and real-time testing. This allows for immediate feedback and adjustments during the development process.
- **Unity Hub:** This tool helps in managing multiple Unity projects, version control, and team collaboration, ensuring that the development process is streamlined and organized.
- **C# Scripting:** Unity uses C# for scripting, which is employed to handle game logic, AI behaviors, and user input. Scripts are crucial for creating interactive and dynamic gameplay elements.

6.3 2D Development in Unity

Unity's 2D toolset is essential for crafting the vibrant and lively world of "Clash of Bubble." Techniques and tools used include:

- **Sprite Management:** Sprites are the primary visual components of the game. Using the Sprite Editor, developers can slice and manage textures efficiently, creating smooth animations and interactions.
- **Tilemaps:** Tilemaps are used for level design, allowing developers to build complex and dynamic environments with ease. This tool enables the creation of detailed and varied levels that enhance the player's experience.
- **Physics Engine:** Unity's 2D physics engine is used to implement realistic bubble interactions and collisions, adding depth to the gameplay. This makes the bubble-popping mechanics feel natural and responsive.

6.4 Customizable AI

To provide a tailored experience, "Clash of Bubble" incorporates customizable AI difficulty levels:

- **AI Behavior Trees:** These are used to design complex and adaptive AI behaviors. By structuring AI logic in behavior trees, developers can create AI that reacts intelligently to player actions.
- **Difficulty Settings:** Players can choose from easy, hard, and very hard modes. These settings adjust the AI's strategies and response times, ensuring that players of all skill levels can enjoy the game. For instance, in harder modes, the AI might predict player movements more accurately and react faster.

6.5 Multiplayer Functionality

Multiplayer capabilities enhance the social and competitive aspects of the game:

- **Unity's Networking API:** This API enables real-time multiplayer matches. It includes features such as player matchmaking, lobbies, and synchronous gameplay, allowing players to compete against each other in real-time.
- **Photon Unity Networking (PUN):** For advanced networking needs, PUN is used to provide reliable and scalable multiplayer experiences. It ensures that players can connect and interact seamlessly, regardless of their location.

6.6 User Interface Design

A well-designed user interface (UI) is critical for user engagement:

- **Unity UI Tools:** These tools are used to create menus, buttons, and other interactive elements. The UI is designed to be responsive, ensuring it looks good and functions well on various devices and screen sizes.
- **Canvas and UI Elements:** The Canvas component in Unity is used to manage all UI elements, ensuring they are properly layered and aligned. This includes health bars, score counters, and in-game menus, all designed to be intuitive and visually appealing.

6.7 Visual and Performance Optimization

Ensuring smooth performance across platforms is essential:

- **Optimization Techniques:** These include sprite atlasing (grouping sprites into a single texture to reduce draw calls), level of detail (LOD) management (adjusting the detail of objects based on their distance from the camera), and efficient memory usage to ensure the game runs smoothly.
- **Performance Testing:** Regular testing is conducted to identify and address performance bottlenecks. This involves profiling the game to monitor frame rates, memory usage, and

load times, ensuring the game provides a seamless experience.

6.8 Testing and Iteration

Continuous testing and iteration are crucial for delivering a high-quality game:

- **Playtesting:** Involves having diverse groups of players test the game to gather feedback on gameplay mechanics, balance, and overall enjoyment. This feedback is essential for making informed improvements.
- **Bug Tracking and Fixes:** Tools like Unity Cloud Build and Unity Analytics are used to track bugs and performance issues. This allows developers to quickly identify and fix problems, improving the game's stability and performance.
- **Agile Development:** An agile approach is adopted to allow for iterative improvements. This means regularly updating the game based on player feedback and industry trends, ensuring it remains engaging and up-to-date.

Chapter 7: Summary and Conclusion

7.1 Summary

"Clash of Bubble" is a vibrant and engaging game crafted with Unity, aiming to provide players with a visually captivating and customizable gaming experience. Set in a colorful world, the game revolves around the simple yet addictive mechanic of popping bubbles across various levels. Its intuitive controls ensure that players of all skill levels can enjoy the game without feeling overwhelmed. Developed to address the common issue of games lacking personalization and suitable challenges, "Clash of Bubble" offers a tailored experience for each player. Its objectives are ambitious, ranging from creating a game compatible with both web browsers and mobile devices to incorporating customizable AI difficulty levels and multiplayer functionality for real-time matches. In a market populated by similar bubble-popping games like Bubble Witch Saga and Bubble Shooter Classic, "Clash of Bubble" distinguishes itself by offering features such as adjustable difficulty levels and multiplayer options, enhancing player engagement and interaction. Leveraging Unity's extensive toolkit, the development process involves utilizing tools like the Unity Editor, Unity Hub, C# scripting, sprite management, tilemaps, physics engine, AI behavior trees, and networking APIs like Photon Unity Networking (PUN) to bring the game to life. Overall, "Clash of Bubble" promises an immersive and enjoyable gaming experience tailored to individual preferences and skill levels.

7.2 Conclusion

The development of "Clash of Bubble" has been a detailed and methodical process, focusing on creating a game that stands out in the crowded puzzle gaming market. Key conclusions from this project include:

- **Unity's Versatility:** Unity's extensive capabilities in 2D game development, AI customization, and multiplayer integration have proven invaluable. Its robust toolset has enabled the development of a game that is not only engaging but also scalable across different platforms.
- **Customizable Gameplay:** The integration of customizable AI difficulty levels ensures that the game can cater to a wide range of player skills. This feature allows players to set the challenge level according to their preferences, maintaining engagement and preventing frustration or boredom.

- **Multiplayer Functionality:** Real-time multiplayer functionality enhances the social aspect of the game, enabling players to compete and interact with friends or other players globally. This feature adds a competitive edge and fosters community building within the game.
- **User Interface and Experience:** An intuitive and visually appealing user interface significantly enhances the gaming experience. The responsive design ensures the game functions well on various devices, providing a seamless user experience.
- **Visual and Performance Optimization:** Optimization techniques, such as sprite atlasing, level of detail (LOD) management, and efficient memory usage, ensure smooth performance across platforms. Regular performance testing and bug fixes are integral to maintaining a high-quality gaming experience.
- **Continuous Improvement:** The commitment to continuous testing, player feedback, and agile development has been crucial in refining the game. This iterative process ensures that the game remains balanced, engaging, and in line with player expectations and industry trends.

Chapter 8: User Manual

Main Screen

When you open the game, you are greeted with the main screen. This screen includes:

- **Play Button:** Tapping this will start the game.
- **Settings Icon:** Located at the bottom left corner, this allows you to access the game settings.
- **Characters and Animations:** Various characters and elements that make the main screen lively and engaging.



Figure 93 Main Screen

Game Map

The game map shows your progress through different levels. Each level is represented by a node, and you can track which levels you have completed and which ones are locked.

- **Lives:** Displayed at the top left with a heart icon.
- **Coins:** Displayed at the top right, showing the number of coins you have collected.
- **Current Level:** Indicated by the character icon on the map.



Figure 94 Game Map

Game Mode

1. VS CPU

Select this option to play against the computer. This mode allows you to practice your skills against an AI opponent.

2. SINGLE

Choose this option to play a single-player game. This mode is designed for solo play, where you can progress through various levels and challenges.

3. MULTIPLAYER

This option allows you to play with other players. You can compete against friends or other online players in this mode.

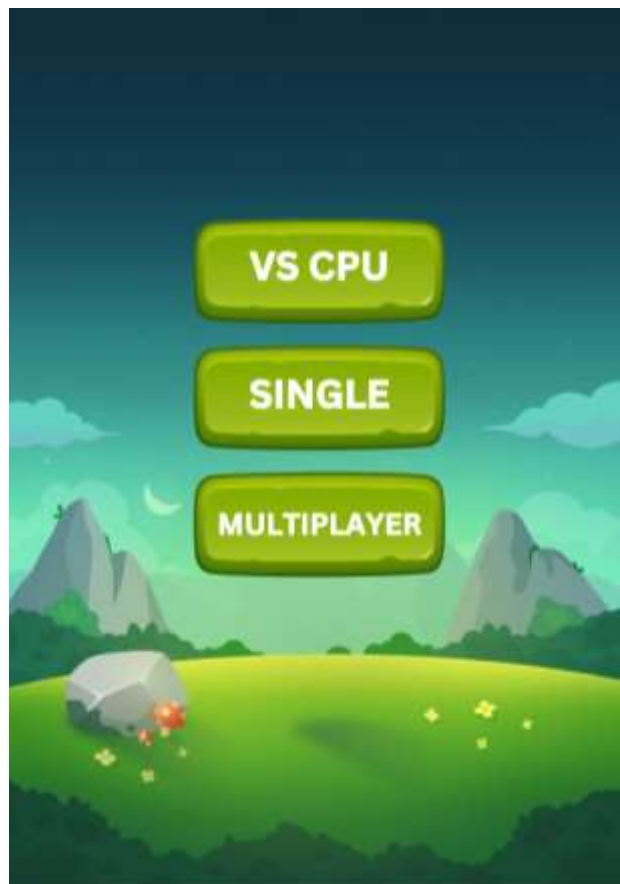


Figure 95 Game Mode

Gameplay

During gameplay, you will see a screen filled with bubbles that you need to shoot and match to clear the level.

- **Aiming:** Use the bottom shooter to aim and fire bubbles at the cluster above.
- **Score and Stars:** Your score and star rating for the level are displayed at the top.
- **Special Bubbles:** Some bubbles have special abilities or require specific strategies to clear.

How to Play

- **Aim:** Drag your finger to aim the shooter towards the bubbles.
- **Shoot:** Release your finger to shoot the bubble.
- **Match:** Try to match three or more bubbles of the same color to clear them from the board.
- **Clear the Level:** Clear all the bubbles or achieve the target score to complete the level.
- **Progress:** Move to the next level and continue your adventure.



Figure 96 Game Play

Multiplayer or Vs CPU Gameplay Interface

1. Objectives Panel (Top Left):

- Orange Tiger Heads: 60
- Purple Hippos: 60
- Green Leaves: 10

These are the items you and your opponent need to collect to win the match. The first to complete the objectives wins.

2. Score Display (Top Right):

- Player Scores: Displays the scores of both players (you and your opponent/CPU).
- Stars: Represent each player's progress and achievement rating.

3. Game Board:

- Contains various colored animal faces, some encased in ice.
- You need to clear these by matching at least three of the same type.
- Each player has their own board.

4. Current Shooter (Bottom Center for each player):

- The blue character with a crown represents the current shooter for Player 1.
- The animal face (purple hippo) in front of the shooter is the next bubble Player 1 will shoot.
- The opponent (or CPU) has a similar shooter on their side of the screen.

5. Shooter Control (Bottom Left for each player):

- The black circle button is likely the control for shooting the next bubble.
- Each player has their own control button.

6. Opponent's Board (Top Center):

- Displays the game board of the opponent or CPU.
- Shows their progress in clearing animal faces and achieving the objectives.

7. Status Indicators (Center Top):

- Displays real-time status updates, such as who is leading, and any power-ups or special moves activated.



Figure 97 Multiplayer or Vs CPU Gameplay Interface

Pause Mode

1. Continue

Tap this button to resume your game from where you left off.

2. Restart

Select this option if you wish to start the current level over. This is useful if you want to improve your performance or correct mistakes.

3. Quit

Use this button to exit the current game session. This will take you back to the main menu.

4. Additionally, at the bottom of the pause menu, there are two icons for sound settings:

5. Music Icon

Tap to toggle the background music on or off.

6. Sound Icon

Tap to toggle the game sound effects on or off.

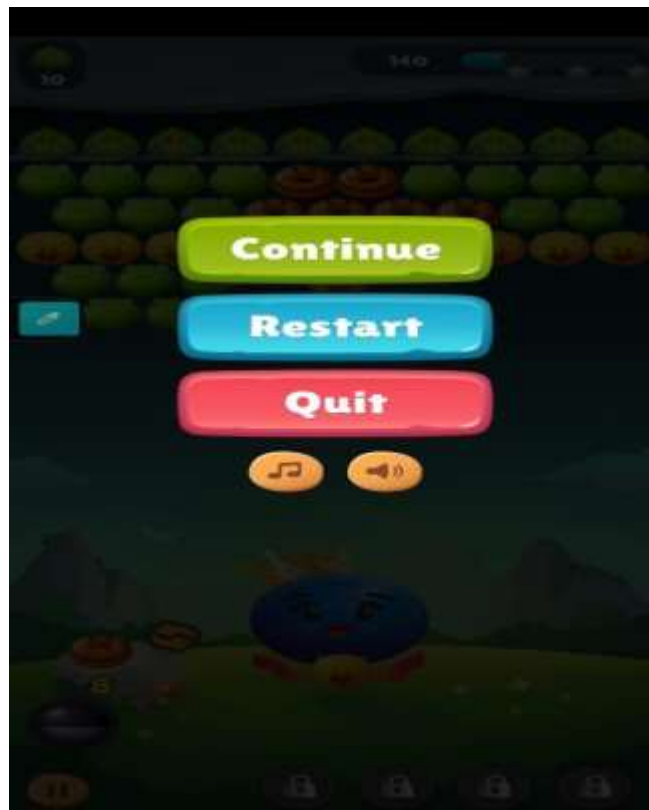


Figure 98 Pause Mode

Level Selection and Objectives

1. Level Indicator:

- Shows the current level.

2. Stars Display:

- Indicates a three-star rating system for level completion. Achieving all three stars is the goal.

3. Objectives and Resources:

- **Bubbles:** 10 number of bubbles you can shoot in this level
- **Collect these items:** Green Leaves – 10, the main objective is to collect 10 green leaves

4. Play Button:

- Starts the level when clicked.

5. Status Indicators:

- **Hearts:** Indicates the number of lives you have left.
- **Coins:** 19,300 your current coin balance.



Figure 99 Level Selection and Objectives

Setting:

In the settings menu, you can customize your gaming experience:

- **Music:** Toggle the game music on or off.
- **Sound:** Toggle the game sound effects on or off.
- **Reset Progress:** Reset your game progress to start over.



Figure 100 Setting

Chapter 9: Lessons Learned and Future Work

9.1 Lessons Learned

The development of "Clash of Bubble" has been a rich learning experience, offering numerous insights and valuable lessons that will inform future projects. Key lessons include:

1. **The Importance of Planning and Documentation:** Comprehensive planning and thorough documentation were crucial for the successful development of "Clash of Bubble." Detailed project plans, clear objectives, and well-maintained documentation ensured that all team members were aligned and could contribute effectively.
2. **Effective Use of Unity:** Utilizing Unity's extensive toolset effectively was a critical factor in the project's success. The versatility and power of Unity's 2D capabilities, AI customization tools, and networking features enabled the creation of a polished, high-quality game. Familiarity with Unity's workflow and best practices significantly improved development efficiency.
3. **Iterative Development and Agile Methodologies:** Adopting an iterative development process and agile methodologies allowed the team to respond quickly to feedback and make continuous improvements. Regular playtesting sessions provided valuable insights into gameplay balance, user experience, and performance issues, leading to a more refined final product.
4. **Player Feedback is Invaluable:** Engaging with players early and often through beta testing and feedback sessions was instrumental in shaping the game's final form. Understanding player preferences, challenges, and suggestions helped prioritize features and improvements that enhanced the overall gaming experience.
5. **Optimizing for Performance:** Ensuring smooth performance across various devices was a significant challenge. Techniques such as sprite atlasing, efficient memory management, and rigorous performance testing were essential in minimizing loading times and preventing lag. This optimization work was crucial for delivering a seamless experience on both web browsers and mobile devices.
6. **Balancing Innovation with Familiarity:** While introducing new features like customizable AI and multiplayer functionality, it was important to maintain the core mechanics that make bubble shooter games enjoyable. Striking the right balance between innovation and familiarity ensured that the game was both fresh and accessible to players.

9.2 Future Work

The development of "Clash of Bubble" is an ongoing journey, with several areas identified for future improvement and expansion. The following are key aspects that will be the focus of future work:

1. **Expanding Game Content:** Adding new levels, challenges, and bubble types will keep the game fresh and engaging for players. Regular content updates will encourage continued play and attract new users.
2. **Enhanced Customization Options:** Expanding the range of customizable options for AI difficulty and player preferences will allow for an even more personalized gaming experience. Introducing features like user-generated content or level editors could also enhance player engagement.
3. **Advanced Multiplayer Features:** Building on the existing multiplayer functionality, future updates could include additional modes such as cooperative gameplay, tournaments, and leaderboards. These features will foster a stronger community and provide new ways for players to interact and compete.
4. **Cross-Platform Synchronization:** Implementing features that allow players to sync their progress across different devices will enhance the user experience. This could involve cloud saving and cross-platform accounts, ensuring a seamless transition between playing on mobile devices and web browsers.
5. **Localization and Accessibility:** Expanding the game's reach by localizing it into multiple languages and ensuring it is accessible to players with disabilities will open it up to a broader audience. Implementing features like adjustable text sizes, colorblind modes, and alternative input methods can make the game more inclusive.
6. **Monetization Strategies:** Exploring sustainable monetization options, such as in-app purchases for cosmetic items, premium levels, or ad-supported free versions, will help fund ongoing development while respecting the player experience.
7. **Integration of Advanced Analytics:** Utilizing advanced analytics to track player behavior, preferences, and performance will provide deeper insights into how the game is played. This data can inform future updates and improvements, ensuring the game continues to meet player needs and expectations.
8. **Community Engagement and Support:** Building a strong community through active

engagement on social media, forums, and within the game itself will be crucial for the game's long-term success. Providing timely support, listening to player feedback.

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